TPS92662A-Q1
SLUSDL7A – OCTOBER 2019 – REVISED JANUARY 2020

TPS92662A-Q1 High Brightness LED Matrix Manager with EMI Mitigation Techniques

1 Features

- AEC-Q100 qualified for automotive applications
  - Grade 1: –40°C to 125°C ambient operating temperature
  - Device HBM classification level H1C
  - Device CDM classification level C5
- Input voltage range: 4.5 V to 60 V
- 12 Integrated bypass switches
- Multi-drop UART communication interface
  - Compatible with TPS92662-Q1 and TPS92663-Q1 devices
  - Compatible with CAN physical layer
- Eight-bit ADC with two MUXed inputs
- Programmable crystal oscillator driver and clock buffer strength
  - Improved radiated and conducted EMI performance
- External EEPROM I^2C interface
- Programmable 10-bit PWM dimming
- LED open and short detection and protection
- Supports multi-LMM synchronization for more than 12-LED string via time division multiplexing

2 Applications

- Automotive headlight systems
- ADB or glare-free high beam
- Sequential turn/animated daytime running lights

3 Description

The TPS92662A-Q1 LED matrix manager device enables fully dynamic adaptive lighting solutions by providing individual pixel-level LED control. The device includes four sub-strings of three series-connected integrated switches for bypassing individual LEDs. The individual sub-strings allow the device to accept either single or multiple current sources.

The TPS92662A-Q1 features a programmable Pierce crystal oscillator driver. Optimal performance is achieved by selecting the driver strength based on the quartz crystal or ceramic resonator manufacturers recommendations. The device also incorporates a selectable drive strength clock buffer. The rise and fall times and the EMI generated by clock signal is controlled by varying the drive strength of the clock buffer. When necessary, the clock buffer can be disabled to eliminate the noise generated by the clock signal and provide the lowest EMI performance.

The TPS92662A-Q1 supports the multi-drop universal asynchronous receiver transmitter (UART) serial interface and is compatible with TPS92662-Q1 and TPS92663-Q1 devices. The I^2C communication interface can be used to read from and write to an external EEPROM that can store system calibration data.

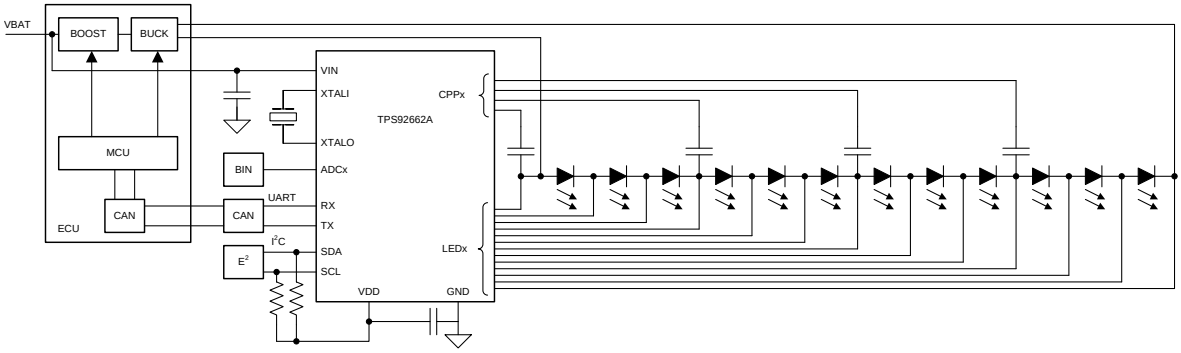
An on-board 8-bit ADC with two multiplexed inputs can be used for system temperature compensation and used to measure a binning value which allows for LED binning and coding.

Device Information(1)

PART NUMBER	PACKAGE	BODY SIZE (NOM)
TPS92662A-Q1	PHP (48)	7.00 mm × 7.00 mm

(1) For all available packages, see the orderable addendum at the end of the data sheet.

Simplified Application



An IMPORTANT NOTICE at the end of this data sheet addresses availability, warranty, changes, use in safety-critical applications, intellectual property matters and other important disclaimers. PRODUCTION DATA.

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4 Revision History

NOTE: Page numbers for previous revisions may differ from page numbers in the current version.

Changes from Original (October 2019) to Revision A	Page
• Changed device status from Advance Information to Production Data	1

5 Description (continued)

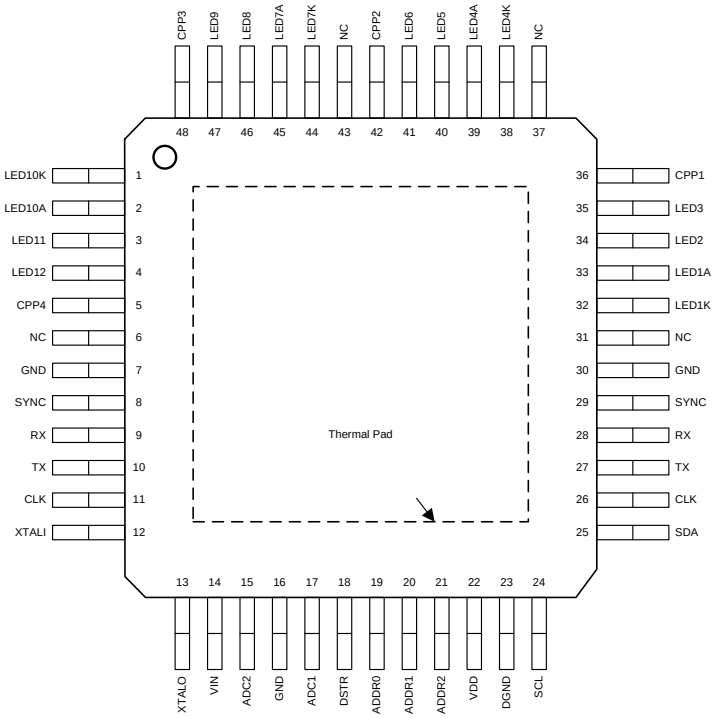
The internal charge pump rail supplies the gate drive voltage for the LED bypass switches. The low on-resistance ($R_{DS(on)}$) of the bypass switch minimizes conduction loss and power dissipation.

The TPS92662A-Q1 and TPS92662-Q1 both incorporate identical register settings for programming phase shift and pulse width of each individual LED in the string and for reporting LED open and short circuit faults.

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6 Pin Configuration and Functions

PHP Package
48-Pin TQFP with PowerPAD
Top View



Pin Functions

PIN		TYPE	DESCRIPTION
NAME	NO.		
ADC1	17	I	First ADC input. Range is from ground to VDD. Result of conversion is read at register address A0h.
ADC2	15	I	Second ADC input. Range is from ground to VDD. Result of conversion is read at register address A1h.
ADDR0	19	I	Least significant bits of the TPS92662A-Q1 device address. Connecting a resistor divider between VDD, GND, and ADDR0 determines the LSBs of the UART address. If eight or fewer TPS92662A-Q1s exist on the bus, it can be directly tied to VDD ('1') or ground ('0'). This pin functions as a digital input by default.
ADDR1	20	I	MSb-1 of the TPS92662A-Q1 device address. Tie to VDD ('1') or to the ground ('0').
ADDR2	21	I	Most significant bit (MSb) of the TPS92662A-Q1 device address. Tie to VDD ('1') or to the ground ('0').
CLK	11, 26	I/O	System clock. This is the primary clock for the device. An external clock can be provided to this pin, or the clock can be generated by a crystal or oscillator attached to the XTALI / XTALO pins. Voltage on DSTR pin determines the CLK drive strength or may disable the CLK output.
CPP1	36	O	Charge pump output. Bypass with a ceramic capacitor with a minimum value of 0.1 μF to LED3 pin.
CPP2	42	O	Charge pump output. Bypass with a ceramic capacitor with a minimum value of 0.1 μF to LED6 pin.
CPP3	48	O	Charge pump output. Bypass with a ceramic capacitor with a minimum value of 0.1 μF to LED9 pin.
CPP4	5	O	Charge pump output. Bypass with a ceramic capacitor with a minimum value of 0.1 μF to LED12 pin.
DGND	23	G	Digital ground. All GND pins and the DGND pin MUST be connected for proper operation.
DSTR	18	I	Drive Strength setting. Connecting a resistor divider between VDD, GND, and DSTR determines the drive strength of the CLK, TX, and SYNC output signals and can also disable the CLK output. Bypass with a ceramic capacitor with a maximum value of 10 nF to GND.
GND	7, 16, 30	G	Device system ground. All GND pins and the DGND pin MUST be connected for proper operation.
NC	6, 31, 37, 43	—	No connection
LED1K	32	I/O	Connect to cathode of LED1.

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Pin Functions (continued)

PIN		TYPE	DESCRIPTION
NAME	NO.		
LED1A	33	I/O	Connect to anode of LED1 and cathode of LED2.
LED2	34	I/O	Connect to anode of LED2 and cathode of LED3.
LED3	35	I/O	Connect to anode of LED3. Connect 10-nF capacitor to GND.
LED4K	38	I/O	Connect to cathode of LED4.
LED4A	39	I/O	Connect to anode of LED4 and cathode of LED5.
LED5	40	I/O	Connect to anode of LED5 and cathode of LED6.
LED6	41	I/O	Connect to anode of LED6. Connect 10-nF capacitor to GND.
LED7K	44	I/O	Connect to cathode of LED7.
LED7A	45	I/O	Connect to anode of LED7 and cathode of LED8.
LED8	46	I/O	Connect to anode of LED8 and cathode of LED9.
LED9	47	I/O	Connect to anode of LED9. Connect 10-nF capacitor to GND.
LED10K	1	I/O	Connect to cathode of LED10
LED10A	2	I/O	Connect to anode of LED10 and cathode of LED11.
LED11	3	I/O	Connect to anode of LED11 and cathode of LED12.
LED12	4	I/O	Connect to anode of LED12. Connect 10-nF capacitor to GND.
RX	9, 28	I/O	Received data pins. Connect one RX pin of first device to microcontroller unit TX output and use second pin to connect to a RX pin of the second device. All other devices use both pins to route the RX line through each device.
SCL	24	O	I^2C clock. Connect a 10-kΩ pullup resistor to VDD if EEPROM is used, otherwise connect as mentioned in *Failsafe Mode At Power-Up*.
SDA	25	I/O	I^2C data. Connect a 10-kΩ pullup resistor to VDD if EEPROM is used, otherwise connect as mentioned in *Failsafe Mode At Power-Up*.
SYNC	8, 29	I/O	Synchronization pins. Allows synchronization of multiple TPS92662A-Q1 devices on the same network. Can be driven by the microcontroller unit, or one TPS92662A-Q1 device can be programmed via the serial interface to provide synchronization pulse. Only one device can drive this signal. Can be left unconnected if not used. Voltage on DSTR pin determines the SYNC drive strength.
TX	10, 27	I/O	Transmitted data pins. Connect one TX pin of the first device to microcontroller unit RX input and use the second pin to connect to a TX pin of the second device. All other devices use both pins to route the TX line through each device. Voltage on DSTR pin determines the TX drive strength.
VDD	22	O	Output of the on-board 5 V LDO. Connect two ceramic output capacitors to DGND. Place capacitors as close as possible to pins. Recommended capacitors for best performance are 0.1 µF (0603 package) and 10 nF (0402 package).
VIN	14	I	Input voltage. Bypass with a ceramic capacitor with a minimum value of 0.1 µF to GND. If a 5-V input is used, connect VIN to VDD to bypass the internal regulator.
XTALI	12	I	Input to the Pierce oscillator inverter. Can also be used to buffer an externally generated clock. Connect to external crystal circuit. Connect to GND if not used.
XTALO	13	O	Output of the Pierce oscillator inverter. Connect to external crystal circuit. XTALDS[2:0] determines XTALO drive strength.
Thermal Pad		G	Connect to system ground.

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7 Specifications

7.1 Absolute Maximum Ratings

over operating free-air temperature range (unless otherwise noted)(1)

		MIN	MAX	UNIT
Input Voltage	VIN	–0.3	60	V
Input Voltage	VDD	–0.3	5.5	V
Input Voltage	CPPx to GND, DGND(2)	–0.3	69	V
Input Voltage	CPP4 to LED12, CPP3 to LED9, CPP2 to LED6, CPP1 to LED3	–0.3	8.8	V
Input Voltage	LEDx to GND, DGND(2)	–0.3	62	V
Input Voltage	LEDx to LED(x-1)	–0.3	20	V
Input Voltage	RX, SDA, SCL, ADC1, ADC2, ADDR0/1/2, DSTR to GND, DGND(2)	–0.3	5.5	V
Input Voltage	XTALI to GND, DGND(2)	–0.3	8.8	V
Input Voltage	SYNC, CLK, TX, XTALO to GND, DGND(2)	–0.3	VDD	V
Operating Junction temperature	T_J	–40	150	°C
Lead temperature	Soldering, 10s		260	
Storage temperature	T_{stg}	–40	150	

(1) Stresses beyond those listed under *Absolute Maximum Rating* may cause permanent damage to the device. These are stress ratings only, which do not imply functional operation of the device at these or any other conditions beyond those indicated under *Recommended Operating Condition*. Exposure to absolute-maximum-rated conditions for extended periods may affect device reliability.
(2) GND and DGND shorted together.

7.2 ESD Ratings

				VALUE	UNIT
	V(ESD) Electrostatic discharge	Human body model (HBM), per AEC Q100-002(1)		±2000	V
		Charged device model (CDM), per AEC Q100-011		±750	

(1) AEC Q100-002 indicates that HBM stressing shall be in accordance with the ANSI/ESDA/JEDEC JS-001 specification.

7.3 Recommended Operating Conditions

over operating free-air temperature range (unless otherwise noted)

		MIN	MAX	UNIT
VIN	Input Voltage	4.5	60	V
V_{LEDx} to V_{LEDx-1}	Input Voltage range per channel		16.2	V
$I_{FET(DC)}$	Switch continuous current		1.5	A
$I_{FET(PULSE)}$	Switch Pulse current, pulse duration = 20 µs, Frequency = 400 Hz(1)		10	A
t_{SYNC}	SYNC input pulse width	$2/f_{CLK}$		s
t_{r-RX}	Maximum rise time of RX		$3/f_{CLK}$	s
t_{f-RX}	Maximum fall time of RX		$3/f_{CLK}$	s
f_{CLK}	Primary IC clock frequency		16	MHz
t_{CLK}	Primary IC clock period	62.5		ns
t_{r-CLK}	Primary IC clock rise Time		12.5	ns
t_{f-CLK}	Primary IC clock fall Time		12.5	ns
t_{W-CLKH}	Primary IC clock pulse duration high time	18.8		ns

(1) The maximum allowable switch current must be set such that the total power loss on the device does not increase the junction temperature, TJ, beyond the value defined in the Absolute Maximum Ratings. For more detail refer to *Internal Switch Resistance*.

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Recommended Operating Conditions (continued)

over operating free-air temperature range (unless otherwise noted)

		MIN	MAX	UNIT
t_{W-CLKL}	Primary IC clock pulse duration low time	18.8		ns
f_{SCL}	I2C SCL frequency, set by I2CTICK register		400	kHz
R_{SCL}, R_{SDA}	I^2C external pull-ups when used with External EEPROM	1.5		kΩ
T_A	Operating Ambient temperature	–40	125	°C
T_J	Operating Junction temperature	–40	150	°C

7.4 Thermal Information

THERMAL METRIC(1)		TPS92662A PHP 48 PINS	UNIT
$R_{\theta JA}$	Junction-to-ambient thermal resistance	26.2	°C/W
$R_{\theta JC(top)}$	Junction-to-case (top) thermal resistance	13.2	
$R_{\theta JB}$	Junction-to-board thermal resistance	5.4	
Ψ_{JT}	Junction-to-top characterization parameter	0.2	
Ψ_{JB}	Junction-to-board characterization parameter	5.5	
$R_{\theta JC(bot)}$	Junction-to-case (bottom) thermal resistance	0.6	

(1) For more information about traditional and new thermal metrics, see the Semiconductor and IC Package Thermal Metrics application report.

7.5 Electrical Characteristics

T_J = –40°C to 150°C, V_{VIN} = 14 V, For digital outputs, C_{LOAD} = 20 pF. (unless otherwise noted).

PARAMETER		TEST CONDITIONS	MIN	TYP	MAX	UNIT
VIN, VDD						
I_Q	Quiescent bias current	V_{LEDx} = 0 V		4.5		mA
		V_{LEDx} = 10 V		2		
$I_{VIN-MAX-OP1}$	Input operating current	All LED outputs grounded. Max frequency. XTAL is not running (external 16MHz applied to CLK pin). UART communication max rate		6.5	10	mA
$VIN_{(UVLO-RISE)}$	UVLO rising threshold	V_{VIN} rising		4.1	4.5	V
$VIN_{(UVLO-FALL)}$	UVLO falling threshold	V_{VIN} falling	3.65	4		V
$VDD_{(POR-RISE)}$	VDD POR rising threshold	V_{VDD} rising, V_{VIN} = 5 V			4.1	V
$VDD_{(POR-FALL)}$	VDD POR falling threshold	V_{VDD} falling, V_{VIN} = 5 V	2.58			V
VDD	Regulated 5-V voltage	0 mA ≤ I_{VDD} ≤ 25 mA	4.7	5	5.3	V
I_{VDD}	VDD current limit			36		mA
VDD_{DRP}	VDD Drop-out voltage	0 mA ≤ I_{VDD} ≤ 25 mA		275	750	mV
VCCPx, INTERNAL OSCILLATOR						
V_{CPP}	Charge pump operating voltage	0 V ≤ V_{LEDx} ≤ 60 V	5		8.5	V
f_{CPP}	Charge pump oscillator frequency		3	5.5	8	MHz
I_{CPP}	Charge pump current	V_{LED3} = V_{LED6} = V_{LED9} = V_{LED12} = 20 V		0.76	1	mA
LED MATRIX SWITCHES						
$R_{DS(ON)}$	LED switch on-resistance(1)			200		mΩ
$R_{DS(ON)-3LEDs}$	3 switches in series on-resistance		230	500	1060	mΩ

(1) Single-channel on-resistance (RDS(ON)) measurement includes internal bond wires. 3×series-connected switches on-resistance (RDS(ON)-3LEDs) must be used for power loss calculations.

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Electrical Characteristics (continued)

T_J = –40°C to 150°C, V_{VIN} = 14 V, For digital outputs, C_{LOAD} = 20 pF. (unless otherwise noted).

PARAMETER		TEST CONDITIONS	MIN	TYP	MAX	UNIT
$I_{DS(off)}$	OFF state FET bias and leakage current	V_{DS} = 20 V, V_{LEDx} = 1 V			100	µA
V_{TH-S}	LED short threshold voltage	V_{LEDx-1} = 0 V, or V_{LEDx-1} = 60 V	0.66		1.37	V
V_{TH-01}	LED OPEN threshold voltage	V_{LEDx-1} = 0 V, or V_{LEDx-1} = 55 V	5	6	6.9	V
V_{TH-02}	LED OPEN threshold voltage	V_{LEDx-1} = 0 V, or V_{LEDx-1} = 48V	10.2	12	13.8	V
V_{TH-03}	LED OPEN threshold voltage	V_{LEDx-1} = 0 V, or V_{LEDx-1} = 42 V	16.2	18	20	V
t_{TO-O}	LED OPEN detection and correction delay			70	175	ns
$t_{RISE1(LEDx)}$	LEDx drain voltage rise time	I_{LED} = 400 mA and V_{LED} = 3.5 V. t_{RISE}: time for V_{DS} to change from 10% to 90%. Register setting: Fast.		2		µs
$t_{FALL1(LEDx)}$	LEDx drain voltage fall time	I_{LED} = 400 mA and V_{LED} = 3.5 V. t_{FALL}: time for V_{DS} to change from 10% to 90%. Register setting: Fast.		2		µs
$t_{RISE2(LEDx)}$	LEDx drain voltage rise time	I_{LED} = 400 mA and V_{LED} = 3.5 V. t_{RISE}: time for V_{DS} to change from 10% to 90%. Register setting: Medium.		10		µs
$t_{FALL2(LEDx)}$	LEDx drain voltage fall time	I_{LED} = 400 mA and V_{LED} = 3.5 V. t_{FALL}: time for V_{DS} to change from 10% to 90%. Register setting: Medium.		10		µs
$t_{RISE3(LEDx)}$	LEDx drain voltage rise time	I_{LED} = 400 mA and V_{LED} = 3.5 V. t_{RISE}: time for V_{DS} to change from 10% to 90%. Register setting: Slow.		20		µs
$t_{FALL3(LEDx)}$	LEDx drain voltage fall time	I_{LED} = 400 mA and V_{LED} = 3.5 V. t_{FALL}: time for V_{DS} to change from 10% to 90%. Register setting: Slow.		20		µs
DIGITAL IO						
V_{IH-TH}	High-level input voltage threshold (SYNC, RX, CLK)	V_{VDD} = 5 V	2.95			V
V_{IL-TH}	Low-level input voltage threshold (SYNC, RX, CLK)	V_{VDD} = 5 V			1.85	V
$V_{IH-ADDRx}$	High-level input voltage (ADDR0/1/2)	V_{VDD} = 5 V	2.0		VDD	V
$V_{IL-ADDRx}$	Low-level input voltage (ADDR0/1/2)	V_{VDD} = 5 V	0		0.75	V
V_{OH}	High-level output voltage (TX, SYNC)	Strength= 0.2x I_{SOURCE} = 2 mA, V_{VDD} = 5 V	3.71			V
V_{OH}	High-level output voltage (TX, SYNC)	Strength= 0.4x I_{SOURCE} = 2 mA, V_{VDD} = 5 V	4.2			V
V_{OH}	High-level output voltage (TX, SYNC)	Strength= 0.85x I_{SOURCE} = 2 mA, V_{VDD} = 5 V	4.5			V
V_{OH}	High-level output voltage (TX, SYNC)	Strength= 1x I_{SOURCE} = 2 mA, V_{VDD} = 5 V	4.53			V
V_{OL}	Low-level output voltage (TX, SYNC)	Strength= 0.2x I_{SINK} = 2 mA, V_{VDD} = 5 V			1.18	V
V_{OL}	Low-level output voltage (TX, SYNC)	Strength= 0.4x I_{SINK} = 2 mA, V_{VDD} = 5 V			0.67	V
V_{OL}	Low-level output voltage (TX, SYNC)	Strength= 0.85x I_{SINK} = 2 mA, V_{VDD} = 5 V			0.375	V
V_{OL}	Low-level output voltage (TX, SYNC)	Strength= 1x I_{SINK} = 2 mA, V_{VDD} = 5 V			0.34	V
V_{OH-CLK}	High-level output voltage (CLK when driven from XTALI)	Strength= 0.2x I_{SOURCE} = 2 mA, V_{VDD} = 5 V	3.71			V
V_{OH-CLK}	High-level output voltage (CLK when driven from XTALI)	Strength= 0.4x I_{SOURCE} = 2 mA, V_{VDD} = 5 V	4.2			V
V_{OH-CLK}	High-level output voltage (CLK when driven from XTALI)	Strength= 0.85x I_{SOURCE} = 2 mA, V_{VDD} = 5 V	4.5			V
V_{OH-CLK}	High-level output voltage (CLK when driven from XTALI)	Strength= 1x I_{SOURCE} = 2 mA, V_{VDD} = 5 V	4.53			V

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Electrical Characteristics (continued)

T_J = –40°C to 150°C, V_{VIN} = 14 V, For digital outputs, C_{LOAD} = 20 pF. (unless otherwise noted).

PARAMETER		TEST CONDITIONS	MIN	TYP	MAX	UNIT
V_{OL-CLK}	Low-level output voltage (CLK when driven from XTALI)	Strength= 0.2x I_{SINK} = 2 mA, V_{VDD} = 5 V			1.18	V
V_{OL-CLK}	Low-level output voltage (CLK when driven from XTALI)	Strength= 0.4x I_{SINK} = 2 mA, V_{VDD} = 5 V			0.67	V
V_{OL-CLK}	Low-level output voltage (CLK when driven from XTALI)	Strength= 0.85x I_{SINK} = 2 mA, V_{VDD} = 5 V			0.375	V
V_{OL-CLK}	Low-level output voltage (CLK when driven from XTALI)	Strength= 1x I_{SINK} = 2 mA, V_{VDD} = 5 V			0.34	V
T_{r-DO}	Maximum rise time of digital output (TX, SYNC, CLK when driven from XTALI)	C_{LOAD}= 50 pF, V_{VDD} = 5 V			15	ns
T_{f-DO}	Maximum fall time of digital output (TX, SYNC, CLK when driven from XTALI)	C_{LOAD}= 50 pF, V_{VDD} = 5 V			15	ns
$I_{IN(HIGH)}$	Input leakage current (RX, ADDR0, ADDR1, ADDR2, ADC1, ADC2, SCL, SDA)	V_{VDD} = 5.5 V, V_{PIN} = 5.5 V	-1		1	µA
IOS-src	Output short circuit current, source (TX, SYNC, CLK)	Strength= 1x V_{VDD} = 5.5 V, V_{PIN}= 0 V		55	100	mA
IOS-snk-pp	Output short circuit current, sink (TX, SYNC, CLK)	Strength= 1x V_{VDD} = 5.5 V, V_{PIN} = 5.5 V		55	100	mA
R_{PASS}	Resistance from SYNC1 to SYNC2, RX1 to RX2, TX1 to TX2, CLK1 to CLK2				28	Ω
R_{SPCLK}	Internal CLK pull-down			200		kΩ
R_{SPSYNC}	Internal SYNC pull-down			500		kΩ
$t_{TO-TX(FALL)}$	CLK fall to TX output valid	V_{VDD} = 5.0 V		30		ns
$t_{TO-TX(RISE)}$	CLK rise to TX output valid	V_{VDD} = 5.0 V		30		ns
t_{TZ-TX}	CLK rise to TX output tri-state	V_{VDD} = 5.0 V		42		ns
$t_{TO-SYNC(RISE)}$	CLK rise to SYNC output valid	V_{VDD} = 5.0 V		27		ns
$t_{TO-SYNC(FALL)}$	CLK fall to SYNC output valid	V_{VDD} = 5.0 V		28		ns
$t_{TZ-SYNC}$	CLK rise to SYNC output tri-state	V_{VDD} = 5.0 V		25		ns
XTAL						
V_{T+}	Positive going threshold (XTALI)		2.2		3.05	V
V_{T-}	Negative going threshold (XTALI)		1.82		2.85	V
$XTALI_{HYST}$	XTALI Hysteresis			295		mV
R_{XTAL}	Resistance between XTALI and XTALO			1		MΩ
$V_{XTALO-OH}$	High-level output voltage (XTALO)	XTALDS= 000, I_{SOURCE} = 2 mA, V_{VDD} = 5 V	4.77	4.80		V
$V_{XTALO-OH}$	High-level output voltage (XTALO)	XTALDS= 001, I_{SOURCE} = 2 mA, V_{VDD} = 5 V	4.75	4.79		V
$V_{XTALO-OH}$	High-level output voltage (XTALO)	XTALDS= 010, I_{SOURCE} = 2 mA, V_{VDD} = 5 V	4.73	4.78		V
$V_{XTALO-OH}$	High-level output voltage (XTALO)	XTALDS= 011, I_{SOURCE} = 2 mA, V_{VDD} = 5 V	4.71	4.77		V
$V_{XTALO-OH}$	High-level output voltage (XTALO)	XTALDS= 100, I_{SOURCE} = 2 mA, V_{VDD} = 5 V	4.30	4.40		V
$V_{XTALO-OH}$	High-level output voltage (XTALO)	XTALDS= 101, I_{SOURCE} = 2 mA, V_{VDD} = 5 V	3.99	4.18		V
$V_{XTALO-OH}$	High-level output voltage (XTALO)	XTALDS= 110, I_{SOURCE} = 2 mA, V_{VDD} = 5 V	3.09	3.60		V
$V_{XTALO-OH}$	High-level output voltage (XTALO)	XTALDS= 111, I_{SOURCE} = 100 µA, V_{VDD} = 5 V	4.07	4.5		V
$V_{XTALO-OL}$	Low-level output voltage (XTALO)	XTALDS= 000, I_{SINK} = 2 mA, V_{VDD} = 5 V		0.16	0.21	V
$V_{XTALO-OL}$	Low-level output voltage (XTALO)	XTALDS= 001, I_{SINK} = 2 mA, V_{VDD} = 5 V		0.17	0.23	V

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Electrical Characteristics (continued)

T_J = –40°C to 150°C, V_{VIN} = 14 V, For digital outputs, C_{LOAD} = 20 pF. (unless otherwise noted).

PARAMETER		TEST CONDITIONS	MIN	TYP	MAX	UNIT
$V_{XTALO-OL}$	Low-level output voltage (XTALO)	XTALDS= 010, I_{SINK} = 2 mA, V_{VDD} = 5 V		0.11	0.25	V
$V_{XTALO-OL}$	Low-level output voltage (XTALO)	XTALDS= 011, I_{SINK} = 2 mA, V_{VDD} = 5 V		0.12	0.28	V
$V_{XTALO-OL}$	Low-level output voltage (XTALO)	XTALDS= 100, I_{SINK} = 2 mA, V_{VDD} = 5 V		0.5	0.69	V
$V_{XTALO-OL}$	Low-level output voltage (XTALO)	XTALDS= 101, I_{SINK} = 2 mA, V_{VDD} = 5 V		0.7	1	V
$V_{XTALO-OL}$	Low-level output voltage (XTALO)	XTALDS= 110, I_{SINK} = 2 mA, V_{VDD} = 5 V		1.35	1.91	V
$V_{XTALO-OL}$	Low-level output voltage (XTALO)	XTALDS= 111, I_{SINK} = 100 µA, V_{VDD} = 5 V		0.15	0.85	V
$V_{XTALO-IL}$	XTALO low-level threshold to input CLK buffer	XTALDS= 111, V_{VDD} = 5 V, V_{XTALI} = 2.5 V		1.01		V
$V_{XTALO-IH}$	XTALO high-level threshold to input CLK buffer	XTALDS= 111, V_{VDD} = 5 V, V_{XTALI} = 2.5 V		4.26		V
IOS-src	Output short circuit current, source (PIN: XTALO)	XTALDS= 000, V_{VDD} = 5.5 V, V_{PIN} = 0 V		55	100	mA
IOS- sink-pp	Output short circuit current, sink (PIN: XTALO)	XTALDS= 000, V_{VDD} = 5.5 V, V_{PIN} = 5.5 V		55	100	mA
I2C						
$V_{IH-TH(I2C)}$	High-level input voltage threshold (SDA, SCL)	V_{VDD} = 5 V	3.15			V
$V_{IL-TH(I2C)}$	Low-level input voltage threshold (SDA, SCL)	V_{VDD} = 5 V			1.35	V
V_{I-HYST}	Input hysteresis voltage		0.215			V
$V_{OL(I2C)}$	Low-level output voltage (SDA, SCL)	I_{SINK} = 3 mA, V_{VDD} = 5 V			0.5	V
IOS-snk-od	Output short circuit current, sink (PIN: SDA, SCL)	V_{VDD} = 5.5 V, V_{PIN} = 5.5 V		36	100	mA
8-BIT ADC						
ADCres	ADC Resolution			8		Bits
DNL	Differential linearity error(2)		–1		1	LSB
INL	Integral linearity error (best fit)(3)		–1		1	LSB
E_{ZEO}	Zero offset error(4)			0.5		LSB
T_{conv}	Conversion-time + wait time(5)				320/f_{CLK}	s

(2) Deviation in code width from the ideal 1LSB code width.
(3) Maximum deviation of a transition point from the corresponding point of the ideal transfer curve.
(4) Difference between the first measured transition point (lowest in voltage) and the first ideal transition point.
(5) Verified by design. Not production tested.

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7.6 Typical Characteristics

T_A = 25°C free air, V_{IN} = 14 V, V_{LEDx} = 0 V unless otherwise specified

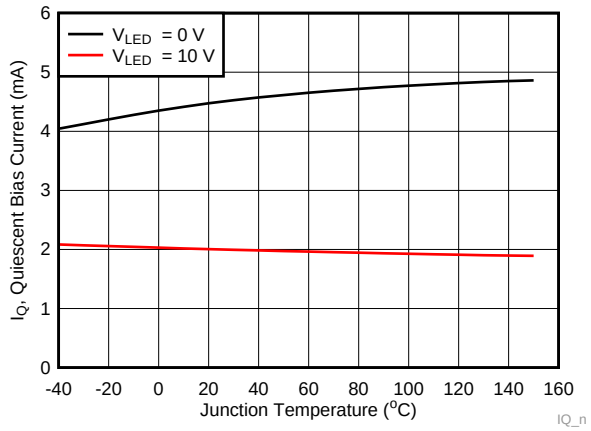


Figure 1. Input Quiescent Current Versus Junction Temperature

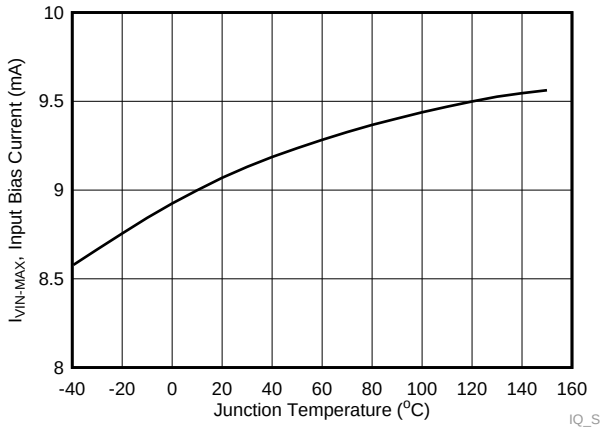


Figure 2. Input Bias Current Versus Junction Temperature

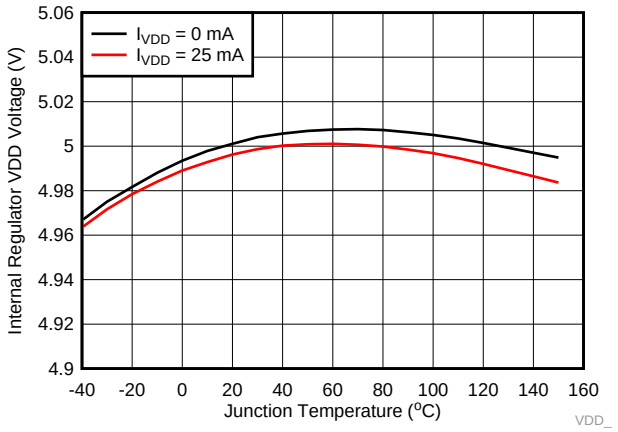


Figure 3. Regulated VDD Voltage Versus Junction Temperature

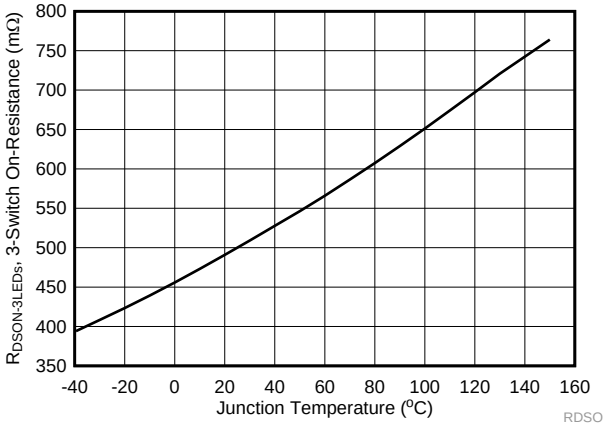


Figure 4. Three Series-Connected Switches On-Resistance Versus Junction Temperature

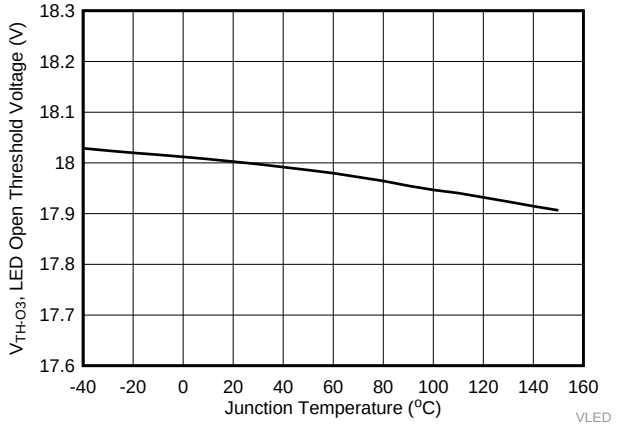


Figure 5. Channel Open Threshold for V_{TH-O3} Versus Junction Temperature

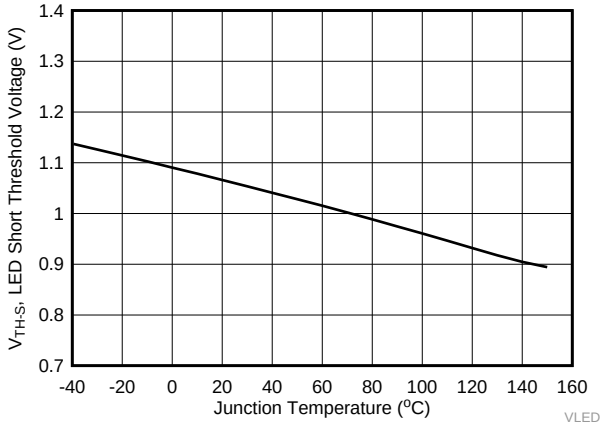


Figure 6. Channel Short Threshold Versus Junction Temperature

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Typical Characteristics (continued)

T_A = 25°C free air, V_{IN} = 14 V, V_{LEDx} = 0 V unless otherwise specified

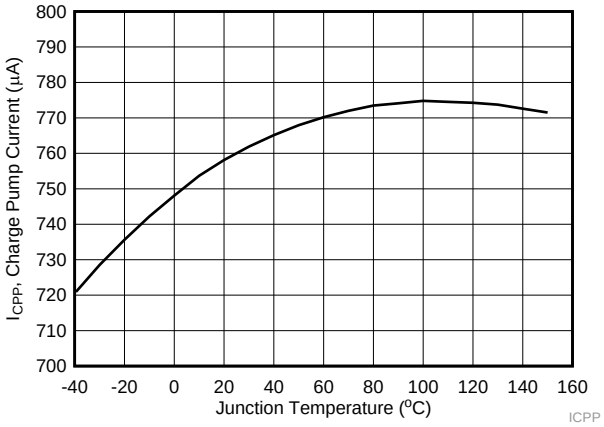


Figure 7. Charge Pump Current Versus Junction Temperature

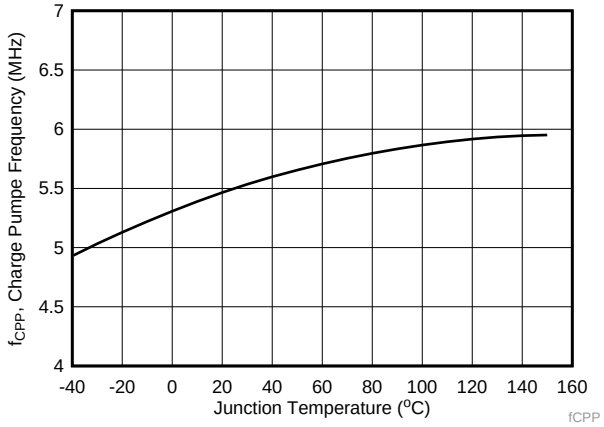


Figure 8. Charge Pump Frequency Versus Junction Temperature

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8 Detailed Description

8.1 Overview

The TPS92662A-Q1 lighting matrix manager device, in conjunction with buck switching current regulators, enables a fully dynamic matrix beam solution where each LED can be individually controlled. This type of control of an LED array is ideally suited for dynamic headlight applications where adaptive beam forming requires pixel-level control of the array.

The TPS92662A-Q1 device configures four sub-strings of three-series-connected integrated switches (MOSFETs) for bypassing individual LEDs. The series-connected internal switches can float up to 62 V above ground potential, as long as the absolute maximum voltage across the drain or source of each switch is not violated. Refer to the *Absolute Maximum Ratings*. Each switch connects in parallel to one (or multiple series-connected) LED or LEDs, thereby creating individual shunt paths across each LED. For high-power LED devices, the TPS92662A-Q1 device can be programmed to operate in parallel configuration, such that multiple of the internal switches are configured in parallel mode, shunting one single LED. Furthermore, LED strings with fewer LEDs can be used. Nevertheless, the unused channels must be physically shorted externally to reduce unnecessary internal power consumption.

Each switch has an individual driver, programmable overvoltage protection circuit, and diagnostics circuit referenced to the source of that switch. This configuration allows for fully dynamic operation with the switches above it and below it. The device monitors overvoltage conditions on each switch and automatically protects them in the event of an open LED connection. The device detects open LED conditions as well as shorted LED conditions and reports them through the fault reporting network.

All twelve internal bypass switches can be individually pulse width modulated (PWM) at a programmed frequency, phase-shift, and duty cycle. This PWM dimming topology provides inherent phase shifting capability. In addition, the switch transitions during PWM dimming have a programmable slew rate limit to mitigate any EMI concerns due to the di/dt and dv/dt of the switching action.

The TPS92662A-Q1 device also provides multi-drop UART communications capability between a host microcontroller unit (MCU) and up to 31 slave TPS92662A-Q1 devices. The UART receives data corresponding to the desired PWM information for all 12 internal switches. In addition, it can send back fault and other diagnostic data to the host MCU. Hardware connections on the three address pins allow addressing of up to 31 devices. An application can use the TPS92662A-Q1 device and the TPS92663-Q1 device on the same bus.

The TPS92662A-Q1 device includes an on-board 8-bit ADC with two multiplexed general purpose inputs, which can be used for system temperature compensation, binning and coding, and so forth. Furthermore, one of the address pins of the device is connected to the input of the ADC to extend the addressability of the TPS92662A-Q1 from 8 to 31 devices.

The internal regulator accepts battery voltage on input pin VIN (as high as 60 V) and generates a 5.0-V regulated output for the internal digital and analog circuitry. The internal regulator can be bypassed by shorting VIN to VDD if an external regulator is being used.

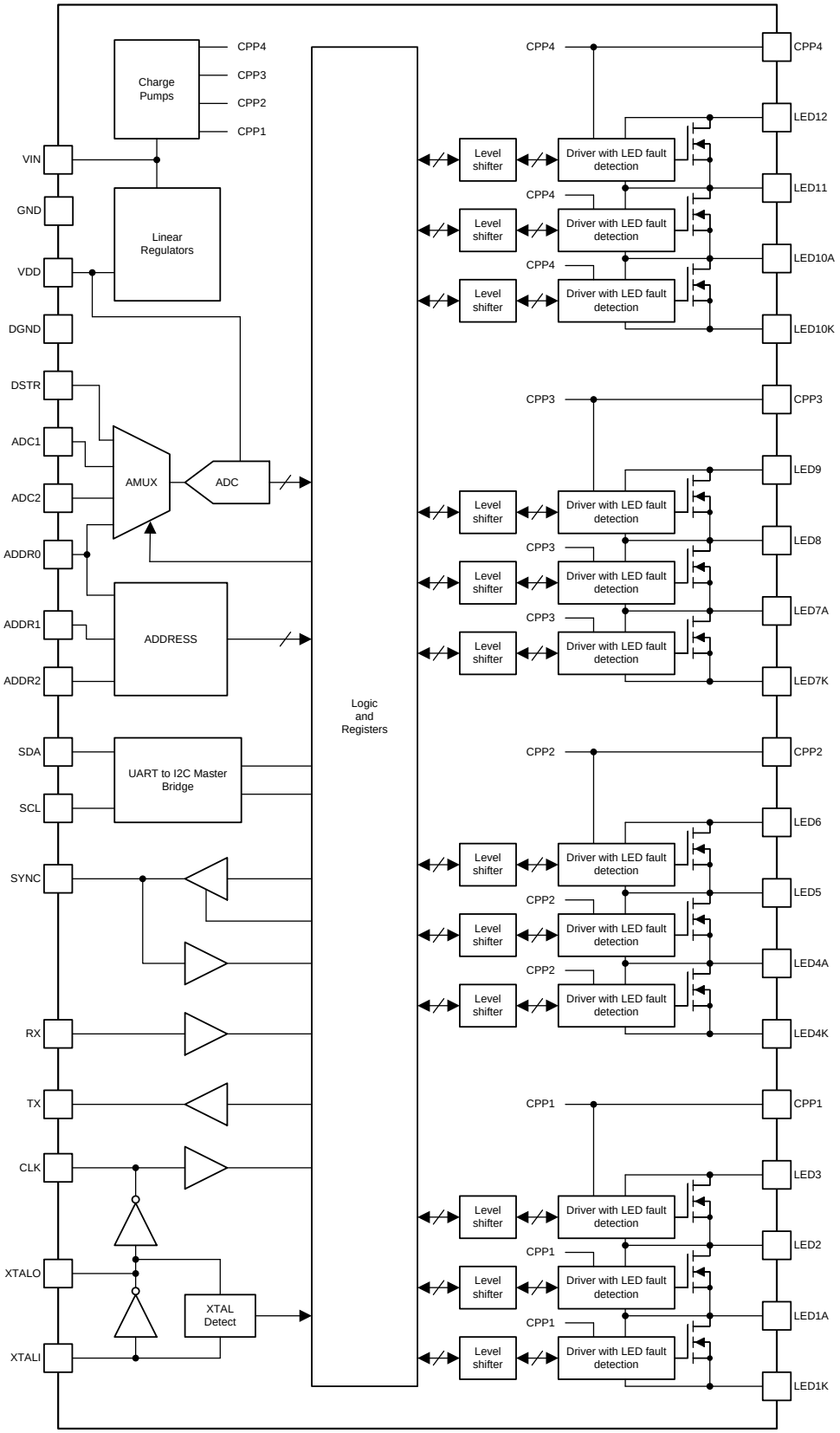
The combination of features in the TPS92662A-Q1 device provides the ideal interface for individual control of high current LED arrays. As shown in the *Application Information* section, the electronic control unit (ECU), usually attached to the outside of the headlight, contains the master MCU, a boost pre-regulator stage that takes the variable battery voltage and steps it up to a stable DC voltage rail. The dynamic headlight application usually includes multiple channels of buck current regulators to provide a stable current for each series string of high brightness LEDs. For the best performance, the TPS92662A-Q1 device resides on the LED load board where it is as close as possible to the LEDs to which it directly connects. This location has two major benefits:

- The close proximity minimizes distributed inductance and parasitic capacitance associated with the cable connection between ECU board and LED load board. When PWM dimming with a parallel shunt FET is performed, placing the switch close to the LED prevents large ringing during each transition.
- The close proximity offers better thermal connection.

Ultimately, the TPS92662A-Q1 device enables an optimal partition for a flexible, high-performing dynamic headlight system.

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8.2 Functional Block Diagram



8.3 Feature Description

8.3.1 Controlling the Internal LED Bypass Switches

The TPS92662A-Q1 device (LED Matrix Manager) consists of four independent three-series-connected bypass switches between terminals LED12 and LED10K, LED9 and LED7K, LED6 and LED4K, and LED3 and LED1K as shown in the *Functional Block Diagram*. Each bypass switch, when driven to an off state, allows the string current to flow through the corresponding parallel-connected LED, turning the LED on. Conversely, driving the bypass switch to an on state shunts the current through the bypass switch and turns the LED off.

8.3.2 Internal Switch Resistance

Each single switch (connected between LEDx and LEDx–1) has a measurable typical $R_{DS(on)}$ value of 200 mΩ. This measurement includes the actual on-resistance ($R_{DS(on)}$) of the switch and the resistance of the two internally-connected bond wires. When multiple series switches are on, the effective resistance is not simply the number of channels multiplied by 200 mΩ because there are not two conducting bond wires for every series connected switch. The on-resistance of the 3×series-connected switches ($R_{DS(on)\text{-}3LEDs}$) is specified in the *Electrical Characteristics*. This value includes three $R_{DS(on)}$ on-resistances and the resistance of the bond wires at each end of the series connected switches.

The dominant power loss mechanism in the TPS92662A-Q1 device is the I^2R loss through the switches. Other power loss sources are typically less than 150 mW. When calculating the power dissipation of the TPS92662A-Q1 device switches, use Table 1 for the best estimation of the power loss. It is important to note that the power dissipation is PWM duty cycle dependent. When PWM dimming is applied to the LEDs with duty cycle D, the effective switch resistance for power dissipation calculation is:

$$R_{EFFECTIVE} = (1 - D) \times R_{DS(on)} \quad (1)$$

Table 1. Total $R_{DS(on)}$ of the Switches

SWITCHES IN SERIES	MAXIMUM TOTAL $R_{DS(on)}$ (mΩ)
3	$1 \times R_{DS(on)\text{-}3LEDs}$
6	$2 \times R_{DS(on)\text{-}3LEDs}$
9	$3 \times R_{DS(on)\text{-}3LEDs}$
12	$4 \times R_{DS(on)\text{-}3LEDs}$

8.3.3 PWM Dimming

The TPS92662A-Q1 device provides 10-bit phase-shifted PWM dimming of each individual LED. The LED phase-shift and pulse-width are separately programmed for each LED via the PHASEx and WIDTHx registers respectively (where x = 1 to 12). Depending on the value of PSON bit (in SYSCFG register), the TPS92662A-Q1 device implements phase shifting of LED turnon time (PSON = 1) or LED turnoff time (PSON = 0). The pulse-width of the LEDs can be set by programming the WIDTHx register value. The free-running 10-bit PWM period counter (TCNT) that increments based on the PWM clock pulse determines the PWM period. The TCNT counts continuously from 0 to 1023 and wraps back to 0 again. The LED PWM dimming period is 1024 times the internally divided, programmable PWM clock period.

Use Equation 2 to calculate the PWM period.

$$t_{PWM} = \frac{DIV1 \times DIV2 \times 1024}{f_{CLK}} \quad (2)$$

where

- DIV1 and DIV2 are the PWM clock divisors

See the *PWM Clock* section for more information on clock divisors.

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8.3.3.1 PWM Dimming with PSON = 1

By setting the PSON bit to “1” in the SYSCFG register, the LED turnon time (switch off-time) event is fixed. When TCNT reaches the programmed PHASEx value for a given LED, the corresponding LED channel turns on (the switch turns off). At the same time, a one-shot counter starts the PWM width counter (WCNT). When WCNT reaches the programmed WIDTHx value, the corresponding LED channel turns off (the switch turns on). The PWM period equals 1024 × PWM-clock period. Figure 9 shows an example of LED PWM, using values of PHASEx = 384 and WIDTHx = 512.

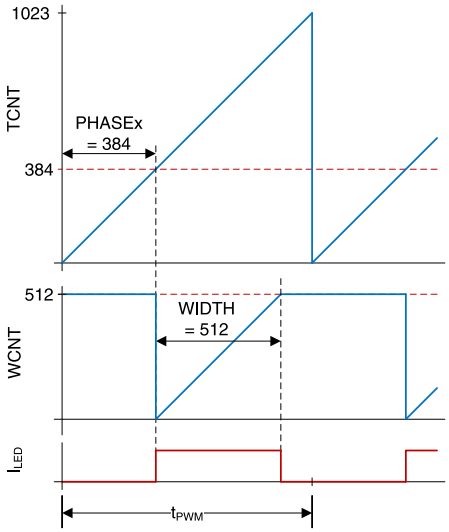


Figure 9. PWM Dimming with PSON = 1

8.3.3.2 PWM Dimming with PSON = 0

With the default value of PSON = 0 in the SYSCFG register, the LED turnoff time event is fixed (switch on-time). When TCNT reaches the programmed PHASEx value for a given LED, the corresponding LED channel turns off (the switch turns on). At the same time, a one-shot counter starts the PWM width counter (WCNT). When WCNT reaches the value of “1023 – WIDTHx”, the corresponding LED channel turns on (the switch turns off). The PWM period equals 1024 x PWM-clock period. Figure 10 shows an example of LED PWM, using values of PHASEx = 256 and WIDTHx = 384.

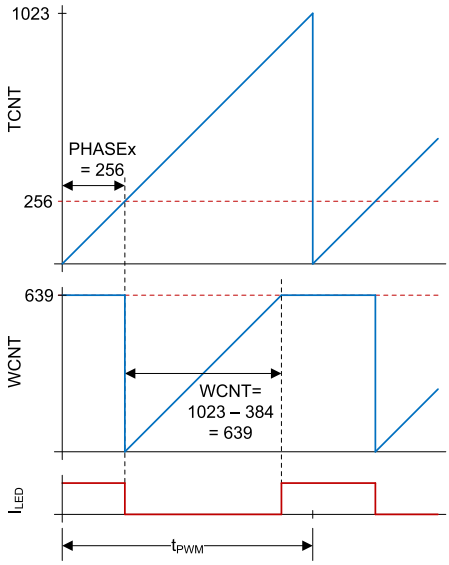


Figure 10. PWM Dimming with PSON = 0

It is important to note that in both cases of PSON = “0” or “1”, the programmed value of WIDTHx specifies the LED on time of the corresponding LEDx.

Furthermore, the PSON bit and PHASE settings are intended to be programmed at initialization and remain static during LED PWM dimming operation.

8.3.4 Duty Cycle = 0% and 100% (LED Fully Off or On)

To achieve a duty cycle of 0% (LED fully off), the register WIDTHx must be set to 0. When TCNT reaches the PHASEx register value, the corresponding LED turns off and remains off.

To achieve a duty cycle of 100% (LED fully on), the register WIDTHx must be set to 1023. As a result, the programmed brightness of 1023/1024 is not available.

Table 2 shows the mapping of the WIDTHx to the resulting duty cycles for duty cycle values close to 100%.

Table 2. Duty Cycle Mapping

DUTY CYCLE (%)	WIDTH [9:0]
99.6	1020
99.7	1021
99.8	1022
99.9	Unavailable
100.0	1023

8.3.5 PWM Clock

The PWM clock that drives TCNT is a divided-down version of the CLK input. The divider is programmed by writing to the PWM Tick Period register (PWMTICK). The divider is comprised of two dividers in series, PTBASE and PTCNT (see Figure 11 and *PWM Tick Period Register (PWMTICK) – Read/Write* for the PWMTICK bitmap). Upon power up, the PWM clock divider is programmed to a divisor value of 16.

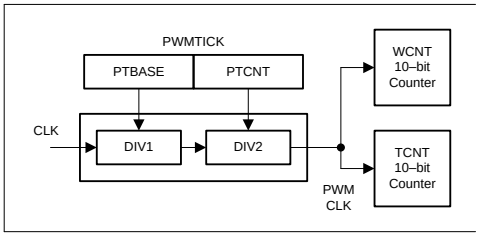


Figure 11. TPS92662A-Q1 PWM Clock

8.3.6 PWM Synchronization

Upon power up, the TCNT counter is reset to 0. The TCNT counter is incremented by the PWM clock. To synchronize multiple TPS92662A-Q1 devices, two conditions must be met:

- All devices must be clocked by the same clock on the CLK or XTALI pins.
- All devices must be programmed with the same PWMTICK value.

Assuming the above conditions are met, the TPS92662A-Q1 devices can be synchronized by either *Hardware Synchronization* or *Software Synchronization*.

8.3.6.1 Hardware Synchronization

As shown in Figure 12, the TPS92662A-Q1 device has a synchronization input and output pin (SYNC) and two synchronization register bits (SYNCOEN and SYNCPEN) in the System Configuration Register (SYSCFG). If SYNCOEN is set to “1”, the TSP92662 drives the output buffer on the SYNC pin, otherwise, the SYNC pin driver is tri-stated and operates as an input.

If SYNCOEN is set to “0”, a high pulse on the SYNC pin for at least two CLK periods resets both TCNT and the PWM clock to 0. SYNC is a feedthrough signal that can be tied to the next TPS92662A-Q1 devices to synchronize multiple devices with respect to each other.

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If SYNCOEN and SYNCPEN are both set to “1”, the TPS92662A-Q1 device generates a high pulse that is one half of a PWM period on SYNC when TCNT and the PWM clock are about to roll over to 0. The resulting SYNC signal can be applied to other TPS92662A-Q1 devices.

In practice, the hardware synchronization method can be applied using one of the following two methods:

- The MCU drives the SYNC with SYNCOEN set to “0” for all TPS92662A-Q1 devices on the bus.
- One TPS92662A-Q1 device drives the SYNC with both SYNCOEN and SYNCPEN set to “1” and the rest of the TPS92662A-Q1 devices on the network have SYNCOEN set to “0”.

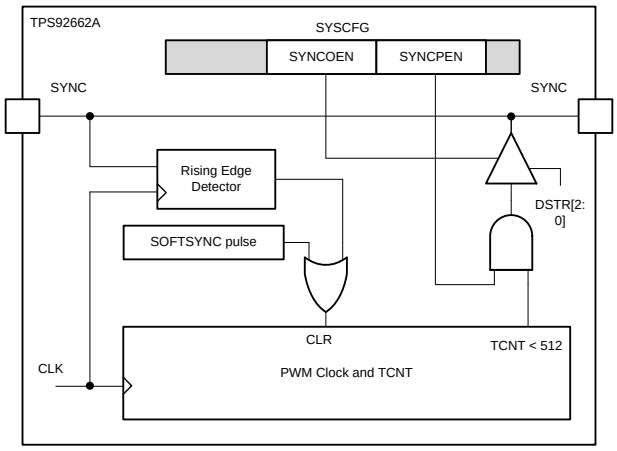


Figure 12. TPS92662A-Q1 PWM Synchronization

If the hardware synchronization feature is not implemented, a strong pulldown is recommended to minimize noise on the SYNC signal. This is implemented on the device when SYNCOEN is set to "1" and SYNCPEN is set to "0". As shown in Table 3, the setup is defined for master and slave devices in a multi-device application. For a single device, setting the SYNCOEN bit reduces the possibility of false synchronization.

Table 3. Master and Slave Setup for Hardware Synchronization

SYSCFG	MASTER	SLAVE	SINGLE DEVICE OR HARDWARE SYNCHRONIZATION NOT ENABLED
SYNCOEN	1	0	1
SYNCPEN	1	0	0

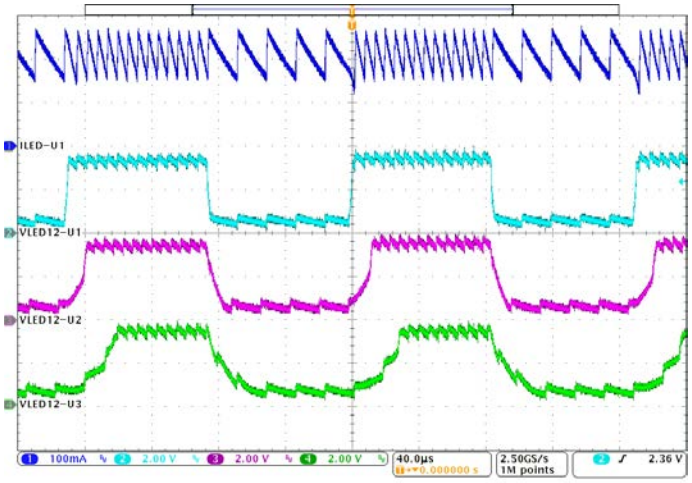
8.3.6.2 Software Synchronization

The TCNT counter and the PWM clock can both be reset to 0 at any time by issuing a broadcast write synchronization command (writing a 1 to the SSYNC bit of the SOFTSYNC register). Due to UART bit sampling variability, the synchronization is achieved within four CLK cycles between TPS92662A-Q1 devices.

8.3.7 Switch Slew Rate Control

The switching slew rate of the series-connected LED bypass switches can be controlled and programmed by the SLEWRATE register. The slew rate of the switches is controlled by changing the gate driver strength. The slew rate of each LED string (3×series-connected switches) can be programmed through two bits, which provide fast ($t_{RISE1/FALL1}$), medium ($t_{RISE2/FALL2}$), and slow ($t_{RISE3/FALL3}$) slew rate (see the *Electrical Characteristics* for more details).

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CH1 (Blue): I_{LED} CH2 (Cyan): V_{LED-U1} CH3 (Pink): V_{LED-U2} CH4 (Green): V_{LED-U3}

Figure 13. Slew Rate Control

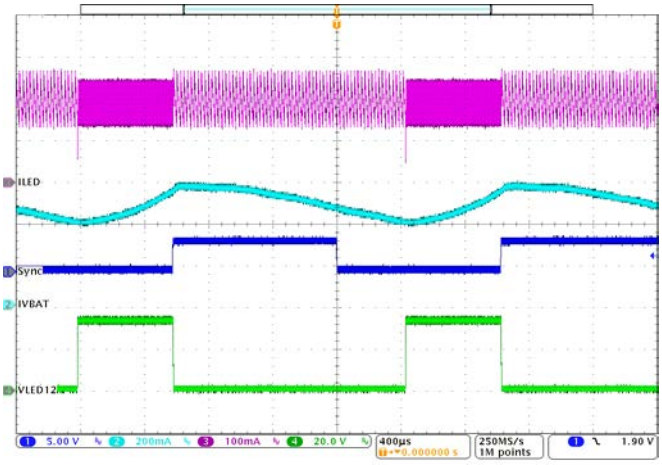
Figure 13 shows the effect of different slew rate settings on the turnon and turnoff edges of the LEDs when PWM dimming is applied. The slew-rate of the TPS92662A-Q1-U1 device is set to the fast slew rate (SLEW_X_Y = 11), the TPS92662A-Q1-U2 device is set to the medium slew rate (SLEW_X_Y = 01), and the TPS92662A-Q1-U3 device is set to the slow slew rate (SLEW_X_Y = 00). For more details on slew rate setting, see the *Slew Rate Register*.

8.3.8 PWM Phase Shift

As discussed in the *PWM Dimming* section, the LED turnon (or turnoff) timer can independently be programmed for each LED by the PHASEx registers. The amount of phase shift between LED turnon (or turnoff) times affects the instantaneous power drawn from the BUCK switching converters on the ECU. It is recommended that the phase-shift applied to each switch to be such that the turnon (or turnoff) times of all the LEDs are evenly distributed in one PWM period.

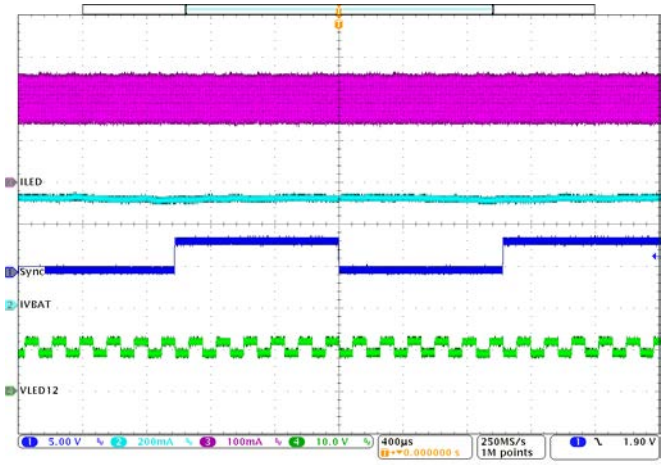
Figure 14 and Figure 15 show the system input current (ECU I_{VBAT} in cyan and the top of the LED string voltage in green) for different phase-shift settings. As shown in Figure 14, when the PHASEx=0 is set for all the LEDs, large variation on the system input current is observed. In Figure 15, the PHASEx for each LEDs is set so that the turnoff times of the individual LEDs are evenly distributed in one PWM period (1023/12 approximates to 85, for 12 series-connected LEDs in the string).

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CH1 (Blue): Sync Signal
CH2 (Cyan): I_{VBAT}
CH3 (Pink): I_{LED}
CH4 (Green): V_{LED12}

Figure 14. No Phase-Shift Between the Turnoff Times



CH1 (Blue): Sync Signal
CH2 (Cyan): I_{VBAT}
CH3 (Pink): I_{LED}
CH4 (Green): V_{LED12}

Figure 15. Phase-Shift = 85 Between the Turnoff Times

In the *SYSCFG Register* of the TPS92662A-Q1 device, the phase shift can be configured to be applied to the LED turn-on (PSON = 1) or turn-off (PSON = 0) time. In particular, the phase shifting of the LED turn-off times (PSON=0) has the benefit of decreasing the LED current overshoot when significant capacitance exists across the LED strings. See the *Application Curves* section for more details.

8.3.9 LED Fault Detection and Protection

Each individual bypass switch is driven by a floating driver which is powered by the charge pump (see the *Functional Block Diagram*). The floating analog circuitry provides continuous protection and monitoring of LED open and short events. In the event of an OPEN LED failure, an internal comparator monitors the drain-to-source voltage of the internal switch. If the voltage exceeds the programmable threshold voltage of V_{TH-O}, the device overrides the switch programmed WIDTHx value and turns on the switch. This action maintains current flow in the rest of the LED string in the presence of a faulty or damaged LED, and protects the internal switch from Electrical Overstress (EOS). The internal latch holds this state until a subsequent LED turnoff event. The protection circuit also sets the corresponding bit in the *Fault Registers*. The MCU can poll this register to determine whether a fault has occurred.

The threshold voltage, V_{TH-O}, can be programmed independently for each LED sub-string (3×series-connected switches) via a 2-bit value in the OVLMT register. The programmed value provides low (V_{TH-O1}), medium (V_{TH-O2}), and high (V_{TH-O3}) threshold values. The programmability of the overvoltage thresholds helps applications where multi-chip LEDs with higher forward voltage drops are used. Furthermore, in applications where significant ringing occurs during switching events that can result in an undesired overvoltage triggering, higher OVLMT thresholds can help mitigate the problem.

When the device is configured as described in the *Parallel Switch Setting* section, digital circuitry of the device turns on all of the corresponding switches, if one of the parallel switches detects an overvoltage condition. This allows all of the paralleled switches to share the shunted current instead of only a single switch handling the full LED current.

Similar to LED open detection, LED short is detected by monitoring the drain-to-source voltage of the internal switch. If the voltage does not exceed the V_{TH-S} threshold by the end of the PWM cycle, the same fault register bit is set. Both short and open events are reported by the same register as each event has the equivalent outcome, which is the LED is shorted.

The synchronized fault inputs are latched into the FAULT registers on the falling edge of the requested LED ON time. This allows the MCU to determine which LEDs had an open or short fault at the end of the LED ON pulse. The MCU can clear the fault bits in the FAULT register by writing a 0 to the corresponding bit, provided that the corresponding fault input is no longer active.

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In the special case of 100% duty cycle (LED fully turned on), there is no PWM falling edge available on which to latch the fault conditions. Therefore, any LED channel that is set to 100% duty cycle latches the incoming faults every 1024 PWM clocks, when TCNT = 0. In the case of 0% duty cycle, faults are never latched into the fault register as the bypass switch is always on.

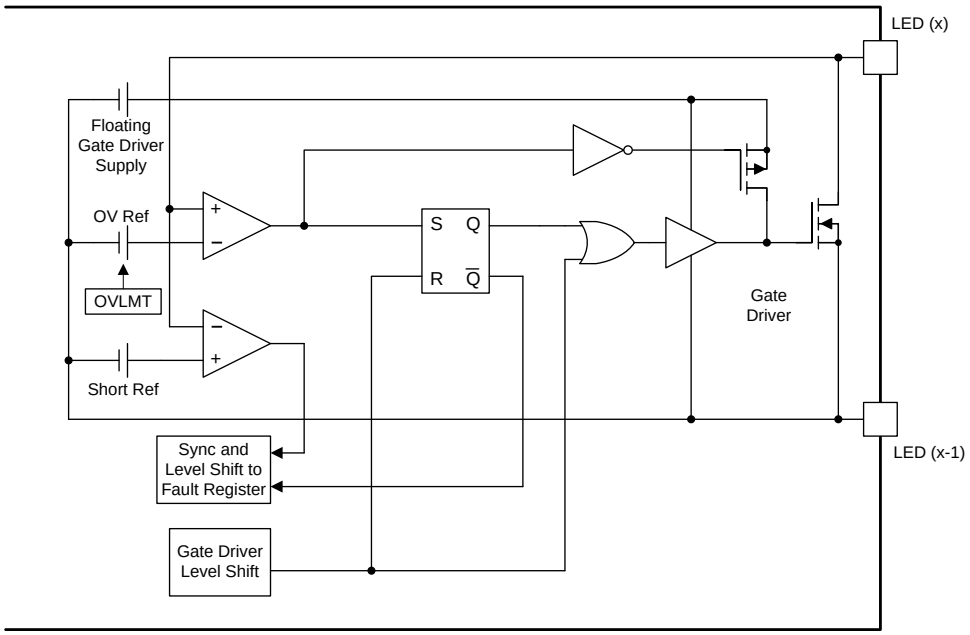
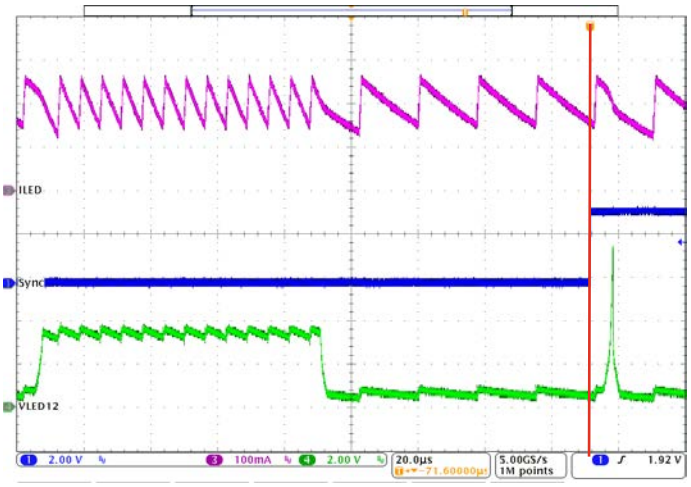


Figure 16. TPS92662A-Q1 Overvoltage Protection and Detection

8.3.9.1 LED Open-Fault Protection Example

Figure 17 shows the operation of the PWM dimming where one of the LEDs is open. When the corresponding shunt switch in the TPS92662A-Q1 device across the open LED is turned off, the current through the LED string causes the voltage across the switch to increase. The voltage across the switch increases until it triggers the OV-Limit comparator at the programmed voltage threshold of V_{TH-O}, causing the switch to immediately turn on. The switch remains on for the duration of WCNT, latching the fault and setting the corresponding FAULTx bit. The switch resets its state when WCNT is complete, but the FAULTx bit remains set until it is cleared by software.



CH1 (Blue): Sync Signal CH3 (Pink): I_{LED} CH4 (Green): V_{LED12}

Figure 17. Overvoltage Protection

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8.3.10 Communications Watchdog Timer

The TPS92662A-Q1 logic has a communications watchdog timer that is operated with the system clock (CLK). The watchdog timer must be enabled in order to be used. This is accomplished by setting the CMWEN bit to 1 in the SYSCFG register and programming the communications watchdog timer tap point by writing the desired value to the CMWTAP register.

The tap point defines the timing of the communications watchdog timer (a 24-bit counter). By default, the tap point is set to bit 21. More specifically, the device requires 2^{21} clock cycles for the communication watchdog timer to time out. Refer to the CMWTAP register for more information.

If a transaction with a correct CRC is not received for the programmed number of system clock cycles, the communications watchdog timer times out and the LEDs are pulse width modulated according to the programmed values in the PHASE and DEFWIDTH registers. The normal PWM duty cycles are resumed based on the PHASE and WIDTH register settings, only after a valid UART command frame with a correct CRC is received. The communications watchdog timer is not reset by a communication reset command.

8.3.11 Parallel Switch Setting

The TPS92662A-Q1 can be configured to operate when the switches are connected in parallel. The parallel switch configuration enables operation of high current LEDs and decreasing power loss and heat. It is important to note that the physical parallel connection must be performed external to the TPS92662A-Q1. The parallel configuration of the TPS92662A-Q1 enables simultaneous PWM switching and fault handling of parallel switches. The TPS92662A-Q1 device includes four sub-strings of S1, S2, S3, and S4 with 3×series-connected switches, as shown in Figure 18. The combinations of LED strings that can be paralleled are:

- None (each sub-string of bypass switches is independent from the others)
- One pair (S1 and S2 are connected in parallel, S3 and S4 are independent)
- Two pairs (S1 and S2 are connected in parallel, S3 and S4 are connected in parallel)
- All (S1, S2, S3, and S4 are all connected in parallel)

At the end of the power-up sequence, the default parallel setting is none; each LED sub-string is independent. The controller must program the PARLED register to set the appropriate paralleling configuration. When using parallel operation, refer to Table 19 for programming WIDTHx values.

Parallel operation also affects the fault management. When sub-strings of LEDs are paralleled, there is a possibility that one switch senses an overvoltage condition and turns on before the other parallel switches do. If this situation were not addressed, the switch that clamps the overvoltage condition continued being the only switch conducting a potentially large current. Therefore, the programming of the PARLED register also configures the combinations of faults, which causes the gate drivers of all parallel switches to turn on in an overvoltage condition. There is a small delay due to synchronizing the incoming faults to the internal clock domain, after which all parallel switches that have an overvoltage condition are immediately turned on to share the LED currents.

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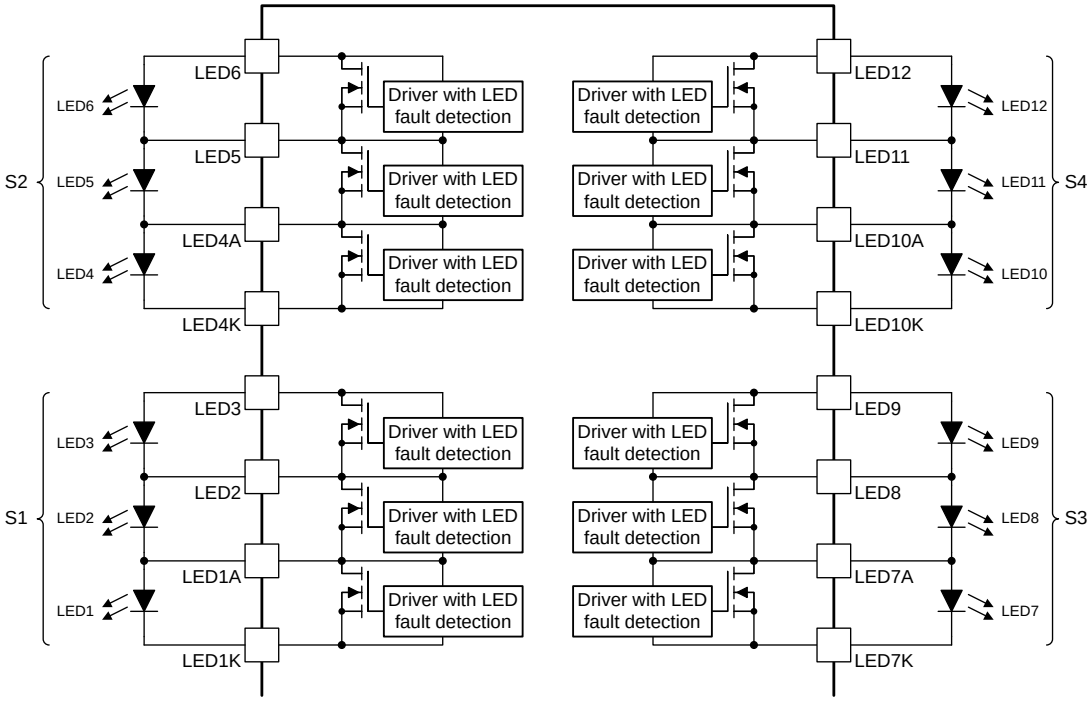


Figure 18. TPS92662A-Q1 Sub-Strings

8.3.12 ADC Operation and Settings

The TPS92662A-Q1 device includes an on-board 8-bit ADC with two multiplexed general purpose inputs, which can be used for system temperature compensation, binning (coding), and so forth. Furthermore, one of the address pins of the device is connected to the input of the ADC to extend the addressability of the TPS92662A-Q1 from 8 to 31 devices. The fourth input to the ADC comes from the DSTR pin, which is sampled at start-up to determine the drive strength of the CLK. Refer to *Drive Strength Register – Read-Only* for details on DSTR.

The two ADC inputs of ADC1 and ADC2 are periodically sampled in turn every 320 CLK cycles. The two 8-bit results are stored in ADC1 and ADC2 registers. The reference voltage for the ADC is VDD = 5.0 V (19.5 mV/LSB) nominal.

The third input to the internal ADC is ADDR0. This input is sampled by the ADC only after the TPS92662A-Q1 device receives a broadcast write to the ADCID register. If ADCID is written to a value of 1 or 2, then the ADDR0 input is selected and the ADDR0 input is sampled continuously until there are at least four consecutive ADC samples with the same value (at a resolution of two bits if ADCID is written to 1, or at a resolution of three bits if ADCID is written to 2). After the four matching values are obtained, the logic begins sampling the ADC1 and ADC2 inputs in an alternating fashion. The UART protocol state machine is suspended until the sampling of ADDR0 is successfully completed.

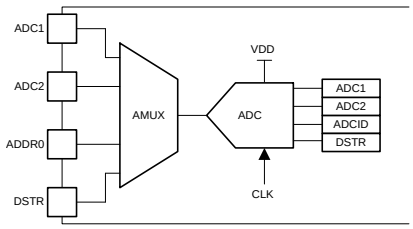


Figure 19. TPS92662A-Q1 Internal ADC Configuration

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8.4 Device Functional Modes

The TPS92662A-Q1 device operates in one of several modes as shown in Figure 20:

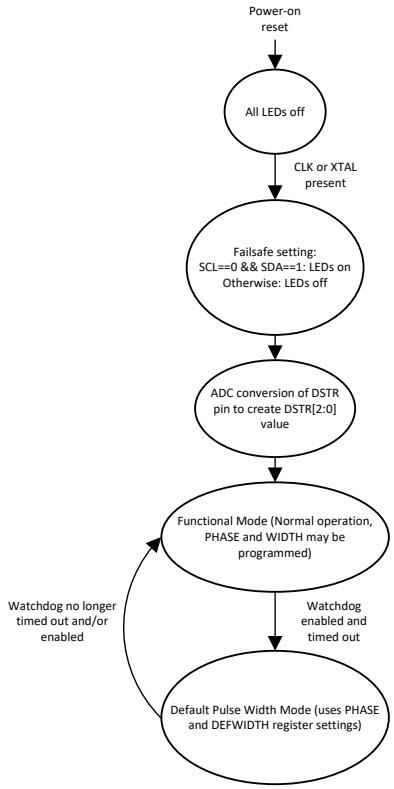


Figure 20. TPS92662A-Q1 Functional Mode

8.4.1 Failsafe Mode At Power-Up

The TPS92662A-Q1 device has a failsafe LED operation mode at power up. Upon exiting power-on reset, the states of the SCL and SDA inputs are checked. If SCL is low and SDA is high, then the duty cycles of all LEDs are set to 100% by setting the WIDTH registers to 1023. Otherwise, the LED duty cycles are all set to 0% by setting the WIDTH registers to 0. The microcontroller can then override this initial setting by writing a new value to the WIDTH register or registers.

8.4.2 Functional Mode

In functional mode, the digital circuitry controls the TPS92662A-Q1 outputs by accepting UART serial commands and by monitoring and responding to realtime events on the inputs associated with the analog and power circuitry. This allows PWMing of the LED outputs using the PHASE and WIDTH register settings. See the *PWM Dimming* section.

8.4.3 Default PWM (Communications Watchdog Timeout)

This mode allows you to program default pulse width settings for the LEDs, which are used only if the communication is broken. If the communications watchdog timer is enabled and subsequently times out (meaning that no successful, error-free UART communications have been received for the number of system clock cycles programmed via the CMWLMT register), then the LED pulse widths are set to the ones programmed in the DEFWIDTH registers. The LED phase shift times are set by the programmed values in the PHASE registers.

8.4.4 Clocks and Reset

The internal logic circuits of the TPS92662A-Q1 operate on the external clock applied to XTALI or CLK pins. The device also includes an on-chip oscillator for the charge pump circuitry.

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Device Functional Modes (continued)

The power-on reset occurs at cold power up when VIN is below the UVLO rising threshold and when VDD is below the VDD POR rising threshold. The power-on reset pulse (POR) is the global reset input to the TPS92663 logic circuitry and initializes the serial port to default values. The power-on reset occurs when either VDD falls below the VDD POR falling threshold or VIN falls below UVLO falling threshold.

The UART and the communications protocol state machine can additionally be reset by transmitting a “communications reset” character on the RX pin.

8.4.4.1 Crystal Oscillator (XTALI and XTALO inputs)

The internal Pierce Oscillator inverter is connected to the XTALI and XTALO pins, which provide the capability to drive crystal oscillators. The XTALO is used by the device as the clock input. Upon power up, the CLK pin is used to clock the UART and the internal logic circuitry. As shown in Figure 21, an on-chip counter connects the XTALO to the internal system clock only after at least 16 rising edges on XTALO. The XTALI input can also be used to buffer an externally generated clock. If there is no crystal connected to the XTALI and XTALO pins, XTALI **must** be connected to GND.

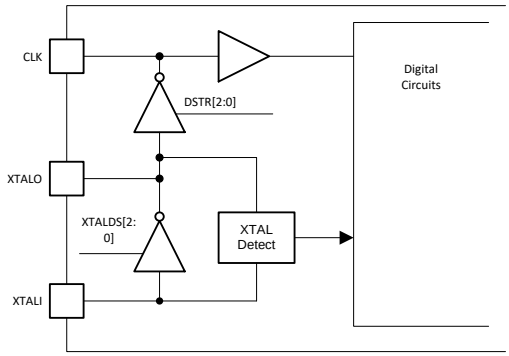


Figure 21. TPS92662A-Q1 Clock Pins

The output drive strength of the Pierce Oscillator inverter is controlled by the XTALDS[2:0] bits (see the *XTAL Inverter Drive Strength Register (XTALDS) – Read/Write* section).

The output drive strength of the CLK inverter is controlled by the DSTR[2:0] bits (see the *Drive Strength Register – Read-Only* section).

8.4.5 UART Interface

The MCU communicates with the TPS92662A-Q1 device using a Universal Asynchronous Receiver and Transmitter (UART). The pairs of TX and RX pins on the TPS92662A-Q1 device are feedthrough pins, and either pin can be used to connect the TPS92662A-Q1 device to the network. Feedthrough pins enable the use of metal core board with single layer routing. The UART communication process uses a command and response protocol mastered by the MCU to write and read the registers to and from each TPS92662A-Q1 device. The registers are mapped into an address space on each device. See the *TPS92662A-Q1 Registers* section for the register list.

8.4.5.1 UART Physical Layer

The TPS92662A-Q1 UART interface uses half-duplex communications (transmit and receive can not overlap). A tri-state buffer drives the TX pin. This allows multiple TPS92662A-Q1 devices to share a common pair of TX and RX signals by connecting all TPS92662A-Q1 TX lines together and all TPS92662A-Q1 RX lines together as shown in Figure 22. This topology requires two independent wires for UART communication. It is recommended to place an external pullup resistor on both the RX input return line of the microcontroller unit and on the TX input to the TPS92662A-Q1 devices. This defines both the TX and RX to the inactive state while the MCU is powering up.

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Device Functional Modes (continued)

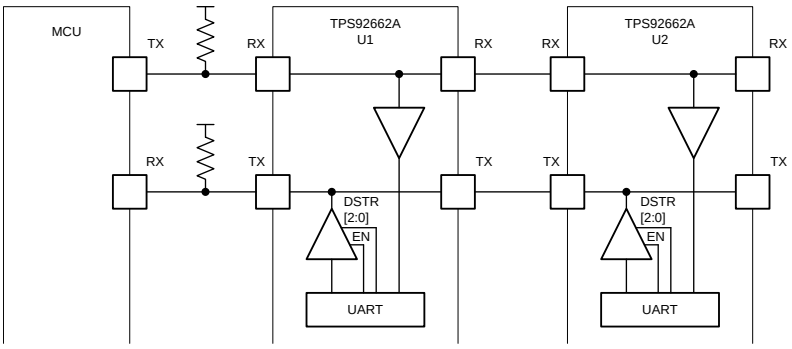


Figure 22. TPS92662A-Q1 UART Connection

Additionally, the physical TX and RX connections can be joined together through a CAN transceiver. This has the added advantage of protection from shorts to battery, shorts to ground on the cables, and harnesses between the MCU board and the TPS92662A-Q1 board. In addition, the CAN physical layer is more immune to EMS/BCI. Figure 23 shows this configuration.

To provide compatibility with CAN transceivers, the UART interface implements a half-duplex protocol. This allows TX and RX to share a single differential pair of wires. Pullup resistors are not required on RX and TX signals because the CAN Xcvr has an internal pullup on the RX output and the TPS92662A-Q1 has an internal pullup on the TX output, but they can be used to define both the TX and RX to the inactive state if issues are seen during test and development.

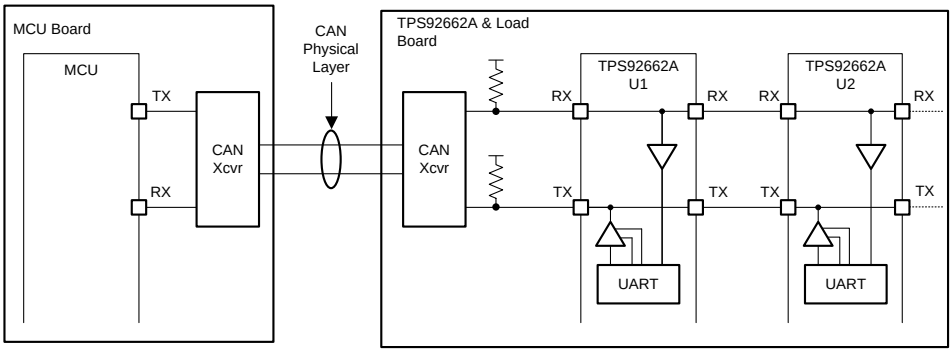


Figure 23. TPS92662A-Q1 UART Connection with CAN Physical Layer

8.4.5.2 UART DATA Format and Timing

The UART operates with eight data bits, one stop bit, and no parity (8N1). Figure 24 shows the waveform for an individual byte transfer on the TTL UART.

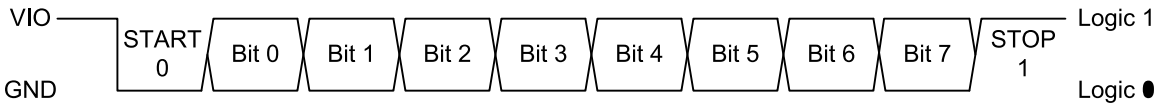


Figure 24. UART Data Byte with Start and Stop Bits

A logic “1” state occurs when the device drives the line to the VDD voltage. A logic “0” state occurs when the device drives the line to system ground. Figure 25 illustrates sending actual data bytes and is intended to represent what an actual UART waveform resembles on an oscilloscope or logic analyzer.

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Device Functional Modes (continued)

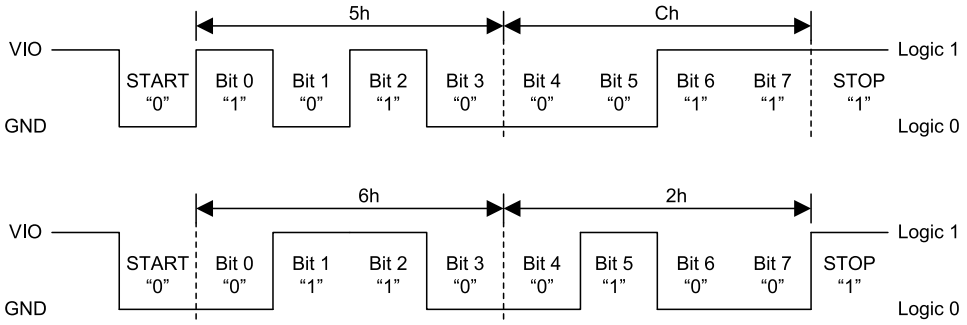


Figure 25. UART DATA Transaction Examples

The baud rate is based on the system CLK and is one-sixteenth of the clock input frequency (see Table 4 for examples). The UART uses 16x over-sampling on the incoming asynchronous RX signal.

Table 4. UART Baud Rate Examples

SYSTEM CLOCK FREQUENCY (MHz)	BAUD RATE (kbps)
6.0	375
8.0	500
16.0	1000

8.4.5.3 TX Bus Parking

The TPS92662A-Q1 device implements parking for the TX bus. This means that all the devices on the bus are actively driving TX high when in idle. This feature provides better immunity to Electro-Magnetic Susceptibility (EMS) in comparison with a floating or tri-stated output.

After a read command or when an acknowledge character after a write command is requested (ACKEN register bit is set to “1”), the TX buffer outputs of the non-addressed TPS92662A-Q1 devices are tri-stated. This allows the addressed device to respond. The required timing when a device releases the TX output is dependent on the configuration of the network. The SEPTR bit in SYSCFG register determines if direct UART connection (SEPTR = “1”) or the CAN transceiver configuration (SEPTR = “0”) is used.

To prevent bus contention between devices in the CAN transceiver topology, padding is added to the beginning and end of all write acknowledges and read responses. All non-addressed devices tristate the TX output for eight clock cycles (equal to one half of the bit time on UART) before the addressed device begins transmitting. Furthermore, a duration of eight input clock cycles is added to the end of the TX tristate time after the final STOP bit is received and before it is driven high. The following list summarizes the sequence for TX bus parking.

1. All devices drive the TX bus high.
2. Non-addressed devices tristate the TX output, and only the addressed device continues driving the TX bus high.
3. After a delay of eight clock cycles, the addressed device begins driving TX low with the first START bit.
4. The non-addressed devices continue tri-stating the TX outputs for eight more clock cycles after the final STOP bit, and the addressed device continues driving TX high.
5. Finally, all devices resume driving the TX bus high.

This sequence pads the response to account for differences in clock frequencies between devices, as well as delay in the loopback path of the CAN transceivers.

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8.4.5.4 Bus Turnaround Time

When considering back-to-back UART data transfer, there are two cases where additional time is required between the end of the previous frame STOP bit and the beginning of the next frame START bit. This time is also known as the minimum required bus turnaround time.

No turnaround time is required in the following cases:

- SEPTR = 1 (direct UART connection) for all back-to-back transfers
- SEPTR = 0 (CAN transceiver configuration) for back-to-back write commands
- SEPTR = 0 and ACKEN=0 between a write command and a read command

Turnaround time is required in the following cases:

- SEPTR = 0 and ACKEN = 1 between a write acknowledge and the next read or write command
- SEPTR = 0 between a read response and the next read or write command frame

In the cases where turnaround time is required, a finite amount of dead time must be inserted between the end of the STOP bit of the previous frame and the beginning of the START bit of the next frame. There must be at least 0.5 bit times of space inserted with the TX line of the MCU high following the final STOP bit of the previous frame before the falling edge of the first START bit of the next frame. This is the minimum delay allowed for the TPS92662A-Q1 UART to prepare for the next incoming frame. However, this dead time is also dependent upon the delay of the loopback path through the CAN transceiver and, therefore, can need to be extended, depending on the timing specifications of the CAN transceiver. Figure 26 illustrates this transaction.

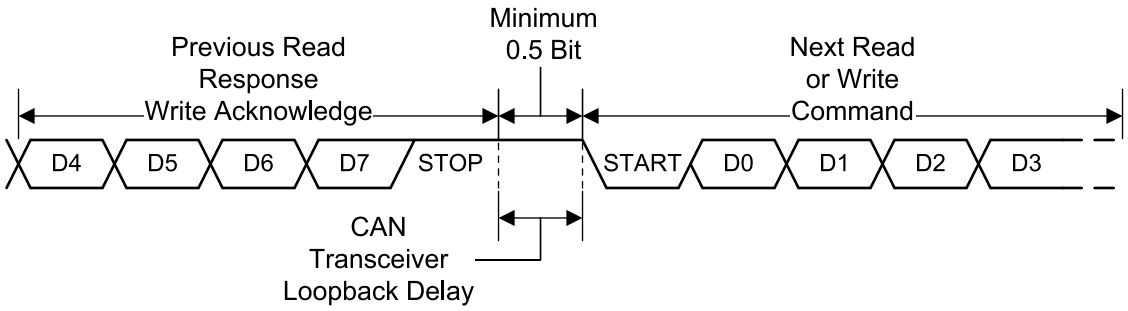


Figure 26. UART Data Transaction with CAN Loopback Delay

8.4.5.5 UART Communications Reset

The MCU unit can reset the device UART and protocol state machine at any time by holding the RX input low for a period of at least 12 bit times (16 × 12 CLK periods). This period signifies a break in communications and causes the TPS92662A-Q1 devices on the network to reset to a known-good state for receiving the next command frame. The device immediately aborts any response frames in progress.

NOTE

A communications reset does not reset the registers and does not halt normal LED PWM operation.

8.4.5.6 UART Communications Protocol

The UART communication process uses a command and response protocol mastered by the MCU to write and read the registers to and from each TPS92662A-Q1 device. This means that the TPS92662A-Q1 device never initiates traffic onto the network. The protocol maps the registers into an address space on each device. All of the registers are read-write (R/W). See the *TPS92662A-Q1 Registers* section for a register list.

The MCU uses the protocol to initiate a communication transaction by sending a command frame. This command frame addresses either one TPS92662A-Q1 device directly or broadcasts to all TPS92662A-Q1 devices on the network. This addressing can cause a response frame to be sent back from the slave TPS92662A-Q1 device, depending on the command. There are three types of command frames:

- **Broadcast Write Command:** To all TPS92662A-Q1 devices (1, 2, 3, 4, 12, 16, or 32 bytes)
- **Single Device Write Command:** To a specific TPS92662A-Q1 device (1, 2, 3, 4, 12, 16, or 32 bytes)

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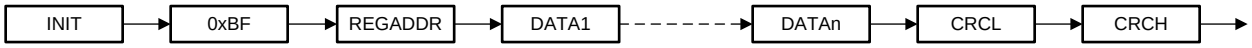
- **Single Device Read Command:** To a specific TPS92662A-Q1 device (1, 2, 3, 4, 12, 16, or 32 bytes)

There is no response frame given following a broadcast write command frame. Therefore, only two types of response frames exist that a TPS92662A-Q1 device sends back to the MCU:

- Single Device Write Acknowledge (if enabled)
- Single Device Read Response of the following bytes of data:
  - 1 byte
  - 2 bytes
  - 3 bytes
  - 4 bytes
  - 12 bytes
  - 16 bytes
  - 32 bytes

8.4.5.6.1 Broadcast Write Command

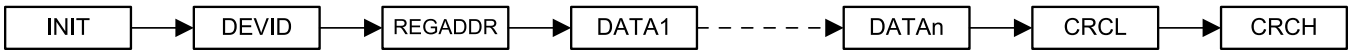
The broadcast write command is comprised of the following sequence:



There is no response given following a broadcast write command.

8.4.5.6.2 Single Device Write Command

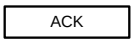
The single device write command sequence is:



If the ACKEN bit is set in the SYSCFG register, the addressed device transmits an acknowledge back to the MCU upon a successful write.

8.4.5.6.3 Single Device Acknowledge

The single device write acknowledge (“ACK”) consists of a single byte (ACK = 0x7F):

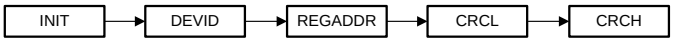


NOTE

ACK is transmitted by the addressed device only if the ACKEN bit is set in the SYSCFG register and only if the write is successful. A successful write yields no CRC checksum error or parity errors. The ACKEN bit must be the same for all devices on the bus. Therefore, a broadcast write to SYSCFG must be done during the initialization to set ACKEN (and SEPTR as well).

8.4.5.6.4 Single Device Read Command

The single device read command sequence is as follows:



A successfully-addressed device sends back the appropriate single device read response.

8.4.5.6.5 Single Device Read Response

The single device read response is as follows:



After the addressed device transmits the requested number of data bytes, the device then transmits the corresponding CRC for these bytes.

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8.4.5.7 Transaction Frame Description

The command and response frames include the following byte types:

- Frame Initialization Byte (INIT)
- Device Identification Byte (DEVID)
- Starting Register Address Byte (REGADDR)
- N Data Byte(s) (N = 1, 2, 3, 4, 12, 16, or 32)
- Cyclic Redundancy Code (CRC) error checking (two bytes)
- Acknowledge Byte (ACK)

8.4.5.7.1 Frame Initialization Byte

The INIT byte identifies the transaction as either a read command or a write command. In addition, the INIT byte indicates how many data bytes are being read or written. The number of data bytes to be read or written can be the following:

- 1
- 2
- 3
- 4
- 12
- 16
- 32

The specific transaction type is encoded into the INIT byte using Table 5.

Table 5. Frame Initialization Byte

INIT TYPE	FRAME DESCRIPTION	ENCODED INIT BYTE
1	Write Command of 1 Byte	0x87
2	Write Command of 2 Bytes	0x99
3	Write Command of 3 Bytes	0x1E
4	Write Command of 4 Bytes	0xAA
5	Write Command of 12 Bytes	0x2D
6	Write Command of 16 Bytes	0x33
7	Write Command of 32 Bytes	0xB4
8	Read Command of 1 Byte	0x4B
9	Read Command of 2 Bytes	0xCC
10	Read Command of 3 Bytes	0xD2
11	Read Command of 4 Bytes	0x55
12	Read Command of 12 Bytes	0xE1
13	Read Command of 16 Bytes	0x66
14	Read Command of 32 Bytes	0x78

8.4.5.7.2 DEVID Byte

The DEVID byte holds the device ID of the destination. Table 6 shows the encoding used for the DEVID byte. There are five dev_id bits–one reserved bit and two parity bits.

Table 6. DEVID Byte

BIT 7	BIT 6	BIT 5	BIT 4	BIT 3	BIT 2	BIT 1	BIT 0
p[1]	p[0]	1	dev_id[4]	dev_id[3]	dev_id[2]	dev_id[1]	dev_id[0]

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The parity bits for the DEVID byte are calculated with the following equations:

p[1] = ~(dev_id[1] ^ dev_id[3] ^ dev_id[4] ^ 1)

p[0] = dev_id[0] ^ dev_id[1] ^ dev_id[2] ^ dev_id[4]

This results in the following mapping for the DEVID byte:

Table 7. DEVID Byte with Parity

dev_id (DEVID[4:0])		DEVID BYTE (WITH PARITY)	
HEX	DEC	HEX	DEC
0x00	0	0x20	32
0x01	1	0x61	97
0x02	2	0xE2	226
0x03	3	0xA3	163
0x04	4	0x64	100
0x05	5	0x25	37
0x06	6	0xA6	166
0x07	7	0xE7	231
0x08	8	0xA8	168
0x09	9	0xE9	233
0x0A	10	0x6A	106
0x0B	11	0x2B	43
0x0C	12	0xEC	236
0x0D	13	0xAD	173
0x0E	14	0x2E	46
0x0F	15	0x6F	111
0x10	16	0xF0	240
0x11	17	0xB1	177
0x12	18	0x32	50
0x13	19	0x73	115
0x14	20	0xB4	180
0x15	21	0xF5	245
0x16	22	0x76	118
0x17	23	0x37	55
0x18	24	0x78	120
0x19	25	0x39	57
0x1A	26	0xBA	186
0x1B	27	0xFB	251
0x1C	28	0x3C	60
0x1D	29	0x7D	125
0x1E	30	0xFE	254
0x1F	31	0xBF	191

The device ID of each TPS92662A-Q1 device is determined by three address pins of ADDR0, ADDR1, and ADDR2. The mapping of the address pins to the device ID are explained in the following.

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8.4.5.7.2.1 Up to Eight TPS92662A-Q1 Devices on One Bus

The TPS92662A-Q1 device includes three address pins of ADDR0, ADDR1 and ADDR2. The address pins can be connected directly to GND or VDD to provide addresses for up to eight TPS92662A-Q1 devices on the same communication network. Upon power up, these address pins are used to map the dev_id. They are digitally mapped to the three bits of the device ID shown in Table 8.

Table 8. Device ID at Power-Up (ADCID = 0x00)

dev_id[4]	dev_id[3]	dev_id[2]	dev_id[1]	dev_id[0]
0	0	ADDR2	ADDR1	ADDR0

Following power-on-reset, each TPS92662A-Q1 occupies a device ID from 0x00 through 0x07.

The special dev_id value of 0x1F is reserved for broadcast writes. Any write to dev_id = 0x1F (with parity bits, DEVID = 0xBF) is considered a broadcast write to all devices. All TPS92662A-Q1 devices on the bus are written on a broadcast write.

8.4.5.7.2.2 Up to 16 TPS92662A-Q1 Devices on One Bus

If there are more than eight TPS92662A-Q1s on the network, a broadcast write to the ADCID register is required to allow each device to acquire an own unique device ID based on an ADC input. For example, if there are between nine and 16 TPS92662A-Q1 devices on the communication network, the ADCID is written to 0x01 via a broadcast write. Subsequently, the two MS-bits of the 8-bit ADC conversion from the ADDR0 are used as the two LSB-bits of the dev_id. These two bits combined with the value of ADDR2 and ADDR1 provide up to 16 unique device IDs. Following the ADC conversions, all the DEVID fields of the transactions are compared against this newly acquired device ID.

Table 9. Device Addressing Following a Broadcast Write of ADCID = 0x01

dev_id[4]	dev_id[3]	dev_id[2]	dev_id[1]	dev_id[0]
0	ADDR2	ADDR1	ADC_OUT[7]	ADC_OUT[6]

8.4.5.7.2.3 Up to 32 TPS92662A-Q1 Devices on One Bus

Similarly, if there are between 17 and 31 TPS92662A-Q1 devices on the communications network, the ADCID is written to 0x02. Subsequently, the three MS-bits of the 8-bit ADC conversion from the ADDR0 are used as the three LS-bits of the dev_id. These three bits combined with the value of ADDR2 and ADDR1 provide up to 31 unique device IDs. Following the ADC conversions, all the DEVID fields of the transactions are compared against this newly acquired device ID.

Table 10. Device Addressing Following a Broadcast Write of ADCID = 0x02

dev_id[4]	dev_id[3]	dev_id[2]	dev_id[1]	dev_id[0]
ADDR2	ADDR1	ADC_OUT[7]	ADC_OUT[6]	ADC_OUT[5]

The device ID of 0x1F (DEVID = 0xBF after adding the parity bits) is reserved for broadcast writes. The resistor divider on ADDR0 together with ADDR2 and ADDR1 connections must be chosen to avoid the following address:

ADDR2, ADDR1, ADC_OUT[7:5] = “11111”

Table 11 shows the summary of the DEVID byte values for a different number of the TPS92662A-Q1 devices on the bus.

Table 11. DEVID Byte Summary

NUMBER OF DEVICES	ADCID	DEVID BYTE							
		BIT 7	BIT 6	BIT 5	BIT 4	BIT 3	BIT 2	BIT 1	BIT 0
Up to 8	0x00 (default) or 0x03	p[1]	p[0]	1	0	0	ADDR2	ADDR1	ADDR0
Up to 16	0x01	p[1]	p[0]	1	0	ADDR2	ADDR1	ADC_OUT[7]	ADC_OUT[6]
Up to 31	0x02	p[1]	p[0]	1	ADDR2	ADDR1	ADC_OUT[7]	ADC_OUT[6]	ADC_OUT[5]

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8.4.5.7.3 REGADDR

The starting register address is a single byte. The protocol allows writing or reading up to 32 successive register locations starting from the addressed register to be accomplished with a single command. The starting register address byte identifies the first TPS92662A-Q1 register being written or read, as described in the *TPS92662A-Q1 Registers* section.

The starting register address byte is present in only write and read command transactions, not in read response transactions. Multi-byte transfers that go beyond address 0xFF automatically wrap back to address 0x00 and continue.

The following two register addresses are handled differently than the other register addresses.

- I2CFIFODATA: Once a multi-byte write or read reaches the I2CFIFODATA address, the register address-pointer is no longer incremented, allowing for multiple bytes to be transferred to and from the I^2C data FIFO using a multi-byte transfer UART frame.
- ADCID: The ADCID register address is also a special location. Any write to this location must be a single-byte broadcast write. The UART protocol state machine is reset after processing the ADCID broadcast write. Any multi-byte or single-device write frames that encompass the ADCID register are considered illegal transfers and are completely discarded, and none of the registers are written. Instead, the protocol state machine returns immediately to the reset state. Furthermore, no write acknowledge is transmitted following the illegal write frame.

8.4.5.7.4 DATA Bytes

The frame initialization byte specifies the number of data bytes to be included in the frame.

8.4.5.7.5 CRCL and CRCH Bytes

The CRCL and CRCH bytes (CRC low-byte and CRC high-byte, respectively) are calculated on incoming command frames using CRC-16-IBM and allow detection of errors within the frame. The CRC error count register (CERRCNT) is incremented each time a CRC error occurs on an incoming read or write command (see the *TPS92662A-Q1 Registers* section for details). The CRC bytes are also calculated by the addressed device during the read response. The CRC bytes are then appended to the end of the read data, low-byte first. This allows the master to check the read data coming from the slave device for any transmission errors.

8.4.5.7.6 ACK Byte

Following a successfully received write command, the addressed device sends back an ACK byte (0x7F) if the ACKEN bit is set in the SYSCFG register.

8.4.5.7.7 Examples

The total number of transmitted bytes depends on the specific task being performed. *Example 1: Broadcast Write ADCID = 0x01* through *Example 3: Single Device Read of Two Bytes* show the sequence of transmitted bytes for a given transaction.

8.4.5.7.7.1 Example 1: Broadcast Write ADCID = 0x01

Write to device ID = 0x1F (DEVID with parity = 0xBF): ADCID = 0x01:

INIT	DEVID	REGADDR	DATA	CRCL	CRCH
0x87	0xBF	0x83	0x01	0xB9	0xA0

8.4.5.7.7.2 Example 2: Single Device Write of Three Bytes

Write to device ID = 0x04 beginning at register 0x00: MPHASE01L = 0x55, MPHASE02L = 0x56, and MPHASE03L = 0x57 with the ACKEN bit set:

INIT	DEVID	REGADDR	DATA			CRCL	CRCH
0x1E	0x64	0x00	0x55	0x56	0x57	0x1C	0x38

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Write acknowledge from identified slave:

ACK
0x7F

8.4.5.7.7.3 Example 3: Single Device Read of Two Bytes

Read from device ID = 0x05 beginning at register address 0x20: LPHASE01H and LPHASE02H:

INIT	DEVID	REGADDR	CRCL	CRCH
0xCC	0x25	0x20	0xDB	0x77

Single device read response. Data = 0x11, 0x22:

DATA		CRCL	CRCH
0x11	0x22	0x8C	0x49

8.4.6 I²C Master for External EEPROM

The TPS92662A-Q1 contains a UART-to-I²C bridge that allows the master MCU to access a local EEPROM via the UART interface. This feature of the TPS92662A-Q1 allows the wiring harness between the ECU and the LED load board to have two fewer connections and simplifies the hardware interconnect. The I²C master supports 7-bit addressing mode at SCL frequencies of 400 kHz or less. The I²C master logic is not fully featured and does not support multi-master or clock stretching since it is primarily intended to communicate with an EEPROM device that stores lighting module configuration data.

In order for the I²C master to be used, external pullups must be tied from SDA and SCL to the VDD rail. The I2CTICK register must be programmed to attain the desired I²C bit-rate, and the I2CEN bit must be set to '1' in the SYSCFG register.

Upon issuing a read or write command (by writing the START bit to 1 in the I2CCTRL register), the I²C master first checks the state of the bus by making sure that SDA is high. If SDA is low, the EEPROM can be stuck in an odd state. The I²C master issues up to 15 clock pulses on SCL to clear out the condition, checking the state of SDA at each clock. If SDA is still low after 15 clocks, then the SDALOW bit is set in the I2CSTAT register and the transaction is aborted. If SDA has been released during the attempt to clear the bus, then the requested transaction begins with a START condition.

8.4.6.1 I²C Data Format with 7-Bit Address

Data transfers follow the format shown in Figure 27. After the START condition, a slave address is transmitted. This address is seven bits long followed by an eighth bit, which is a data direction bit ('R/W'). If the R/W bit is zero, it indicates a transmit operation (send). If it is one, it indicates a request for data (receive). A data transfer is always terminated by a STOP condition generated by the master. However, a master can initiate communications with another device on the bus by generating a repeated START condition and addressing another slave without first generating a STOP condition. Various combinations of receive and transmit formats are then possible within a single transfer.

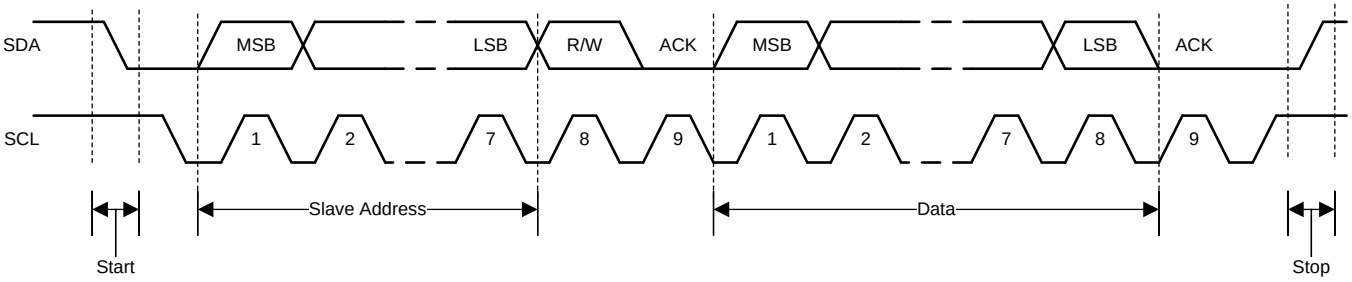


Figure 27. Complete Data Transfer with a 7-Bit Address

The first seven bits of the first byte make up the slave address. The eighth bit determines the direction of the message. A zero in the R/W position of the first byte means that the master transmits (sends) data to the selected slave, and a one in this position means that the master receives data from the slave.

8.4.6.2 I²C Master Initialization

To use the I²C master logic, the MCU must first enable I²C functionality by writing the I2CEN bit to a 1 in the SYSCFG register and program the desired SCL frequency by writing the appropriate value to the I2CTICK register. Once these two writes are performed, the I²C master logic is ready for read or write operation.

8.4.6.3 Single-Byte EEPROM Write

The I²C master is capable of performing single-byte writes to an external I²C EEPROM. Figure 28 shows the EEPROM write sequence. It consists of the I²C master transmitting a START condition, the EEPROM slave address and the R/W bit (cleared to 0). Next, the byte address and data byte is written followed by a STOP condition. Each byte must be acknowledged by the EEPROM in order for the I²C master to continue to the next byte, otherwise the transfer is terminated prematurely with a STOP condition and the appropriate error bit or bits are set in the I2CSTAT register.

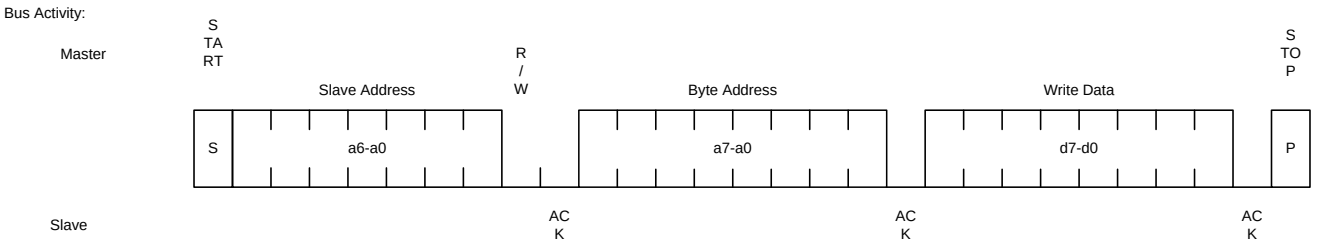


Figure 28. Single-Byte EEPROM Write Sequence

The MCU must program the slave address (I2CSLA), the byte address (I2CADR), and the data byte (I2CDAT) to be sent to the EEPROM. Next, the MCU issues the write command by writing the START bit of the I2CCTRL register to a 1. Following a single-byte EEPROM write command, the MCU can poll the I2CSTAT register to check that the BUSY bit is cleared, indicating that the write is finished. After the BUSY bit is cleared, the SLANACK, ADRNACK, DATNACK, SDALOW, and BUSYERR bits must be checked. If any of these bits are 1, the write command can be re-tried by writing the START bit to 1 again.

Figure 29 shows the write command sequence flow chart.

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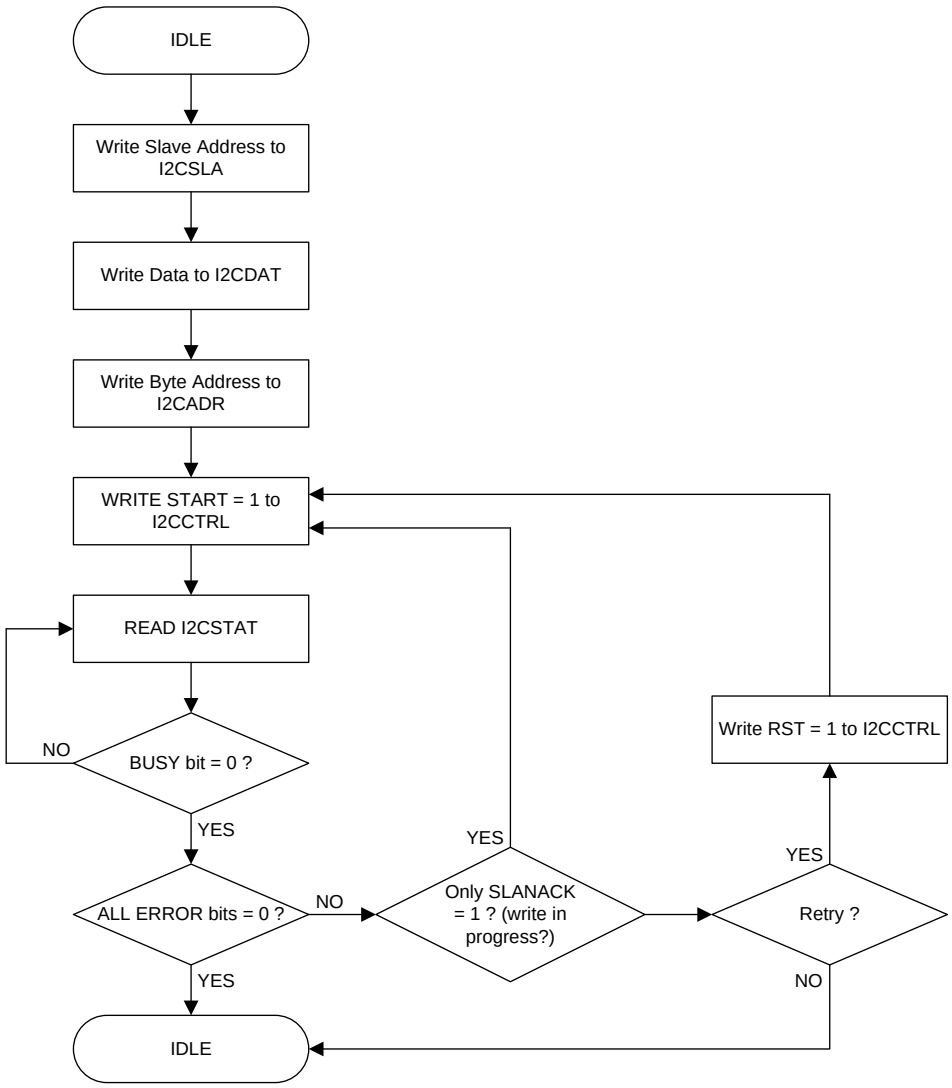


Figure 29. Single Byte EEPROM Write

The ERROR bits are defined as the following:

- BUSYERR
- SLANACK
- ADRNACK
- DATNACK
- SDALOW

8.4.6.4 Single-Byte and Multi-Byte EEPROM Read Operation

The I^2C master is capable of performing single-byte and multi-byte reads of an external I^2C EEPROM. Figure 30 shows the single-byte EEPROM read sequence. Figure 31 shows the difference for multi-byte EEPROM read.

The read sequence begins with the following:

- I^2C master transmitting a START condition
- The slave address
- The R/W bit (cleared to 0)
- The byte address from which to begin reading

This portion of the transaction sets the address pointer in the EEPROM. The I²C master then issues a repeated START condition, the slave address, and the R/W bit (set to 1). Afterwards, the I²C master issues clock pulses on SCL and reads the data coming in on SDA. This sequence continues for the requested read length. The I²C master acknowledges each incoming data byte except the last one, after which it issues a STOP condition to terminate the transfer. If the slave does not acknowledge the slave address or byte address, the transfer is terminated prematurely with a STOP condition and the appropriate error bits are set in the I2CSTAT register.

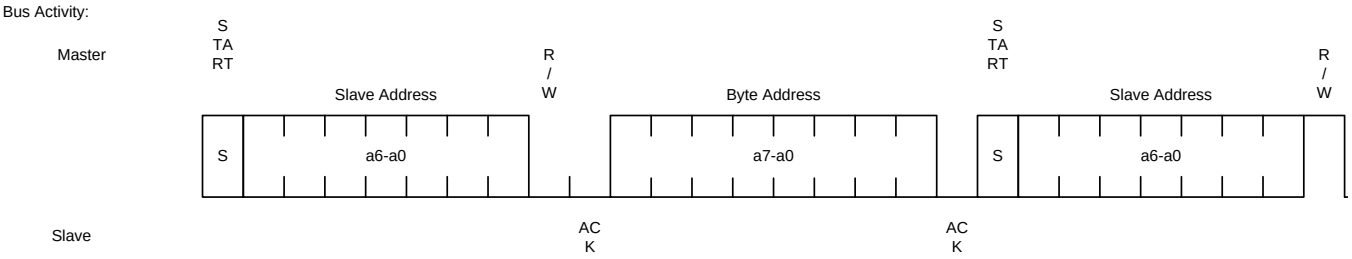


Figure 30. Single-Byte EEPROM Read Sequence

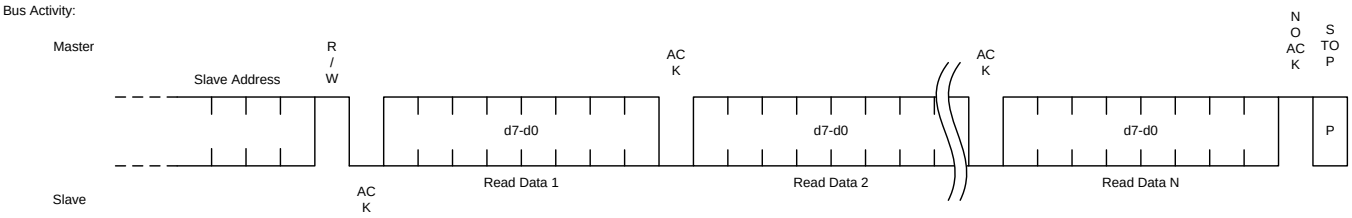


Figure 31. Multi-Byte EEPROM Read Sequence

To perform an I²C EEPROM read, the MCU must program the slave address (I2CSLA), the byte address (I2CADR), and the read length (I2CRLEN). Next, the MCU must issue the read command by writing the START and RD bits of the I2CCTRL register to 1. Once the read command is issued, there are two cases that must be handled differently:

1. Read length greater than or equal to the receive FIFO depth
2. Read length less than the receive FIFO depth

These are described in more detail in the following sections. Note that the receive FIFO depth is 16 bytes.

8.4.6.4.1 Read Length Greater Than or Equal to Receive FIFO Depth

In case the read length is greater than or equal to the receive FIFO depth of 16, the I²C master reads enough data bytes to fill the receive FIFO. When the receive FIFO is full, the RFF bit is set to 1 in the I2CSTAT register and the I²C master pauses and waits for the receive FIFO to become less than full before it continues to read from the EEPROM. In this case, the MCU waits until the RFF bit is 1, and then reads all of the receive FIFO data from the I2CDAT register. The MCU can repeat this sequence of checking for the RFF bit and reading the entire receive FIFO depth until the remainder of data to be read from the EEPROM is less than the receive FIFO depth. At this point, the MCU can use the technique described in the next section to read the remaining data from the FIFO.

8.4.6.4.2 Read Length Less Than Receive FIFO Depth

In case the read length is less than the receive FIFO depth (less than 16), the MCU can simply allow all of the EEPROM reads to take place and wait for the BUSY bit to be cleared, indicating that the entire read sequence is finished. After the BUSY bit is cleared, the SLANACK, ADRNACK, SDALOW, and BUSYERR bits must be checked. If any of these bits are 1, the read command can be re-tried by writing the RST bit to 1 first to reset the EEPROM read state machine and flush the receive FIFO, then writing the START and RD bits to 1 again. If none of these ERROR bits are 1, then the MCU can read the data using an optimized set of UART reads, depending on how many bytes of data are left to read. The data can be read from the I2CDAT register using combinations of 1, 2, 3, 4, or 12 byte UART protocol reads.

Figure 32 shows the read command sequence.

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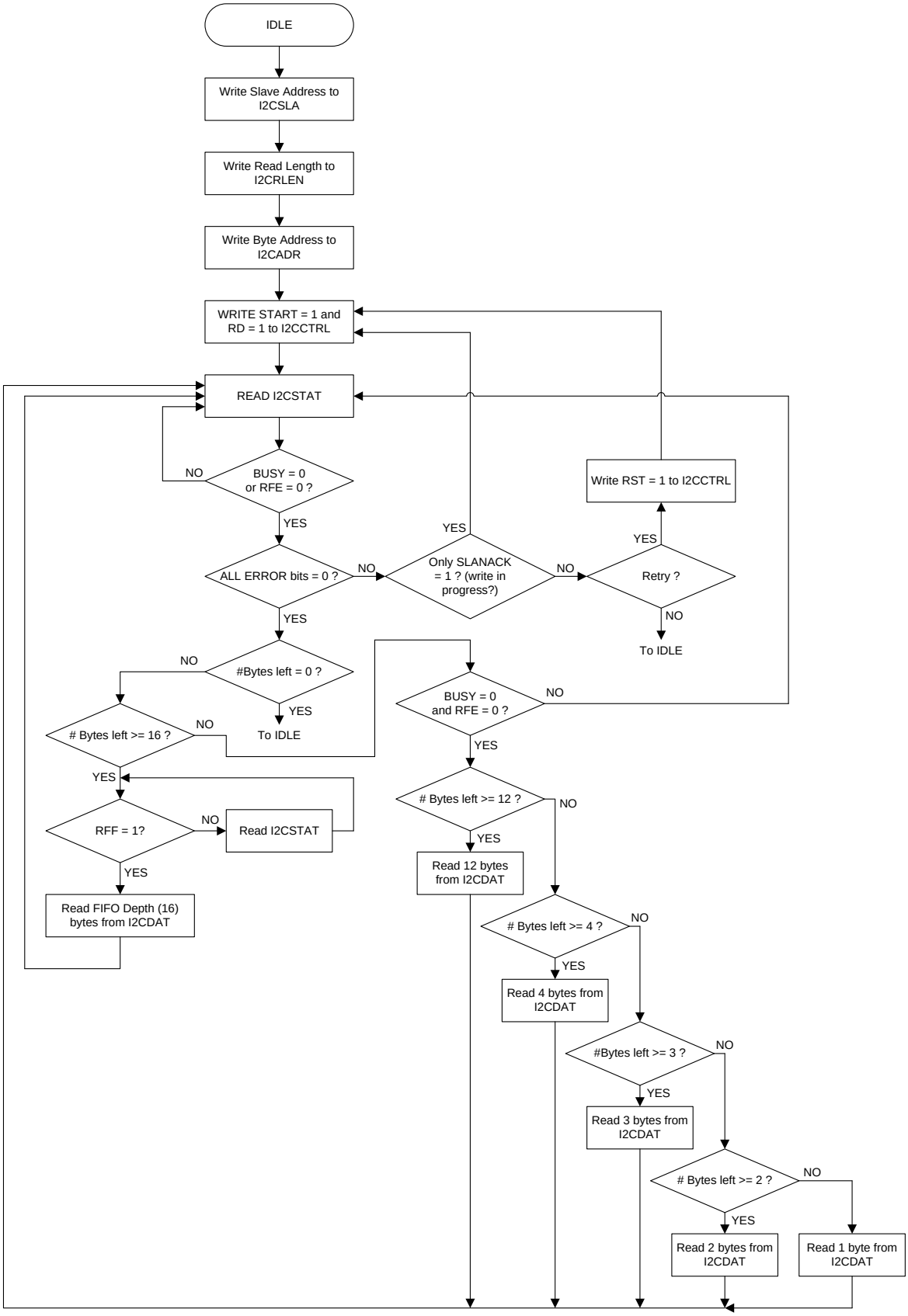


Figure 32. Single-Byte and Multi-Byte EEPROM Read Logic Diagram

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8.5 TPS92662A-Q1 Registers

The registers in the TPS92662A device contain programmed information and operating status. Upon power-up the registers are reset to the default values. Register addresses marked RESERVED or not shown in the register map can be written with any value. Regardless of the data written to it, a read back from a RESERVED register always returns the value of 0. Table 12 lists the memory-mapped registers for the TPS92662A device.

Table 12. TPS92662A-Q1 Register Map

ADDR	REGISTER	D7	D6	D5	D4	D3	D2	D1	D0	DEFAULT
LED PWM Phase-Shift and Width Registers										
00h	**MPHASE01L**	PHASE01[7:0]								00000000
01h	**MPHASE02L**	PHASE02[7:0]								00000000
02h	**MPHASE03L**	PHASE03[7:0]								00000000
03h	**MPHASE03_01H**	RESERVED		PHASE03[9:8]		PHASE02[9:8]		PHASE01[9:8]		00000000
04h	**MPHASE04L**	PHASE04[7:0]								00000000
05h	**MPHASE05L**	PHASE05[7:0]								00000000
06h	**MPHASE06L**	PHASE06[7:0]								00000000
07h	**MPHASE06_04H**	RESERVED		PHASE06[9:8]		PHASE05[9:8]		PHASE04[9:8]		00000000
08h	**MPHASE07L**	PHASE07[7:0]								00000000
09h	**MPHASE08L**	PHASE08[7:0]								00000000
0Ah	**MPHASE09L**	PHASE09[7:0]								00000000
0Bh	**MPHASE09_07H**	RESERVED		PHASE09[9:8]		PHASE08[9:8]		PHASE07[9:8]		00000000
0Ch	**MPHASE10L**	PHASE10[7:0]								00000000
0Dh	**MPHASE11L**	PHASE11[7:0]								00000000
0Eh	**MPHASE12L**	PHASE12[7:0]								00000000
0Fh	**MPHASE12_10H**	RESERVED		PHASE12[9:8]		PHASE11[9:8]		PHASE10[9:8]		00000000
10h	**MWIDTH01L**	WIDTH01[7:0]								00000000
11h	**MWIDTH02L**	WIDTH02[7:0]								00000000
12h	**MWIDTH03L**	WIDTH03[7:0]								00000000
13h	**MWIDTH03_01H**	RESERVED		WIDTH03[9:8]		WIDTH02[9:8]		WIDTH01[9:8]		00000000
14h	**MWIDTH04L**	WIDTH04[7:0]								00000000
15h	**MWIDTH05L**	WIDTH05[7:0]								00000000
16h	**MWIDTH06L**	WIDTH06[7:0]								00000000
17h	**MWIDTH06_04H**	RESERVED		WIDTH06[9:8]		WIDTH05[9:8]		WIDTH04[9:8]		00000000
18h	**MWIDTH07L**	WIDTH07[7:0]								00000000
19h	**MWIDTH08L**	WIDTH08[7:0]								00000000
1Ah	**MWIDTH09L**	WIDTH09[7:0]								00000000
1Bh	**MWIDTH09_07H**	RESERVED		WIDTH09[9:8]		WIDTH08[9:8]		WIDTH07[9:8]		00000000
1Ch	**MWIDTH10L**	WIDTH10[7:0]								00000000
1Dh	**MWIDTH11L**	WIDTH11[7:0]								00000000
1Eh	**MWIDTH12L**	WIDTH12[7:0]								00000000
1Fh	**MWIDTH12_10H**	RESERVED		WIDTH12[9:8]		WIDTH11[9:8]		WIDTH10[9:8]		00000000
20h	**LPHASE01H**	PHASE01[9:2]								00000000
21h	**LPHASE02H**	PHASE02[9:2]								00000000
22h	**LPHASE03H**	PHASE03[9:2]								00000000
23h	**LPHASE04H**	PHASE04[9:2]								00000000
24h	**LPHASE05H**	PHASE05[9:2]								00000000
25h	**LPHASE06H**	PHASE06[9:2]								00000000
26h	**LPHASE07H**	PHASE07[9:2]								00000000
27h	**LPHASE08H**	PHASE08[9:2]								00000000
28h	**LPHASE09H**	PHASE09[9:2]								00000000

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TPS92662A-Q1 Registers (continued)

Table 12. TPS92662A-Q1 Register Map (continued)

ADDR	REGISTER	D7	D6	D5	D4	D3	D2	D1	D0	DEFAULT
29h	LPHASE10H	PHASE10[9:2]								00000000
2Ah	LPHASE11H	PHASE11[9:2]								00000000
2Bh	LPHASE12H	PHASE12[9:2]								00000000
2Ch	LPHASE03_01L	RESERVED		PHASE03[1:0]		PHASE02[1:0]		PHASE01[1:0]		00000000
2Dh	LPHASE06_04L	RESERVED		PHASE06[1:0]		PHASE05[1:0]		PHASE04[1:0]		00000000
2Eh	LPHASE09_07L	RESERVED		PHASE09[1:0]		PHASE08[1:0]		PHASE07[1:0]		00000000
2Fh	LPHASE12_10L	RESERVED		PHASE12[1:0]		PHASE11[1:0]		PHASE10[1:0]		00000000
30h	LWIDTH01H	WIDTH01[9:2]								00000000
31h	LWIDTH02H	WIDTH02[9:2]								00000000
32h	LWIDTH03H	WIDTH03[9:2]								00000000
33h	LWIDTH04H	WIDTH04[9:2]								00000000
34h	LWIDTH05H	WIDTH05[9:2]								00000000
35h	LWIDTH06H	WIDTH06[9:2]								00000000
36h	LWIDTH07H	WIDTH07[9:2]								00000000
37h	LWIDTH08H	WIDTH08[9:2]								00000000
38h	LWIDTH09H	WIDTH09[9:2]								00000000
39h	LWIDTH10H	WIDTH10[9:2]								00000000
3Ah	LWIDTH11H	WIDTH11[9:2]								00000000
3Bh	LWIDTH12H	WIDTH12[9:2]								00000000
3Ch	LWIDTH03_01L	RESERVED		WIDTH03[1:0]		WIDTH02[1:0]		WIDTH01[1:0]		00000000
3Dh	LWIDTH06_04L	RESERVED		WIDTH06[1:0]		WIDTH05[1:0]		WIDTH04[1:0]		00000000
3Eh	LWIDTH09_07L	RESERVED		WIDTH09[1:0]		WIDTH08[1:0]		WIDTH07[1:0]		00000000
3Fh	LWIDTH12_10L	RESERVED		WIDTH12[1:0]		WIDTH11[1:0]		WIDTH10[1:0]		00000000
40h	DPHASE01L	PHASE01[7:0]								00000000
41h	DPHASE01H	RESERVED						PHASE01[9:8]		00000000
42h	DWIDTH01L	WIDTH01[7:0]								00000000
43h	DWIDTH01H	RESERVED						WIDTH01[9:8]		00000000
44h	DPHASE02L	PHASE02[7:0]								00000000
45h	DPHASE02H	RESERVED						PHASE02[9:8]		00000000
46h	DWIDTH02L	WIDTH02[7:0]								00000000
47h	DWIDTH02H	RESERVED						WIDTH02[9:8]		00000000
48h	DPHASE03L	PHASE03[7:0]								00000000
49h	DPHASE03H	RESERVED						PHASE03[9:8]		00000000
4Ah	DWIDTH03L	WIDTH03[7:0]								00000000
4Bh	DWIDTH03H	RESERVED						WIDTH03[9:8]		00000000
4Ch	DPHASE04L	PHASE04[7:0]								00000000
4Dh	DPHASE04H	RESERVED						PHASE04[9:8]		00000000
4Eh	DWIDTH04L	WIDTH04[7:0]								00000000
4Fh	DWIDTH04H	RESERVED						WIDTH04[9:8]		00000000
50h	DPHASE05L	PHASE05[7:0]								00000000
51h	DPHASE05H	RESERVED						PHASE05[9:8]		00000000
52h	DWIDTH05L	WIDTH05[7:0]								00000000
53h	DWIDTH05H	RESERVED						WIDTH05[9:8]		00000000
54h	DPHASE06L	PHASE06[7:0]								00000000
55h	DPHASE06H	RESERVED						PHASE06[9:8]		00000000
56h	DWIDTH06L	WIDTH06[7:0]								00000000

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TPS92662A-Q1 Registers (continued)

Table 12. TPS92662A-Q1 Register Map (continued)

ADDR	REGISTER	D7	D6	D5	D4	D3	D2	D1	D0	DEFAULT
57h	DWIDTH06H	RESERVED						WIDTH06[9:8]		00000000
58h	DPHASE07L	PHASE07[7:0]								00000000
59h	DPHASE07H	RESERVED						PHASE07[9:8]		00000000
5Ah	DWIDTH07L	WIDTH07[7:0]								00000000
5Bh	DWIDTH07H	RESERVED						WIDTH07[9:8]		00000000
5Ch	DPHASE08L	PHASE08[7:0]								00000000
5Dh	DPHASE08H	RESERVED						PHASE08[9:8]		00000000
5Eh	DWIDTH08L	WIDTH08[7:0]								00000000
5Fh	DWIDTH08H	RESERVED						WIDTH08[9:8]		00000000
60h	DPHASE09L	PHASE09[7:0]								00000000
61h	DPHASE09H	RESERVED						PHASE09[9:8]		00000000
62h	DWIDTH09L	WIDTH09[7:0]								00000000
63h	DWIDTH09H	RESERVED						WIDTH09[9:8]		00000000
64h	DPHASE10L	PHASE10[7:0]								00000000
65h	DPHASE10H	RESERVED						PHASE10[9:8]		00000000
66h	DWIDTH10L	WIDTH10[7:0]								00000000
67h	DWIDTH10H	RESERVED						WIDTH10[9:8]		00000000
68h	DPHASE11L	PHASE11[7:0]								00000000
69h	DPHASE11H	RESERVED						PHASE11[9:8]		00000000
6Ah	DWIDTH11L	WIDTH11[7:0]								00000000
6Bh	DWIDTH11H	RESERVED						WIDTH11[9:8]		00000000
6Ch	DPHASE12L	PHASE12[7:0]								00000000
6Dh	DPHASE12H	RESERVED						PHASE12[9:8]		00000000
6Eh	DWIDTH12L	WIDTH12[7:0]								00000000
6Fh	DWIDTH12H	RESERVED						WIDTH12[9:8]		00000000
Slew Rate Registers										
70h	SLEWRATE	SLEW_12_10[1:0]		SLEW_09_07[1:0]		SLEW_06_04[1:0]		SLEW_03_01[1:0]		11111111
Over Voltage Limit Registers										
71h	OVLMT	OVLMT_12_10[1:0]		OVLMT_09_07[1:0]		OVLMT_06_04[1:0]		OVLMT_03_01[1:0]		10101010
Parallel LED String Registers										
72h	PARLED	RESERVED						PARLED[1:0]		00000000
Default Pulse Width Registers										
73h	DEFWIDTH02_01	DEFWIDTH02[3:0]				DEFWIDTH01[3:0]				00000000
74h	DEFWIDTH04_03	DEFWIDTH04[3:0]				DEFWIDTH03[3:0]				00000000
75h	DEFWIDTH06_05	DEFWIDTH06[3:0]				DEFWIDTH05[3:0]				00000000
76h	DEFWIDTH08_07	DEFWIDTH08[3:0]				DEFWIDTH07[3:0]				00000000
77h	DEFWIDTH10_09	DEFWIDTH10[3:0]				DEFWIDTH09[3:0]				00000000
78h	DEFWIDTH12_11	DEFWIDTH12[3:0]				DEFWIDTH11[3:0]				00000000
System Configuration Register										
80h	SYSCFG	SEPTR	I2CEN	ACKEN	PSON	CMWEN	SYNCOEN	SYNCPEN	PWR	00000000
Communications Watchdog Timer Tap Point Register										
81h	CMWTAP	RESERVED					CMWTAP[2:0]			00000101
PWM Tick Period Register										
82h	PWMTICK	PTBASE[1:0]		PTCNT[5:0]						00001100
ADC ID Register										

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TPS92662A-Q1 Registers (continued)

Table 12. TPS92662A-Q1 Register Map (continued)

ADDR	REGISTER	D7	D6	D5	D4	D3	D2	D1	D0	DEFAULT
83h	ADCID	RESERVED						ADCID[1:0]		00000000
SOFTSYNC Register										
84h	SOFTSYNC	RESERVED							SSYNC	00000000
XTAL DRIVE STRENGTH Register										
85h	XTALDS	RESERVED					XTALDS[2:0]			00000000
I2C Tick Period Register										
90h	I2CTICK	RESERVED					I2CTICK[2:0]			00000101
I2C Slave Address Register										
91h	I2CSLA	SA[6:0]							RESERVED	00000000
I2C Read Length Register										
92h	I2CRLEN	I2CRLEN[7:0]								00000000
I2C Byte Address Register										
93h	I2CADR	I2CADR[7:0]								00000000
I2C Master Control Register										
94h	I2CCTRL	RESERVED				NOADR	START	RD	RST	00000000
I2C Data Register										
95h	I2CDAT	I2CDAT[7:0]								00000000
I2C Status Register										
96h	I2CSTAT	RFE	RFF	BUSYERR	SLANACK	ADRNACK	DATNACK	SDALOW	BUSY	10000000
I2C Bus Monitor Register (Read-Only)										
97h	I2CBMON	RESERVED						SDA	SCL	00000000
ADC Registers (Read-Only)										
A0h	ADC1	ADC1[7:0]								00000000
A1h	ADC2	ADC2[7:0]								00000000
DSTR Register (Read-Only)										
A2h	DSTR	RESERVED					DSTR[2:0]			00000000
Fault and CRC Error Count Registers										
B0h	FAULTL	FAULT(8:1)								00000000
B1h	FAULTH	RESERVED				FAULT(12:9)				00000000
B2h	CERRCNT	CERRCNT[7:0]								00000000
IC Identification Register (Read only)										
FFh	ICID	ICID[7:0]								10010100

In the following sub-sections, the descriptions of different registers in Table 12 are provided.

Table 13. Register Summary

ADDRESS	ACRONYM	REGISTER NAME	SECTION
70h	SLEWRATE	Slew Rate Register	Section 7.5.2
71h	OVLMT	Overvoltage Limit Register	Section 7.5.3
72h	PARLED	Parallel LED String Register	Section 7.5.4
73h–78h	DEFWIDTH	Default Pulse Width Register	Section 7.5.5
80h	SYSCFG	System Configuration Register	Section 7.5.6
81h	CMWTAP	Communications Watchdog Timer Tap Point Register	Section 7.5.7
82h	PWMTICK	PWM Tick Period Register	Section 7.5.8

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Table 13. Register Summary (continued)

ADDRESS	ACRONYM	REGISTER NAME	SECTION
83h	ADCID	ADC ID Register	Section 7.5.9
84h	SOFTSYNC	Software Synchronization Register	Section 7.5.10
85h	XTALDS	XTAL Inverter Drive Strength Register	Section 7.5.11
90h	I2CTICK	I^2C Tick Period Register	Section 7.5.12
91h	I2CSLA	I^2C Slave Address	Section 7.5.13
92h	I2CRLEN	I^2C Read Length Register	Section 7.5.14
93h	I2CADR	I^2C Byte Address Register	Section 7.5.15
94h	I2CCTRL	I^2C Master Control Register	Section 7.5.16
95h	I2CDAT	I^2C Data Register	Section 7.5.17
96h	I2CSTAT	I^2C Status Register	Section 7.5.18
97h	I2CBMON	I^2C Bus Monitor Register (Read-Only)	Section 7.5.19
A0h–A1h	ADC	ADC Registers	Section 7.5.20
A2h	DSTR	DSTR Register	Section 7.5.21
B0h	FAULTL	Fault and CRC Error Count Registers	Section 7.5.22
B1h	FAULTH	Fault and CRC Error Count Registers	Section 7.5.22
B2h	CERRCNT	Fault and CRC Error Count Registers	Section 7.5.22
FFh	ICID	IC Identification Register	Section 7.5.23

8.5.1 Phase and Width Registers: LED Turnon and Turnoff Control

There are 10-bit registers to control the phase shift and pulse width of the LEDs with respect to the PWM counter (TCNT). These registers are the PHASE and WIDTH registers, respectively. These registers can be written or read using several register maps. These different maps all communicate with a common set of registers internally. The PHASE and WIDTH values are interpreted as follows:

- PHASE[9:0] defines the count value within the 10-bit TCNT period when a start pulse is sent to a given PWM generator.
- WIDTH[9:0] defines the count value within the 10-bit PWM generator when the LED pulse ends.

0% Duty Cycle: If WIDTH = 0 for a given LED, that LED turns off and remains off (no PWM).

100% Duty Cycle: If WIDTH = 1023 for a given LED, that LED turns on and remains on (no PWM).

The PHASE and WIDTH register addresses are from 0x00 to 0x6F.

8.5.1.1 Most Significant Bit-Packed PHASE/WIDTH Registers (MPHASE and MWIDTH)–Read/Write

The most significant bit-packed PHASE/WIDTH register map groups the most significant bits (MSBs) of the PHASE and WIDTH values together. This makes a total of 16 PHASE registers (at addresses 00h through 0Fh) and 16 WIDTH registers (at addresses 10h through 1Fh). It allows writing the least significant bits of PHASE/WIDTH through a 3-byte write command for subtle light contrast without touching the major brightness change per any LED sub-string.

8.5.1.2 Least Significant Bit-Packed PHASE/WIDTH Registers (LPHASE and LWIDTH) –Read/Write

The least significant bit-packed register map groups the two least significant bits (LSBs) of each LED PHASE and WIDTH value together and places these packed registers at the end of each register set. This makes a total of 16 PHASE registers (at addresses 20h through 2Fh) and 16 WIDTH registers (at addresses 30h through 3Fh). It allows writing the most significant bits of PHASE/WIDTH through a 12-byte write command for major brightness change without considering subtle light contrast for all LEDs.

8.5.1.3 Discrete LED PHASE/WIDTH Registers (DPHASE and DWIDTH) –Read/Write

The discrete LED PHASE/WIDTH register map allows all twenty bits (10 bits per PHASE and 10 bits per WIDTH) to be written or read for a single LED in one four-byte write/read transaction without a need to write or read other LED settings. This makes a total of 48 registers (at addresses 40h through 6Fh).

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8.5.2 Slew Rate Register – Read/Write

The LED bypass switch gate drivers have variable slew rates, one setting per LED sub-string. This is controlled by the SLEWRATE register. There are two bits for each LED sub-string.

Table 14. Slew Rate Registers

ADDR	REGISTER	D7	D6	D5	D4	D3	D2	D1	D0	DEFAULT
70h	SLEWRATE	SLEW_12_10[1:0]		SLEW_09_07[1:0]		SLEW_06_04[1:0]		SLEW_03_01[1:0]		11111111

- SLEW_12_10[1:0]: Slew rate setting for LEDs 10–12
- SLEW_09_07[1:0]: Slew rate setting for LEDs 7–9
- SLEW_06_04[1:0]: Slew rate setting for LEDs 4–6
- SLEW_03_01[1:0]: Slew rate setting for LEDs 1–3

Table 15 shows the mapping of the SLEWRATE bits to the actual slew rate setting.

Table 15. Slew Rate Mapping

SLEW REGISTER BITS	SLEW RATE
0	Slow rate
1	Medium rate
2	Fast rate
3	Fast rate (default)

8.5.3 Overvoltage Limit Register – Read/Write

The LED bypass switches have configurable overvoltage limits, one setting per LED sub-string. This is controlled by the OVLMT register. There are two bits for each LED sub-string.

Table 16. Overvoltage Limit Registers

ADDR	REGISTER	D7	D6	D5	D4	D3	D2	D1	D0	DEFAULT
71h	OVLMT	OVLMT_12_10[1:0]		OVLMT_09_07[1:0]		OVLMT_06_04[1:0]		OVLMT_03_01[1:0]		10101010

- OVLMT_12_10[1:0]: Overvoltage limit for LEDs 10–12
- OVLMT_09_07[1:0]: Overvoltage limit for LEDs 7–9
- OVLMT_06_04[1:0]: Overvoltage limit for LEDs 4–6
- OVLMT_03_01[1:0]: Overvoltage limit for LEDs 1–3

The two-bit overvoltage limits are mapped to three levels of low, medium, and high threshold voltage. Upon power up, the overvoltage limit for each string is set to the high threshold value (V_{TH-O3}).

Table 17. Overvoltage Limit Mapping

OVLMT REGISTER BITS	OVERVOLTAGE LIMIT
0	V_{TH-O1}
1	V_{TH-O2}
2	V_{TH-O3} (default)
3	V_{TH-O3}

8.5.4 Parallel LED String Register – Read/Write

The combinations of LED sub-strings that can be paralleled are as follows:

1. None (each string of bypass switches is independent from the others)
2. One pair (sub-strings of S1 and S2 are connected in parallel, and S3 and S4 are independent)
3. Two pairs (sub-strings of S1 and S2 are connected in parallel, and S3 and S4 are connected in parallel)
4. All (sub-strings of S1, S2, S3, and S4 are all connected in parallel)

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Table 18. Parallel LED Register

ADDR	REGISTER	D7	D6	D5	D4	D3	D2	D1	D0	DEFAULT
72h	PARLED	RESERVED						PARLED[1:0]		00000000

PARLED[1:0] controls the settings for the PWM phase and width of each LED. In addition, this two-bit manages the open LED protection and fault detection for each LED. As shown in Table 19, there are four possible configurations where the used PHASE, WIDTH, SLEWRATE, OVLMT, and DEFWIDTH register bits for each LED and sub-strings are specified.

Table 19. PARLED Register Mapping

PARLED[1:0]	LED SUB-STRINGS IN PARALLEL (||)	USED REGISTERS FOR PHASE/WIDTH/SLEWRATE/OVLMT/DEFWIDTH			
		S1	S2	S3	S4
0	None	LED1-3	LED4-6	LED7-9	LED10-12
1	S1 || S2	LED1-3	LED1-3	LED7-9	LED10-12
2	S1 || S2, S3 || S4	LED1-3	LED1-3	LED7-9	LED7-9
3	S1 || S2 || S3 || S4	LED1-3	LED1-3	LED1-3	LED1-3

8.5.5 Default Pulse Width Registers (DEFWIDTH) – Read/Write

The default pulse width registers are used for PWM dimming of each LED when communications watchdog timeout condition occurs.

Table 20. Default Pulse Width Registers

ADDR	REGISTER	D7	D6	D5	D4	D3	D2	D1	D0	DEFAULT
73h	DEFWIDTH02_01	DEFWIDTH02[3:0]				DEFWIDTH01[3:0]				00000000
74h	DEFWIDTH04_03	DEFWIDTH04[3:0]				DEFWIDTH03[3:0]				00000000
75h	DEFWIDTH06_05	DEFWIDTH06[3:0]				DEFWIDTH05[3:0]				00000000
76h	DEFWIDTH08_07	DEFWIDTH08[3:0]				DEFWIDTH07[3:0]				00000000
77h	DEFWIDTH10_09	DEFWIDTH10[3:0]				DEFWIDTH09[3:0]				00000000
78h	DEFWIDTH12_11	DEFWIDTH12[3:0]				DEFWIDTH11[3:0]				00000000

DEFWIDTHn[3:0]: Default pulse width. This 4-bit value determines the pulse width for LEDn when the communications watchdog timer times out. Note that a value of “0000” means fully OFF, and a value of “1111” means fully ON. In default pulse width operating mode, the values in PHASE registers determine the phase shift. The pulse widths are provided in Table 21. This means in order for the default pulse width mode to operate correctly, both the PHASE and the DEFWIDTH registers must be programmed.

Table 21. DEFWIDTH-to-LED Pulse Width Mapping

DEFWIDTHn[3:0]	LEDn PULSE WIDTH
0	0 (LED is fully off)
1	64
2	128
3	192
4	256
5	320
6	384
7	448
8	512
9	576
10	640
11	704
12	768
13	832

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Table 21. DEFWIDTH-to-LED Pulse Width Mapping (continued)

DEFWIDTHn[3:0]	LEDn PULSE WIDTH
14	896
15	1023 (LED is fully on)

Note that the DEFWIDTH-to-LED pulse width mapping is modified by the PARLED register setting in the watchdog timeout condition.

Example 1:
PARLED[1:0] = 1
PHASE01 and DEFWIDTH01 control LED1 and LED4
PHASE02 and DEFWIDTH02 control LED2 and LED5
PHASE03 and DEFWIDTH03 control LED3 and LED6
PHASE07 and DEFWIDTH07 control LED7
PHASE08 and DEFWIDTH08 control LED8
PHASE09 and DEFWIDTH09 control LED9
PHASE10 and DEFWIDTH10 control LED10
PHASE11 and DEFWIDTH11 control LED11
PHASE12 and DEFWIDTH12 control LED12

Example 2:
PARLED[1:0] = 2
PHASE01 and DEFWIDTH01 control LED1 and LED4
PHASE02 and DEFWIDTH02 control LED2 and LED5
PHASE03 and DEFWIDTH03 control LED3 and LED6
PHASE07 and DEFWIDTH07 control LED7 and LED10
PHASE08 and DEFWIDTH08 control LED8 and LED11
PHASE09 and DEFWIDTH09 control LED9 and LED12

Example 2:
PARLED[1:0] = 3
PHASE01 and DEFWIDTH01 control LED1, LED4, LED7, and LED10
PHASE02 and DEFWIDTH02 control LED2, LED5, LED8, and LED11
PHASE03 and DEFWIDTH03 control LED3, LED6, LED9, and LED12

8.5.6 System Configuration Register (SYSCFG) – Read/Write

The control register SYSCFG allows control of several functions.

Table 22. System Configuration Register

ADDR	REGISTER	D7	D6	D5	D4	D3	D2	D1	D0	DEFAULT
80h	**SYSCFG**	SEPTR	I2CEN	ACKEN	PSON	CMWEN	SYNCOEN	SYNCPEN	PWR	00000000

SEPTR: Separate TX/RX
- 0 : Separate TX/RX disabled (signals on TX appear on RX). Used in a CAN transceiver configuration.
- 1 : Separate TX/RX enabled (signals on TX do not appear on RX). Used in a direct UART connection.

I2CEN: I^2C Enable
- 0 : I^2C function disabled
- 1 : I^2C function enabled

ACKEN: Acknowledge Enable
- 0 : No acknowledge is transmitted following successfully received writes.
- 1 : Acknowledge (0x7F) is transmitted following successfully received writes.

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PSON: Phase Shift LED ON. See the *PWM Dimming* section.

- 0 : LED phase shifts apply to LED turnoff times.
- 1 : LED phase shifts apply to LED turnon times.

CMWEN: Communications Watchdog Timer Enable

- 0 : Communications watchdog timer disabled
- 1 : Communications watchdog timer enabled

SYNCOEN: SYNC Output Enable

- 0 : SYNC output disabled
- 1 : SYNC output enabled

SYNCPEN: SYNC Pulse Enable

- 0 : If SYNCOEN = 1, the SYNC output is driven continuously low (no pulses).
- 1 : If SYNCOEN = 1, the SYNC output is driven with a 50% duty cycle pulse to synchronize other connected TPS92662A devices.

PWR: This bit is reset to 0 upon power-up. It can be written to a 1 by the MCU. Reading this bit allows the MCU to detect when there has been a power cycle.

- 0 : A power cycle has occurred since last write to a '1'.
- 1 : No power cycle has occurred since the last write to a '1'.

8.5.7 Communications Watchdog Timer Tap Point Register (CMWTAP) – Read/Write

Table 23 shows the communications watchdog register setting.

Table 23. Communications Watchdog Timer Tap Point Register

ADDR	REGISTER	D7	D6	D5	D4	D3	D2	D1	D0	DEFAULT
81h	**CMWTAP**	RESERVED					CMWTAP[2:0]			00000101

CMWTAP[2:0]: This 3-bit value selects the tap point (that is the bit number, starting from 0) on the 24-bit communications watchdog timer to establish the timeout condition. Table 24 shows the mapping of these three bits to the tap point.

Table 24. CMWTAP-to-Tap Point Mapping

CMWTAP [2:0]	TAP POINT (BIT NUMBER)
0	16
1	17
2	18
3	19
4	20
5 (default)	21
6	22
7	23

8.5.8 PWM Tick Period Register (PWMTICK) – Read/Write

This register determines the divisor value applied to the input CLK, which generates the PWM clock cycle.

Table 25. PWM Tick Period Register

ADDR	REGISTER	D7	D6	D5	D4	D3	D2	D1	D0	DEFAULT
82h	**PWMTICK**	PTBASE[1:0]		PTCNT[5:0]						00001100

PWMTICK register is comprised of two fields: PTBASE and PTCNT. These two fields program the two divider blocks in series, which are applied to the input CLK and generate the PWM clock.

Table 26 shows the mapped program value from PTBASE[1:0] to the first division on the input CLK.

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Table 26. PTBASE Mapping to the First Division Factor (DIV1)

PTBASE[1:0]	DIVISION FACTOR DIV1
0 (default)	÷1
1	÷50
2	÷125
3	÷200

Table 27 shows the mapped program value from PTCNT[5:0] to the second division on the input CLK.

Table 27. PTCNT Mapping to the Second Division Factor (DIV2)

PTCNT [5:0]	DIVISION FACTOR DIV2	PTCNT [5:0]	DIVISION FACTOR DIV2	PTCNT [5:0]	DIVISION FACTOR DIV2	PTCNT [5:0]	DIVISION FACTOR DIV2
0	÷2	16	÷20	32	÷36	48	÷53
1	÷3	17	÷21	33	÷37	49	÷54
2	÷4	18	÷22	34	÷38	50	÷55
3	÷6	19	÷23	35	÷39	51	÷56
4	÷8	20	÷24	36	÷40	52	÷57
5	÷9	21	÷25	37	÷41	53	÷58
6	÷10	22	÷26	38	÷42	54	÷59
7	÷11	23	÷27	39	÷43	55	÷60
8	÷12	24	÷28	40	÷44	56	÷62
9	÷13	25	÷29	41	÷45	57	÷63
10	÷14	26	÷30	42	÷46	58	÷65
11	÷15	27	÷31	43	÷47	59	÷68
12 (default)	÷16	28	÷32	44	÷49	60	÷71
13	÷17	29	÷33	45	÷50	61	÷74
14	÷18	30	÷34	46	÷51	62	÷78
15	÷19	31	÷35	47	÷52	63	÷85

Example 1: Default mode, PWMTICK[7:0] = '00001100'

- PTBASE[1:0] = 0 → DIV1 = 1
- PTCNT[5:0] = 12 → DIV2 = 16
- PWM_{CLK} = CLK ÷ DIV1 ÷ DIV2
- CLK = 8 MHz → PWM_{CLK} = 500 kHz

Example 2: PWMTICK[7:0] = '11101111'

- PTBASE[1:0] = 3 → DIV1 = 200
- PTCNT[5:0] = 47 → DIV2 = 52
- PWM_{CLK} = CLK ÷ DIV1 ÷ DIV2
- CLK = 16 MHz → PWM_{CLK} = 1.54 kHz

8.5.9 ADC ID Register (ADCID) – Read/Write

Table 28. ADC ID Register

ADDR	REGISTER	D7	D6	D5	D4	D3	D2	D1	D0	DEFAULT
83h	**ADCID**	RESERVED						ADCID[1:0]		00000000

ADCID[1:0]: This two-bit value determines whether to use the ADDR0 analog input as part of the device ID. This register is intended to be written with a single-byte broadcast write. Writing this register to a value of 1 or 2 causes the logic to perform ADC conversions on ADDR0 until four consecutive conversions occurs with the same value. This is described in more detail as follows:

- ADCID[1:0] = 0 or 3: ADDR0 is used as a single-bit binary input value for DEVID mapping (not as an ADC-converted value).

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- ADCID[1:0] = 1: Sample the ADDR0 input as a two-bit binary value coming from the ADC. When the two MSBs of the ADC converted value are the same for four consecutive conversions, the converted value is latched and this value and ADDR2 and ADDR1 inputs are used to generate a four-bit device ID.
- ADCID[1:0] = 2: Sample the ADDR0 input as a three-bit binary value from the ADC. When the three MSBs of the ADC converted value are the same for four consecutive conversions, the converted value is latched and this value and ADDR2 and ADDR1 inputs are used to generate a five-bit device ID.

NOTE

If a write frame is multi-byte (more than one byte) or single-device (that is not a broadcast write), and the address range encompasses the ADCID register address, the entire write frame is discarded and no registers are written. This is considered an illegal frame, so no write acknowledge is transmitted.

8.5.10 Software Synchronization Register (SOFTSYNC) – Write-Only

Table 29. SOFTSYNC Register

ADDR	REGISTER	D7	D6	D5	D4	D3	D2	D1	D0	DEFAULT
84h	SOFTSYNC	RESERVED							SSYNC	00000000

SSYNC: Set this bit to ‘1’ to re-initialize the internal PWM counter (TCNT) to zero. This bit always reads back as a 0. A broadcast write to this register can be used to synchronize all TPS92662A devices on the same UART bus, such that programmed phase shifts are all referenced to a common starting time. Writing this bit to 0 has no effect.

8.5.11 XTAL Inverter Drive Strength Register (XTALDS) – Read/Write

Table 30. XTALDS Register

ADDR	REGISTER	D7	D6	D5	D4	D3	D2	D1	D0	DEFAULT
85h	XTALDS	RESERVED					XTALDS[2:0]			00000000

XTALDS[2:0]: This 3-bit value selects the XTALO drive strength. The default value of 0 is the maximum drive strength. Table 31 shows the drive strength reduction mapping.

Table 31. XTALDS-to-Drive Strength Mapping

XTALDS [2:0]	XTALO DRIVE STRENGTH
0 (default)	100%
1	91.0%
2	82.1%
3	73.1%
4	28.4%
5	19.4%
6	10.5%
7 (1)	NA

(1) This setting is reserved for test use. It is not recommended to use in application.

8.5.12 I²C Tick Period Register (I2CTICK) – Read/Write

Table 32. I²C Tick Period Register

ADDR	REGISTER	D7	D6	D5	D4	D3	D2	D1	D0	DEFAULT
90h	I2CTICK	RESERVED					I2CTICK[2:0]			00000101

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I2CTICK[2:0]: This 3-bit value sets the I²C period. The I²C tick counter generates a clock pulse on the SCL output. Table 33 shows the I2CTICK to SCL divider mapping and provides examples of SCL frequencies. Note that the values in dark gray exceed the maximum allowed frequency for I²C fast mode and must not be used.

Table 33. I2CTICK-to-SCL Clock Divider Mapping

I2CTICK[2:0]	CLK-TO-SCL CLOCK DIVIDER	f_{SCL} (kHz)		
		f_{sys_clk} = 6 MHz	f_{sys_clk} = 8 MHz	f_{sys_clk} = 16 MHz
0	20	300	400	800
1	24	250	333	667
2	28	214	286	571
3	32	188	250	500
4	36	167	222	444
5 (default)	40	150	200	400
6	80	75	100	200
7	160	37.5	50	100

8.5.13 I²C Slave Address (I2CSLA) – Read/Write

This register consists of seven address bits (A6-A0) and a dummy bit in the LSB position. Note that the seven-bit slave address must be left-shifted by one bit before being written.

Table 34. I²C Slave Address Register

ADDR	REGISTER	D7	D6	D5	D4	D3	D2	D1	D0	DEFAULT
91h	I2CSLA	SA [6:0]							RESERVED	00000000

SA[6:0]: I²C Slave Address

- This field specifies bits A6 through A0 of the slave address.

8.5.14 I²C Read Length (I2CRLEN) – Read/Write

This register contains the programmed length of bytes that are transferred during a read request.

Table 35. I²C Read Length Register

ADDR	REGISTER	D7	D6	D5	D4	D3	D2	D1	D0	DEFAULT
92h	I2CRLEN	I2CRLEN[7:0]								00000000

8.5.15 I²C Byte Address (I2CADR) – Read/Write

This register contains the byte address that is transferred following the slave address during an I²C read or write transaction (for example, in the case of a connected EEPROM device or the starting address in memory). This byte is skipped if the NOADR bit is set in the I2CCTRL register for a given transaction.

Table 36. I²C Byte Address Register

ADDR	REGISTER	D7	D6	D5	D4	D3	D2	D1	D0	DEFAULT
93h	I2CADR	I2CADR[7:0]								00000000

8.5.16 I²C Control (I2CCTRL) – Write-Only

This write-only register sets the I²C control bits. The control register configures and launches I²C operation.

Table 37. I²C Master Control Register

ADDR	REGISTER	D7	D6	D5	D4	D3	D2	D1	D0	DEFAULT
94h	I2CCTRL	RESERVED				NOADR	START	RD	RST	00000000

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NOADR: No Byte Address
- 0 = Byte address is transferred.
- 1 = Byte address is skipped.

START: Start I^2C operation
- 0 = No transaction is started.
- 1 = The transaction type defined by the RD bit is started.

RD: Read
- 0 = The I^2C transaction is a write.
- 1 = The I^2C transaction is a read.

RST: Reset
- 0 = The I^2C state machine is not reset and the receive FIFO is not flushed.
- 1 = The I^2C state machine is reset and the receive FIFO is flushed.

Note that RST has priority over START. If both bits are written to 1, the I^2C state machine is reset and the receive FIFO is flushed. Writing RST to 1 does not reset the writeable I^2C registers (for example, I2CTICK, I2CSLA, I2CRLEN, and I2CADR).

8.5.17 I^2C Data (I2CDAT) – Read/Write

This register contains the data to be transmitted when in the master transmit state and the data received when in the master receive state. Multi-byte writes or reads that reach this address location are not auto-incremented to the next address. This allows all of the data bytes present in the receive FIFO to be read out in a single UART read command and response transaction.

Table 38. I^2C Data Register

ADDR	REGISTER	D7	D6	D5	D4	D3	D2	D1	D0	DEFAULT
95h	**I2CDAT**	I2CDAT[7:0]								00000000

DATA[7:0]: This byte contains the data transferred during a transaction.

8.5.18 I^2C Status (I2CSTAT) – Read-Only

This read-only register indicates the state of the I^2C master. The bits indicate the receive FIFO status. If any errors are detected during operation, a general BUSY bit indicates the I^2C controller is currently active and is still processing a transaction.

The RFE, RFF, and BUSY bits are “live” status bits and are set and cleared anytime the associated conditions change. The following bits are only valid when the BUSY bit is no longer active. These bits are only cleared upon reading the I2CSTAT register:
- BUSYERR
- SLANACK
- ADRNACK
- DATNACK
- SDALOW

Table 39. I^2C Status Register

ADDR	REGISTER	D7	D6	D5	D4	D3	D2	D1	D0	DEFAULT
96h	**I2CSTAT**	RFE	RFF	BUSYERR	SLANACK	ADRNACK	DATNACK	SDALOW	BUSY	10000000

RFE: Receive FIFO Empty
- 0 = The receive FIFO is not empty.
- 1 = The receive FIFO is empty.

RFF: Receive FIFO Full
- 0 = The receive FIFO is not full.
- 1 = The receive FIFO is full.

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BUSYERR: Busy Error
- 0 = The I²C controller was not interrupted with a new command during an ongoing transaction.
- 1 = The I²C controller was interrupted with a new command during an ongoing transaction.

SLANACK: Slave Address Not Acknowledged Error
- 0 = The slave address was acknowledged.
- 1 = The slave address was not acknowledged.

ADRNACK: Byte Address Not Acknowledged Error
- 0 = The byte address was acknowledged.
- 1 = The byte address was not acknowledged.

DATNACK: Write Data Not Acknowledged Error
- 0 = The write data was acknowledged.
- 1 = The write data was not acknowledged.

SDALOW: SDA Stuck Low Error
- 0 = The SDA input is not being held at a low state.
- 1 = The SDA input is being held at a low state.

BUSY: I²C Controller Busy
- 0 = The I²C controller is idle and ready to receive new commands via the I2CCTRL START bit.
- 1 = The I²C controller is busy and does not perform new commands issued via the I2CCTRL START bit.

8.5.19 I²C Bus Monitor (I2CBMON) – Read-Only

This register is used to determine the SCL and SDA signal status.

Table 40. I²C Bus Monitor Register (Read-Only)

ADDR	REGISTER	D7	D6	D5	D4	D3	D2	D1	D0	DEFAULT
97h	I2CBMON	RESERVED						SDA	SCL	00000000

SDA: I²C SDA Status
- 0 = The I²C SDA signal is low.
- 1 = The I²C SDA signal is high.

SCL: I²C SCL Status
- 0 = The I²C SCL signal is low.
- 1 = The I²C SCL signal is high.

8.5.20 ADC Registers – Read-Only

The ADC registers hold the results of ADC conversions.

Table 41. ADC Registers

ADDR	REGISTER	D7	D6	D5	D4	D3	D2	D1	D0	DEFAULT
A0h	ADC1	ADC1[7:0]								00000000
A1h	ADC2	ADC2[7:0]								00000000

- ADC1[7:0]: Most recent ADC1 conversion result
- ADC2[7:0]: Most recent ADC2 conversion result

8.5.21 Drive Strength Register – Read-Only

The drive strength register holds the results of ADC conversion on the DSTR pin which occurs during power up.

Table 42. DSTR Register

ADDR	REGISTER	D7	D6	D5	D4	D3	D2	D1	D0	DEFAULT
A2h	DSTR	RESERVED					DSTR[2:0]			00000000

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Table 43. DSTR Pin Voltage-to-Drive Strength Mapping

DSTR PIN VOLTAGE	DSTR [2:0]	CLK INVERTER	TX / SYNC INVERTER
0 to 1/8 VDD	000 (default)	High Z	100%
1/8 to 3/8 VDD	001 & 010	20%	20%
3/8 to 5/8 VDD	011 & 100	40%	40%
5/8 to 7/8 VDD	101 & 110	85%	85%
7/8 VDD to VDD	111	100%	100%

8.5.22 Fault and CRC Error Count Registers – Read/Write

The diagnostic registers hold the results of various faults and status flags for the system. The diagnostic registers exist in the address range B0h to B2h.

Table 44. Fault and CRC Error Count Registers

ADDR	REGISTER	D7	D6	D5	D4	D3	D2	D1	D0	DEFAULT
B0h	**FAULTL**	FAULT(8:1)								00000000
B1h	**FAULTH**	RESERVED				FAULT(12:9)				00000000
B2h	**CERRCNT**	CERRCNT[7:0]								00000000

FAULT(12:1): Fault Register

- FAULT(x) = 0 : An LED fault has not occurred on LEDx
- FAULT(x) = 1 : An LED fault has occurred on LEDx
- The LED open and short fault detection circuitry is sampled just before the corresponding bypass switch is closed. If a fault exists at this sample time, a 1 is latched into the associated FAULT register bit. The FAULT register bits must be cleared manually by writing them back to 0. If an LED fault condition still exists at the next PWM period, the corresponding FAULT register bit is immediately set back to a 1. Writing the FAULT register bits to 1 has no effect.

CERRCNT[7:0]: CRC Error Count Register

- This register value is incremented each time a CRC error is received. This register can be read by the MCU and then written back to 0 to clear the count. The CERRCNT value saturates at FFh; it does not wrap back to 0 when it reaches FFh. The CERRCNT register is not automatically cleared when a communications reset is received. It must be cleared by writing it back to 0. Note that the CERRCNT register can be written to any 8-bit value. This is intended for diagnostic purposes.

8.5.23 IC Identification Register – Read-Only

The ICID register exists at address FFh and is read-only.

Table 45. IC Identification Register (Read-Only)

ADDR	REGISTER	D7	D6	D5	D4	D3	D2	D1	D0	DEFAULT
FFh	**ICID**	ICID[7:0]								10010100

ICID[7:0]: Reads to this register address return the value 0x94. This is intended to be used to identify a TPS92662A device. Writes to this register have no effect.

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9 Application and Implementation

NOTE

Information in the following applications sections is not part of the TI component specification, and TI does not warrant its accuracy or completeness. TI's customers are responsible for determining suitability of components for their purposes. Customers should validate and test their design implementation to confirm system functionality.

9.1 Application Information

The TPS92662A-Q1 device is capable of shunting any combination of 12-series LEDs at high frequency and at variable duty cycles. The device includes four sets of 3×series-connected integrated switches for bypassing individual LEDs. The individual sub-strings make the TPS92662A-Q1 device capable of handling single or multiple current sources, as well as paralleling up to four switches for bypassing high current LEDs. Figure 33 illustrates a dynamic headlight application using multiple TPS92662A-Q1 devices on multiple boards. The electronic control unit (ECU), usually attached to the outside of the headlight, contains the master MCU and a boost pre-regulator stage that takes the variable battery voltage and steps it up to a stable DC voltage rail. The application also includes multiple channels of buck current regulators to provide a stable current for each series string of high brightness LEDs. For the best performance, the TPS92662A-Q1 device must reside on the LED load board where it is as close as possible to the LEDs to which it directly connects.

Figure 33 shows an example of an application implementation where TPS92662A-Q1 devices are implemented in different configurations:

- A standard single string of LEDs
- Parallel switches for high current LEDs
- Multiple LEDs per switch
- Independent sub-strings of LEDs

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Application Information (continued)

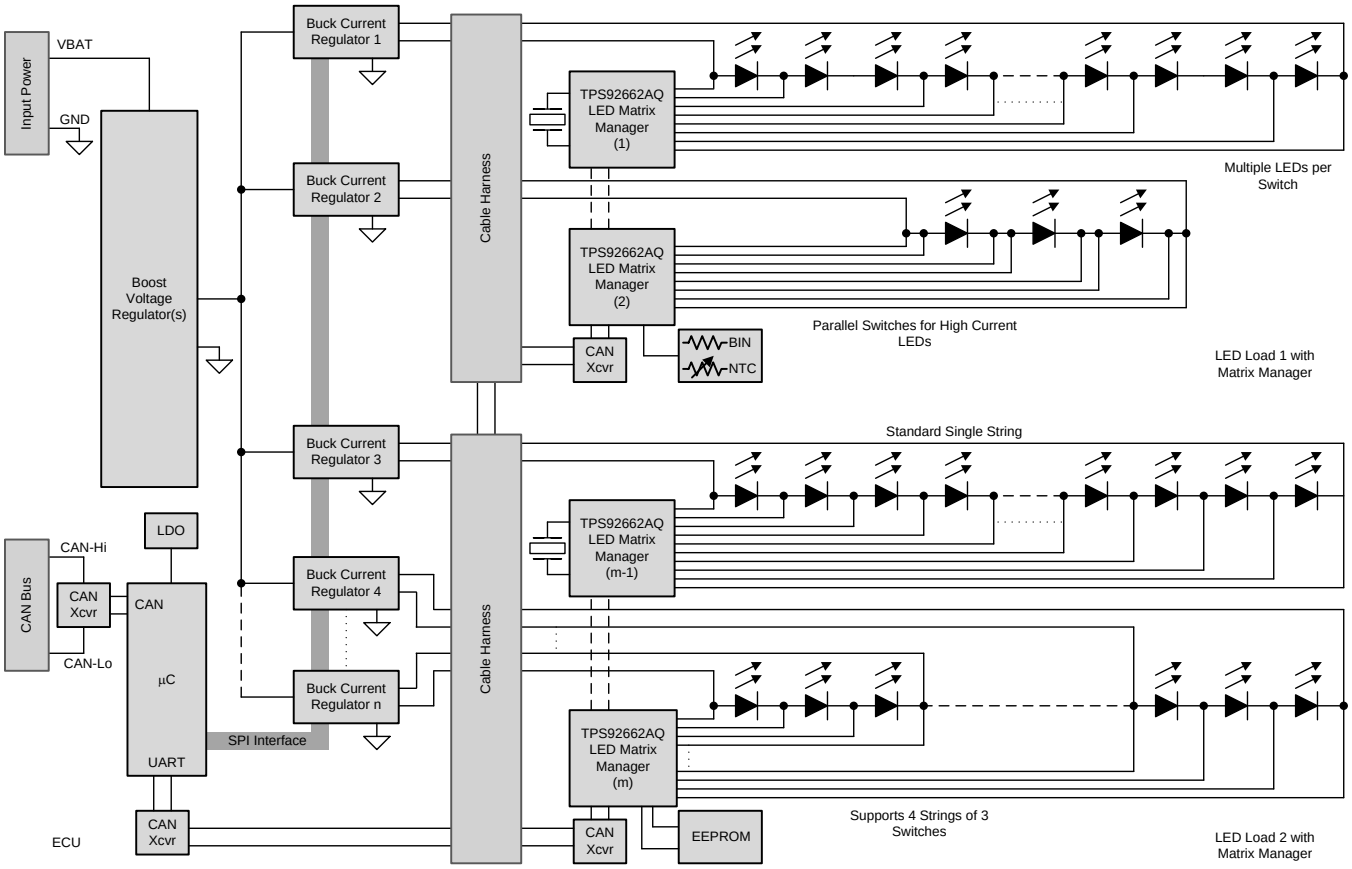


Figure 33. Multi-Board Matrix Manager Application

9.1.1 Guidelines For Current Source

The TPS92662A-Q1 device is developed considering a high-side sensing hysteretic buck current source. This type of high bandwidth current source is recommended to power the LED strings. The recommended guidelines for the current source are as follows:

- Operate the current source switching converter in CCM (continuous conduction mode).
- It is recommended to operate at a switching frequency between 250 times the PWM frequency and 2000 times the PWM frequency.
- Minimize the capacitance at the output of the current source between the top of the string (sub-string) and the GND to avoid excessive overshoot current through the un-shunted LEDs when PWM dimming is applied.
- Minimize capacitance across each LED to avoid excessive overshoot current through the TPS92662A-Q1 shunt switch when PWM dimming is applied.
- Follow the *Device and Documentation Support*.

9.1.2 Address Settings for Multiple Devices on the Bus

The connections for ADDR2/1/0 for a different number of devices on the network are explained in the following sections:

9.1.2.1 Up to Eight TPS92662A-Q1 Devices on the Bus

When only up to eight devices are on the same network, the default dev_id mapping can be used where the mapping of ADDR2/1/0 to dev_id is as follows:

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Application Information (continued)

dev_id[4]	dev_id[3]	dev_id[2]	dev_id[1]	dev_id[0]
0	0	ADDR2	ADDR1	ADDR0

To set the bits of dev_id[2:0] to '1' or '0', the ADDR2, ADDR1, and ADDR0 pins are connected to VDD or GND.

dev_id		ADDR2	ADDR1	ADDR0
DEC	HEX			
0	0x00	GND	GND	GND
1	0x01	GND	GND	VDD
2	0x02	GND	VDD	GND
3	0x03	GND	VDD	VDD
4	0x04	VDD	GND	GND
5	0x05	VDD	GND	VDD
6	0x06	VDD	VDD	GND
7	0x07	VDD	VDD	VDD

9.1.2.2 Up to 16 TPS92662A-Q1 Devices on the Bus

When more than eight and less than 16 TPS92662A-Q1 devices are installed on the network, the ADDR0 voltage is used to define the LS Bits of the DEVID byte. The mapping of the ADDR0 voltage to the DEVID is performed once, following a broadcast write to the ADCID register. See the *DEVID Byte* section for more details.

To set the voltage on ADDR0, a simple resistor divider between VDD and GND pins is recommended. The mapping of the dev_id is as follows:

dev_id[4]	dev_id[3]	dev_id[2]	dev_id[1]	dev_id[0]
0	ADDR2	ADDR1	ADC_OUT[7]	ADC_OUT[6]

Connect ADDR2 and ADDR1 to VDD or GND to set the dev_id[3,2] to '1' or '0'. The two LSBs of the dev_id (dev_id[1,0]) are the two MS-bits of the 8-bit AD-conversion of ADDR0. The circuit in Figure 34 can be used to set the voltage on ADDR0.

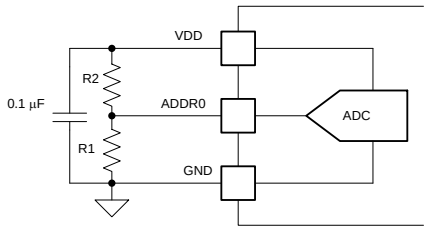


Figure 34. ADDR0 Setting for More than Eight TPS92662A-Q1 on the Bus

To achieve the four binary levels on the 2-MSBs of the 8-bit ADC, the resistor values in Table 46 can be used for R1 and R2 (1%).

Table 46. Resistor Divider Values on ADDR0 for Up to 16 TPS92662A-Q1 on the Bus

ADC_OUT[7,6]	V_{ADDR0} (V)	R1 (kΩ)	R2 (kΩ)
00	GND	0	DNP
01	0.25×VDD	10	30.1
10	0.50×VDD	10	10
11	VDD	DNP	0

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9.1.2.3 Up to 31 TPS92662A-Q1 Devices on the Bus

When up to 31 TPS92662A-Q1 devices are installed on the network, the mapping of the dev_id is as follows:

dev_id[4]	dev_id[3]	dev_id[2]	dev_id[1]	dev_id[0]
ADDR2	ADDR1	ADC_OUT[7]	ADC_OUT[6]	ADC_OUT[5]

Connect ADDR2 and ADDR1 to VDD or GND to set the dev_id[4,3] to '1' or '0'. The three LSBs of the dev_id (dev_id[2:0]) are the three MS-bits of the 8-bit AD-conversion of ADDR0. A similar voltage divider circuit shown in Figure 34 can be used to set the voltage on ADDR0.

The resistor values shown in Table 47 can be used for R1 and R2 (1%) to achieve the eight binary levels on the 3-MSBs of the 8-bit ADC.

Table 47. Resistor Divider Values on ADDR0 for Up to 31 TPS92662A-Q1 on the Bus

ADC_OUT[7:5]	VADDR0 (V)	R1 (kΩ)	R2 (kΩ)
000	GND	0	DNP
001	0.125×VDD	10	69.8
010	0.25×VDD	10	30.1
011	0.375×VDD	10	16.5
100	0.5×VDD	10	10
101	0.625×VDD	10	6.04
110	0.75×VDD	10	3.32
111	VDD	DNP	0

9.1.2.4 CLK, TX, and SYNC Drive Strength

The drive strength of the CLK, TX, and SYNC signals can be adjusted by the voltage applied to the DSTR pin. The voltage on the DSTR pin is sampled by the internal ADC. This occurs following a power-on reset event.

The voltage divider circuit in Figure 35 can be used to set the voltage on DSTR.

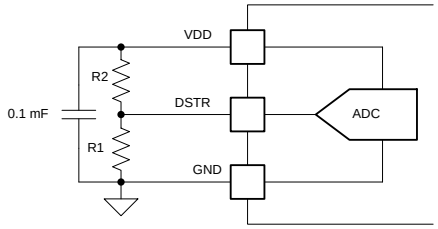


Figure 35. DSTR Setting for Adjusting CLK, TX, and SYNC Drive Strength

To achieve the four drive strength levels on CLK, TX, and SYNC pins and to disable the CLK output, the resistor values shown in Table 48 can be used for R1 and R2 (1%).

Table 48. Resistor Divider Values on DSTR for Adjusting CLK, TX, and SYNC Drive Strength

R1 (kΩ)	R2 (kΩ)	DSTR (V)	DSTR[2:0]	CLK Buffer	TX / SYNC BUFFER
0	DNP	GND	000	High Z	100%
10	30.1	1.25	001 / 010	20%	20%
10	10	2.5	011 / 100	40%	40%
10	3.32	3.75	101 / 110	80%	80%
DNP	0	5.0	111	100%	100%

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9.2 Typical Application

Figure 36 shows a typical application where two TPS92662A-Q1 devices, U1 and U2, are connected to the same communication network.

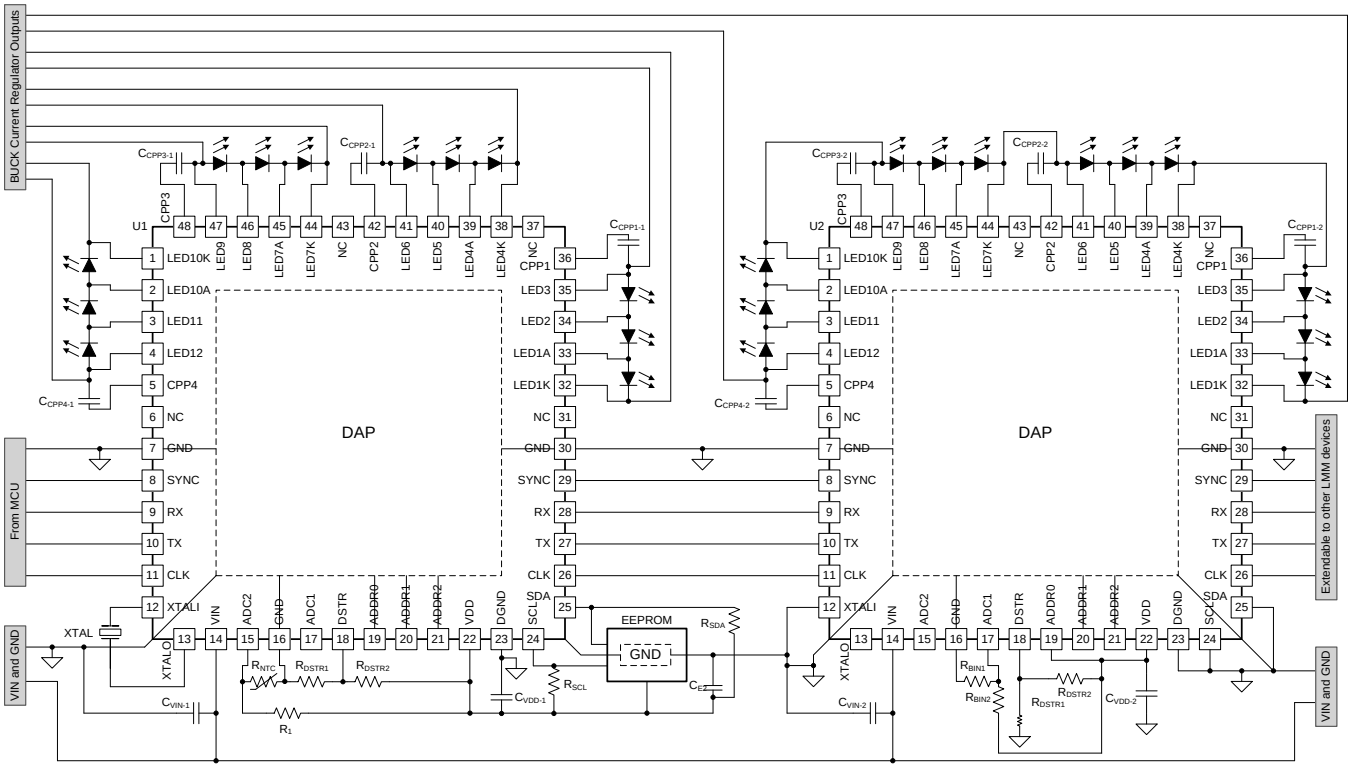


Figure 36. Application Example

9.2.1 Design Requirements

The two devices, U1 and U2, are set to the addresses 0x00 and 0x01, respectively. U1 is configured to support four independent LED sub-strings with $V_{LED-MAX}$ = 13.2 V. The second device is configured to support one 12-series connected LED string with $V_{LED-MAX}$ = 3.3 V. A board temperature measurement is needed to perform LED current foldback at high temperature.

9.2.2 Detailed Design Procedure

All power pins of the TPS92662A-Q1 are bypassed with 0.1-µF capacitors. See the *Power Supply Recommendations* section for more details.

9.2.2.1 Device Addressing

U1 address (dev_id [4:0]) = 0x00

- ADDR0, ADDR1, and ADDR2 are connected to GND.

U2 address (dev_id [4:0]) = 0x01

- ADDR1 and ADDR2 are connected to GND. ADDR0 is connected to VDD.

See the *Address Settings for Multiple Devices on the Bus* section for more details on device addressing.

9.2.2.2 System Clock and Synchronization

In this application example, the clock frequency for U1 and U2 is set to 8 MHz using a ceramic resonator connected to the crystal driver pins of U1, XTALI, and XTALO. The CLK pins of the U1 and U2 are then connected together and the XTALI pin of U2 is connected to GND.

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Typical Application (continued)

To synchronize the two devices, U1 is configured to generate the synchronization pulse by setting the SYNCOEN and SYNCPEN bits of the SYSCFG register on U1 to one. The SYNC pins of the U1 and U2 are then connected together to make them synchronized.

9.2.2.3 Over Voltage Setting

The overvoltage limit of the bypass switches are set according to the $V_{LED-MAX}$. In this application, the OVLMT in the *Overvoltage Limit Register* section is set to 0xAA (18 V typ) for U1 and to 0x00 (6 V typ) for U2.

9.2.2.4 Temperature Measurement

Board temperature measurement is performed using the internal 8-bit ADC of U1. The resistor R_1 and an NTC-thermistor, R_{NTC}, are connected as a voltage divider between the VDD, ADC2, and GND pins of U1. In this application, R_1 = 10 kΩ and R_{NTC} with 25°C characteristic resistance of 100 kΩ are selected. Figure 37 shows the temperature measurement result using the internal 8-bit ADC over the range of -40°C to 150°C.

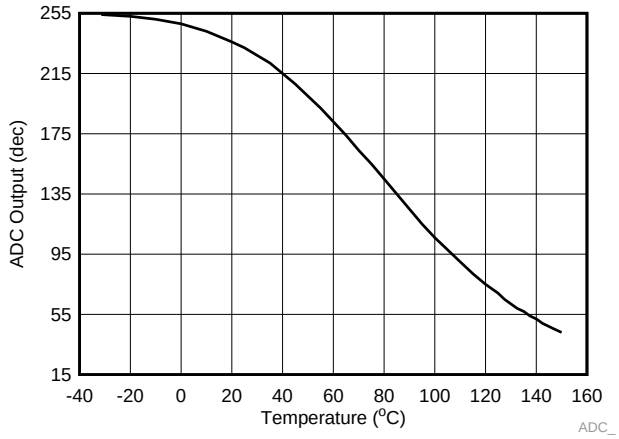


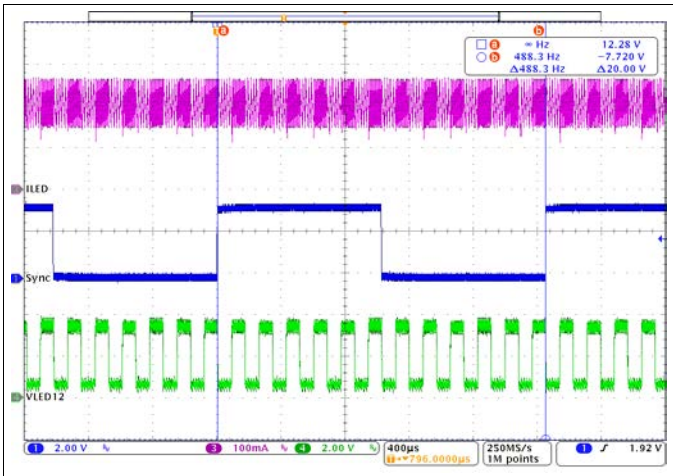
Figure 37. Board Temperature Measurement with 8-Bit ADC

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Typical Application (continued)

9.2.3 Application Curves

Figure 38 and Figure 39 show PWM dimming waveforms when 12 LEDs are connected in series (the device U2 in Figure 36). The WIDTHx is set to the value of 43. The PHASEx for each shunt switch is set such that the turnon (or turnoff) times of the LEDs are separated by 85 counts (out of 1023 count) in one PWM period. The rising edge of the SYNC signal shows the beginning of the PWM period (count 0) and the falling edge shows the end (count 512). Figure 38 shows the waveforms for PSON = 0 where the PHASEx sets the turnoff time of the LEDs. Figure 39 shows the waveforms for PSON = 1 where the PHASEx sets the turnon time of the LEDs.

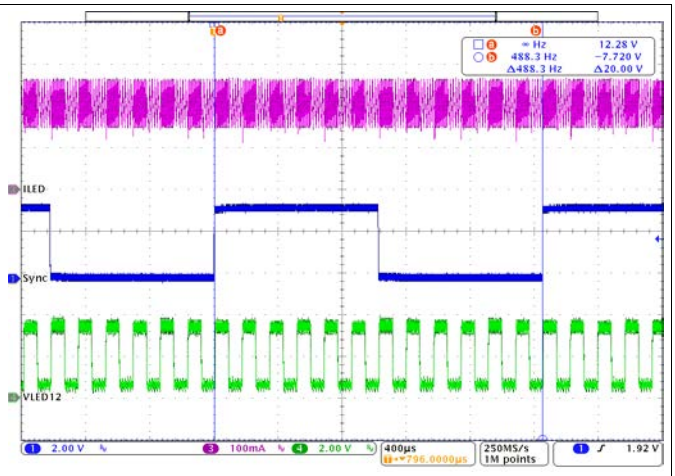


CH1 (Blue) : Sync Signal

CH4 (Green) : V_{LED12}

CH3 (Pink) : I_{LED}

Figure 38. PWM Dimming with PSON = 0



CH1 (Blue) : Sync Signal

CH4 (Green) : V_{LED12}

CH3 (Pink) : I_{LED}

Figure 39. PWM Dimming with PSON = 1

9.3 What to Do and What Not to Do

- Check the TPS92662A-Q1 case and junction temperature during and after prototyping the solution.
- Check the soldering of the device thermal pad (DAP) in production.
- Minimize the capacitance at the output of the current source, between the top of the string (sub-string), and the GND to avoid excessive overshoot current through the un-shunted LEDs when PWM dimming is applied.
- Apply appropriate phase shift between LED turnon (or turnoff) time to minimize the ECU instantaneous input current (see the *PWM Phase Shift* section for more details).

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9.4 Initialization Set Up

Figure 40 outlines the steps required to begin communication with the TPS92662A-Q1 device and PWMing the LEDs.

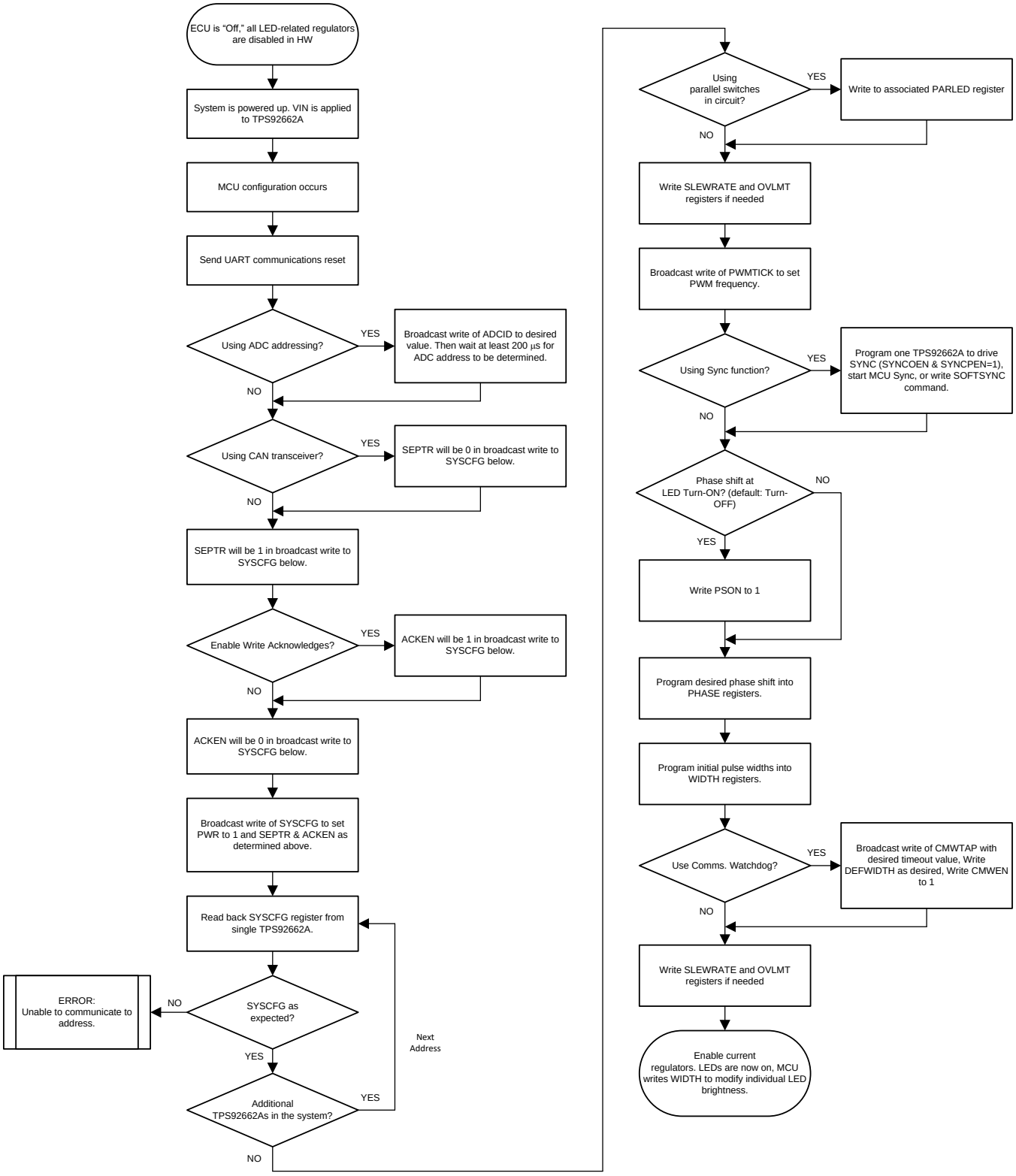


Figure 40. TPS92662A-Q1 Initialization Logic Diagram

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10 Power Supply Recommendations

10.1 General Recommendations

The TPS92662A-Q1 device requires a supply voltage on the VIN pin (V_{IN} range: 4.5 V to 60 V) to power the charge pump, internal logic, and references. The VIN rail generates multiple low voltage rails of VDD = 5.0 V, which is used for the internal analog blocks, digital communications, and the internal ADCs as outlined in the *Internal Regulators* section. The TPS92662A-Q1 device is only compatible with 5-V logic levels.

10.2 Internal Regulators

Figure 41 showsz= the on-board LDOs of the TPS92662A-Q1 device and the associated pins. All the power pins must be bypassed by a 0.1-µF ceramic capacitor.

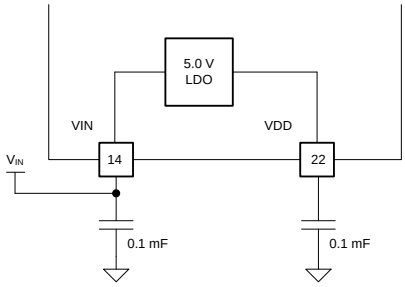


Figure 41. TPS92662A-Q1 Internal Regulator

If the V_{IN} supply is equivalent or less than 5 V, the VIN pin must be connected to the VDD pin. This connection bypasses the internal VDD LDO as shown in Figure 42.

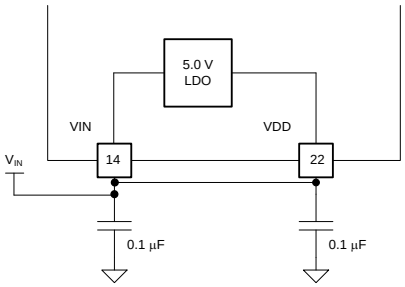


Figure 42. Power Pin Connections for VIN ≤ 5 V

The VDD pin is the positive voltage rail for the digital I/Os of the device.

10.3 Power Up and Reset

VIN must be greater than $VIN_{(UVLO-R)}$ prior to sourcing current through the LED string to ensure a controlled start-up. See the *Failsafe Mode At Power-Up* section for more details on the initial condition of the bypass switches upon power-up.

10.4 V_{IN} Power Consumption

Power consumption increases with increased clock frequency, VIN supply, and temperature. It is always best to select lower V_{IN} and the lowest clock frequency that the system can tolerate.

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11 Layout

11.1 Layout Guidelines

The system-level configuration of an LED matrix application varies greatly. Layout considerations for passive components follow a simple rule: place the VDD and VIN decoupling capacitors close to the device and make the traces between the CPPx pins and the associated bypass capacitors short. The challenge is to develop a careful layout plan that efficiently routes the current source to the LED strings. The communication pins are designed for ease of routing on a single-sided, metal core board. Each communications pin has a duplicate pin on the opposite side of the device, allowing multiple devices to use a daisy chain configuration.

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11.2 Layout Example

Figure 43 shows a TPS92662A-Q1 application layout example.

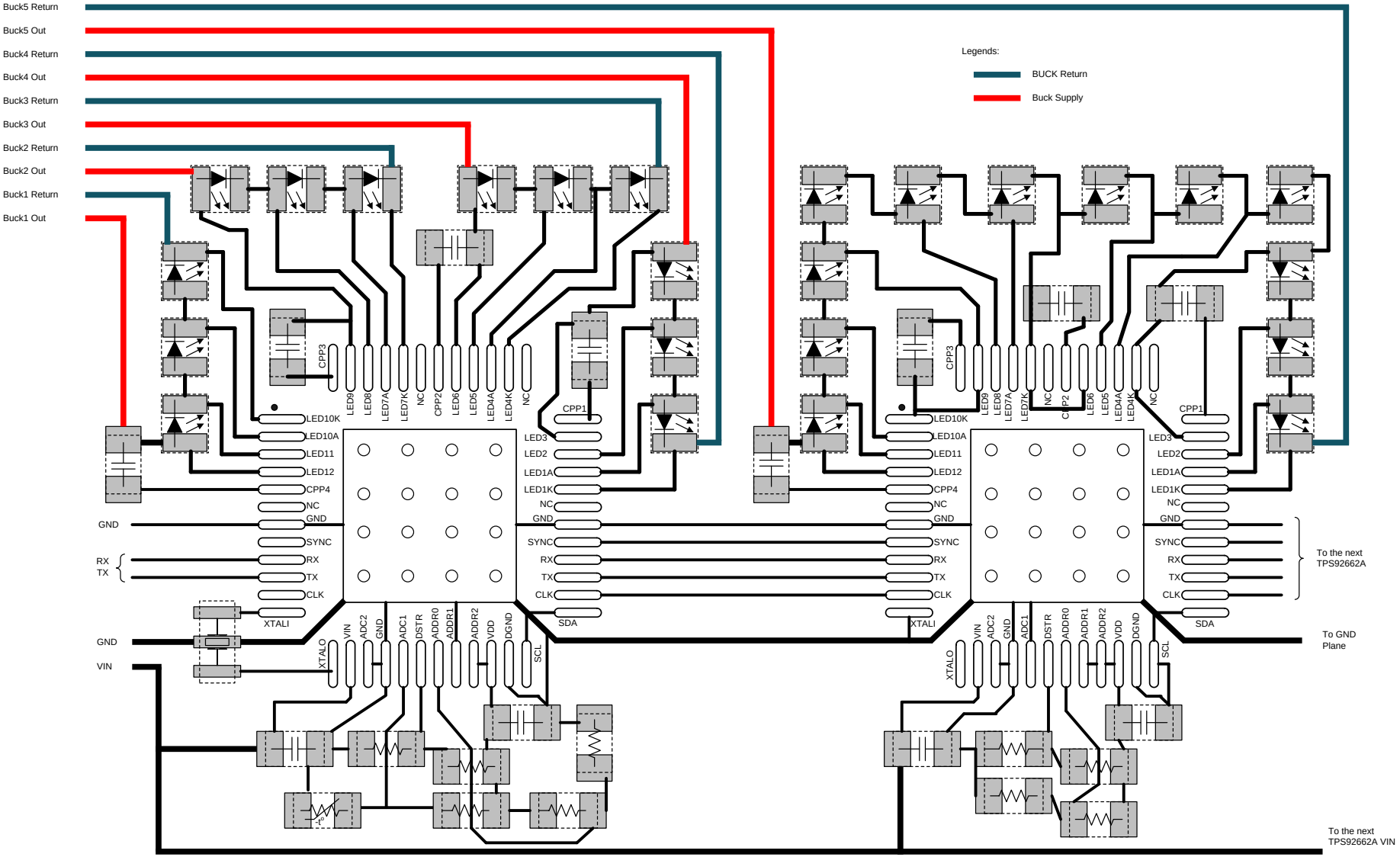


Figure 43. Layout Example

12 Device and Documentation Support

12.1 Support Resources

TI E2E™ support forums are an engineer's go-to source for fast, verified answers and design help — straight from the experts. Search existing answers or ask your own question to get the quick design help you need.

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12.2 Trademarks

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12.3 Electrostatic Discharge Caution



This integrated circuit can be damaged by ESD. Texas Instruments recommends that all integrated circuits be handled with appropriate precautions. Failure to observe proper handling and installation procedures can cause damage.

ESD damage can range from subtle performance degradation to complete device failure. Precision integrated circuits may be more susceptible to damage because very small parametric changes could cause the device not to meet its published specifications.

12.4 Glossary

SLYZ022 — *TI Glossary*.

This glossary lists and explains terms, acronyms, and definitions.

13 Mechanical, Packaging, and Orderable Information

The following pages include mechanical, packaging, and orderable information. This information is the most current data available for the designated devices. This data is subject to change without notice and revision of this document. For browser-based versions of this data sheet, refer to the left-hand navigation.

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13.1 Package Option Addendum

13.1.1 Packaging Information

Orderable Device	Status (1)	Package Type	Package Drawing	Pins	Package Qty	Eco Plan (2)	Lead/Ball Finish (3)	MSL Peak Temp (4)	Op Temp (°C)	Device Marking (5) (6)
TPS92662AQPHPRQ1	PREVIEW	HTQFP	PHP	48	1000	Green (RoHS and no Sb/Br)	CU NIPDAU	Level-3-260C-168 HR	-40 to 125	P92662AQ

(1) The marketing status values are defined as follows:
ACTIVE: Product device recommended for new designs.
LIFEBUY: TI has announced that the device will be discontinued, and a lifetime-buy period is in effect.
NRND: Not recommended for new designs. Device is in production to support existing customers, but TI does not recommend using this part in a new design.
PRE_PROD Unannounced device, not in production, not available for mass market, nor on the web, samples not available.
PREVIEW: Device has been announced but is not in production. Samples may or may not be available.
OBSOLETE: TI has discontinued the production of the device.

(2) Eco Plan - The planned eco-friendly classification: Pb-Free (RoHS), Pb-Free (RoHS Exempt), or Green (RoHS & no Sb/Br) - please check http://www.ti.com/productcontent for the latest availability information and additional product content details.
TBD: The Pb-Free/Green conversion plan has not been defined.
Pb-Free (RoHS): TI's terms "Lead-Free" or "Pb-Free" mean semiconductor products that are compatible with the current RoHS requirements for all 6 substances, including the requirement that lead not exceed 0.1% by weight in homogeneous materials. Where designed to be soldered at high temperatures, TI Pb-Free products are suitable for use in specified lead-free processes.
Pb-Free (RoHS Exempt): This component has a RoHS exemption for either 1) lead-based flip-chip solder bumps used between the die and package, or 2) lead-based die adhesive used between the die and leadframe. The component is otherwise considered Pb-Free (RoHS compatible) as defined above.
Green (RoHS & no Sb/Br): TI defines "Green" to mean Pb-Free (RoHS compatible), and free of Bromine (Br) and Antimony (Sb) based flame retardants (Br or Sb do not exceed 0.1% by weight in homogeneous material)

(3) Lead/Ball Finish - Orderable Devices may have multiple material finish options. Finish options are separated by a vertical ruled line. Lead/Ball Finish values may wrap to two lines if the finish value exceeds the maximum column width.

(4) MSL, Peak Temp. -- The Moisture Sensitivity Level rating according to the JEDEC industry standard classifications, and peak solder temperature.

(5) There may be additional marking, which relates to the logo, the lot trace code information, or the environmental category on the device

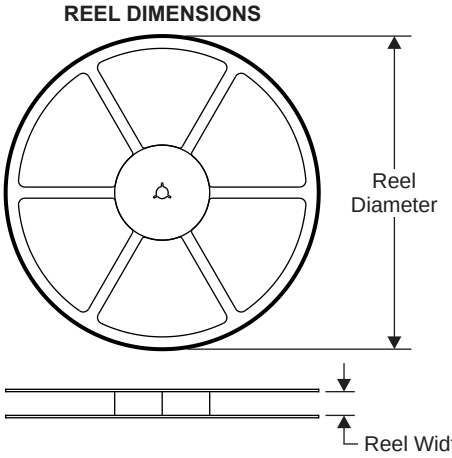
(6) Multiple Device markings will be inside parentheses. Only one Device Marking contained in parentheses and separated by a "~" will appear on a device. If a line is indented then it is a continuation of the previous line and the two combined represent the entire Device Marking for that device.

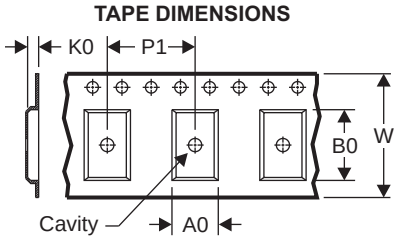
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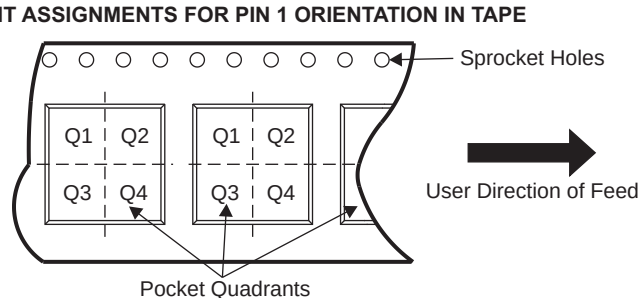
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13.1.2 Tape and Reel Information





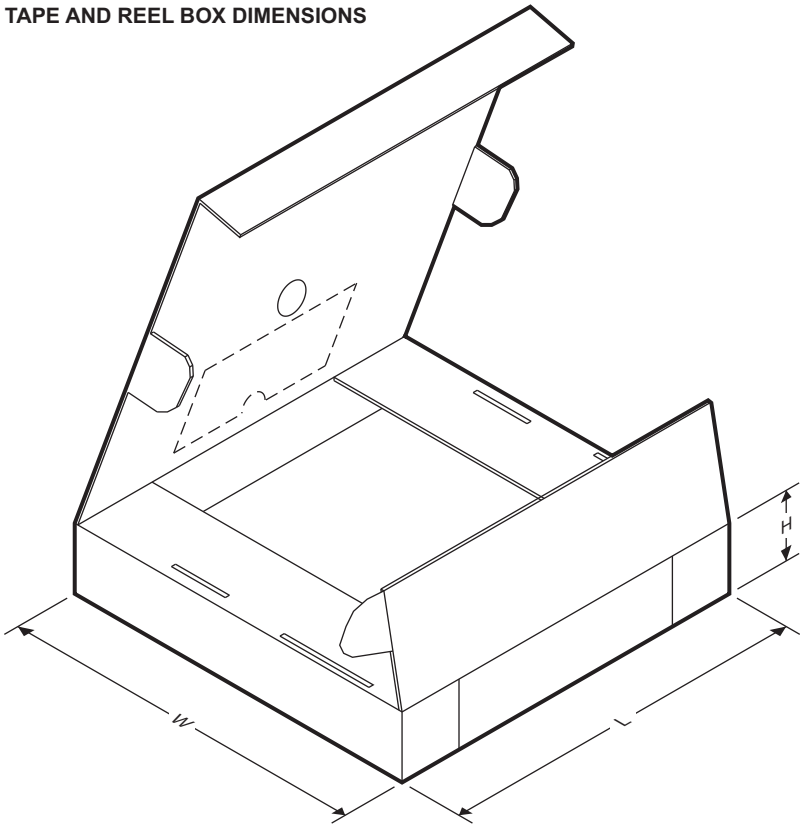
A0	Dimension designed to accommodate the component width
B0	Dimension designed to accommodate the component length
K0	Dimension designed to accommodate the component thickness
W	Overall width of the carrier tape
P1	Pitch between successive cavity centers



Device	Package Type	Package Drawing	Pins	SPQ	Reel Diameter (mm)	Reel Width W1 (mm)	A0 (mm)	B0 (mm)	K0 (mm)	P1 (mm)	W (mm)	Pin1 Quadrant
TPS92662AQPHPRQ1	HTQFP	PHP	48	1000	330	16.4	9.6	9.6	1.5	12	16	Q2

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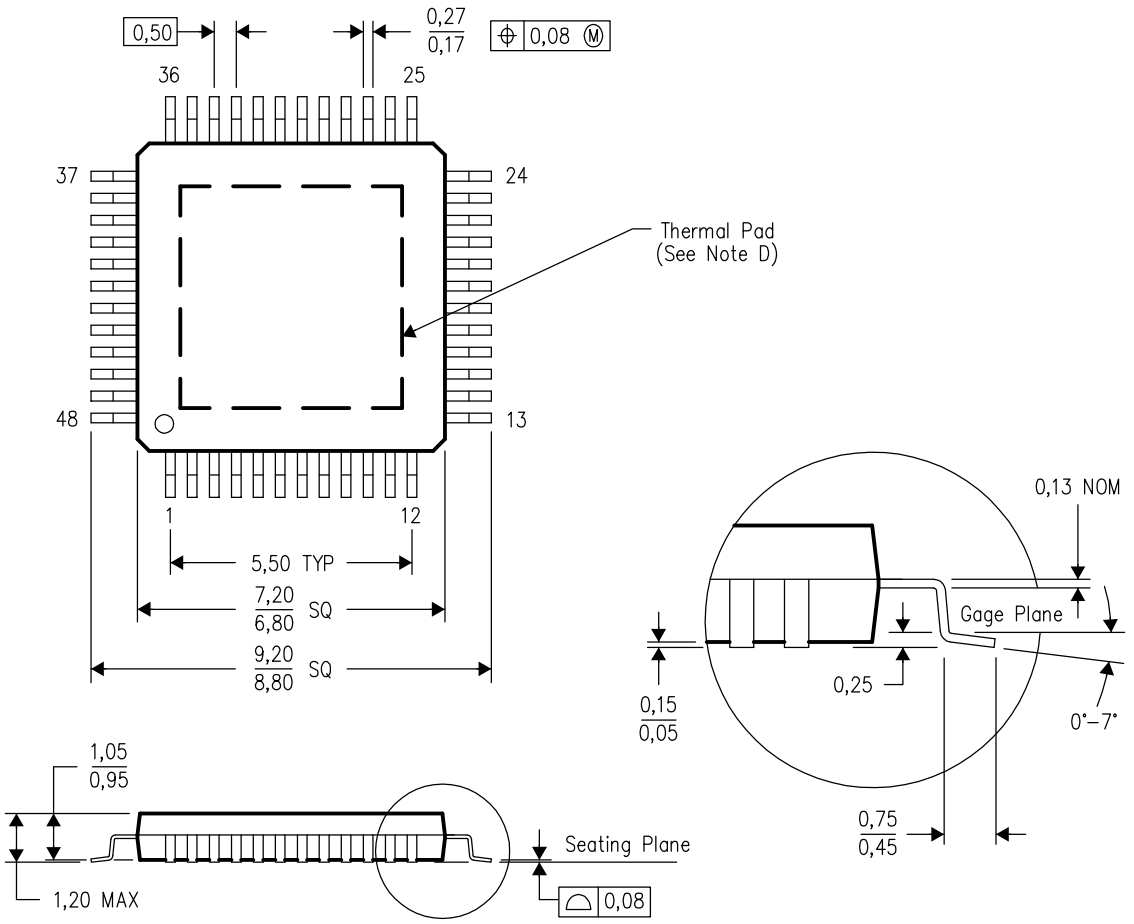
TAPE AND REEL BOX DIMENSIONS



Device	Package Type	Package Drawing	Pins	SPQ	Length (mm)	Width (mm)	Height (mm)
TPS92662AQPHPRQ1	HTQFP	PHP	48	1000	336.6	336.6	31.8

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PHP (S–PQFP–G48) PowerPAD™ PLASTIC QUAD FLATPACK



NOTES:
A. All linear dimensions are in millimeters.
B. This drawing is subject to change without notice.
C. Body dimensions do not include mold flash or protrusion
D. This package is designed to be soldered to a thermal pad on the board. Refer to Technical Brief, PowerPad Thermally Enhanced Package, Texas Instruments Literature No. SLMA002 for information regarding recommended board layout. This document is available at www.ti.com <http://www.ti.com>.
E. Falls within JEDEC MS–026

PowerPAD is a trademark of Texas Instruments.

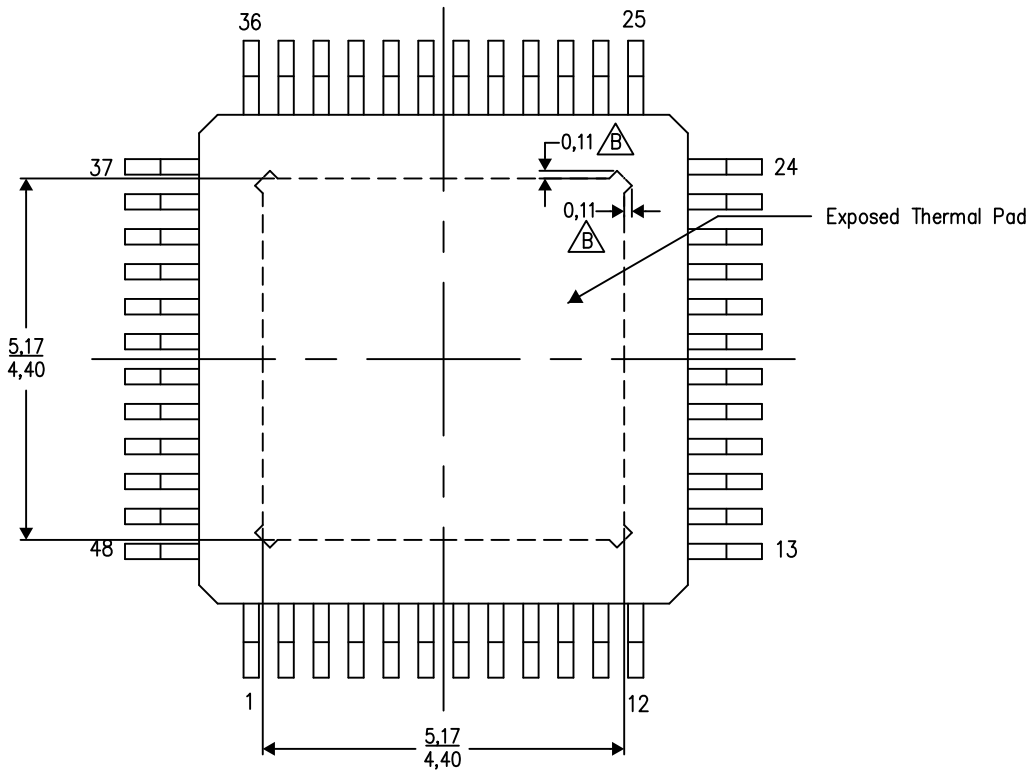
PHP (S–PQFP–G48) PowerPAD™ PLASTIC QUAD FLATPACK

THERMAL INFORMATION

This PowerPAD™ package incorporates an exposed thermal pad that is designed to be attached to a printed circuit board (PCB). The thermal pad must be soldered directly to the PCB. After soldering, the PCB can be used as a heatsink. In addition, through the use of thermal vias, the thermal pad can be attached directly to the appropriate copper plane shown in the electrical schematic for the device, or alternatively, can be attached to a special heatsink structure designed into the PCB. This design optimizes the heat transfer from the integrated circuit (IC).

For additional information on the PowerPAD package and how to take advantage of its heat dissipating abilities, refer to Technical Brief, PowerPAD Thermally Enhanced Package, Texas Instruments Literature No. SLMA002 and Application Brief, PowerPAD Made Easy, Texas Instruments Literature No. SLMA004. Both documents are available at www.ti.com.

The exposed thermal pad dimensions for this package are shown in the following illustration.



Top View

Exposed Thermal Pad Dimensions

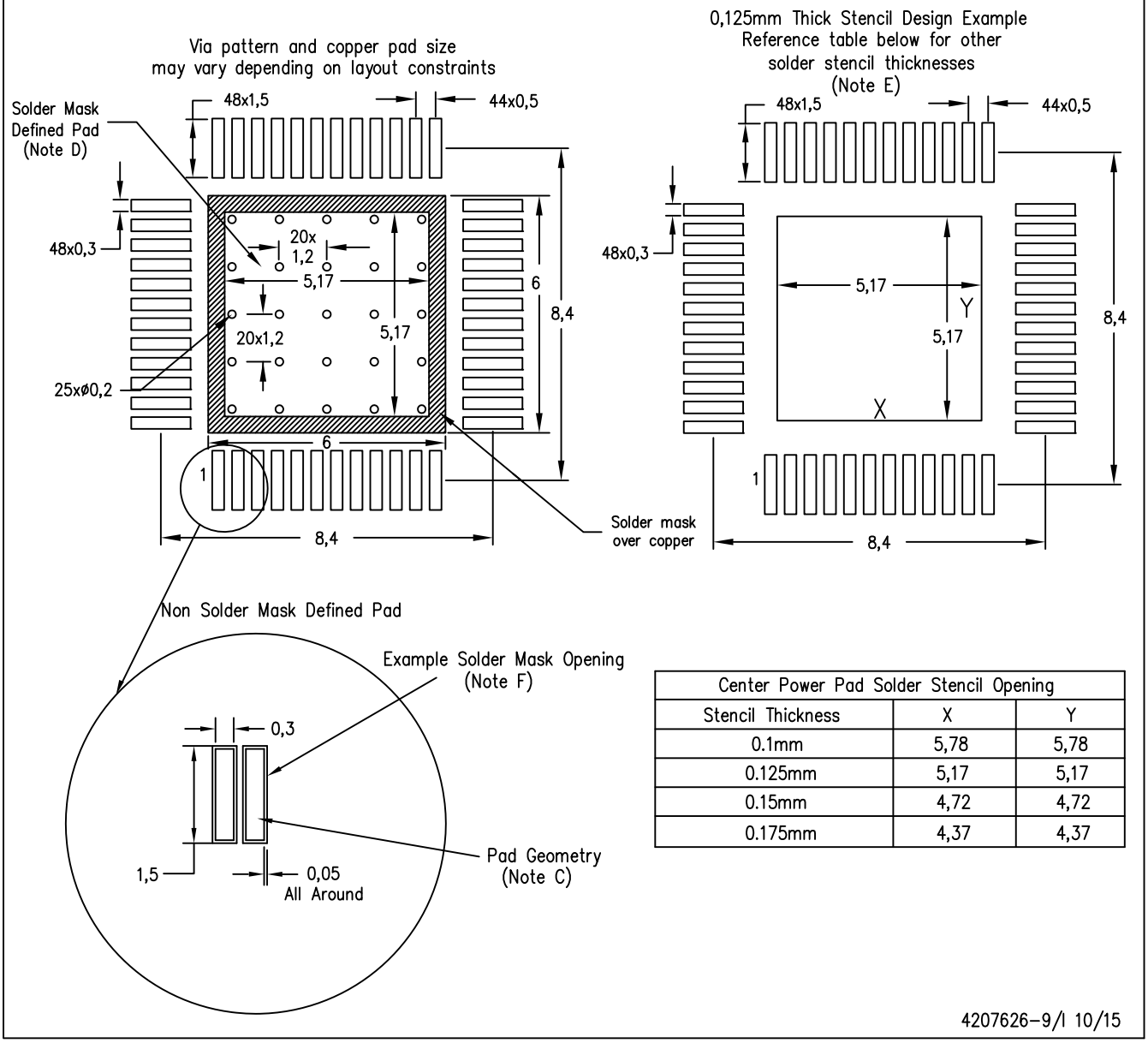
4206329–7/P 03/15

NOTE: A. All linear dimensions are in millimeters

B Tie strap features may not be present.

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PHP (S–PQFP–G48) PowerPAD™ PLASTIC QUAD FLATPACK



Center Power Pad Solder Stencil Opening		
Stencil Thickness	X	Y
0.1mm	5,78	5,78
0.125mm	5,17	5,17
0.15mm	4,72	4,72
0.175mm	4,37	4,37

NOTES:
A. All linear dimensions are in millimeters.
B. This drawing is subject to change without notice.
C. Publication IPC–7351 is recommended for alternate designs.
D. This package is designed to be soldered to a thermal pad on the board. Refer to Technical Brief, PowerPad Thermally Enhanced Package, Texas Instruments Literature No. SLMA002, SLMA004, and also the Product Data Sheets for specific thermal information, via requirements, and recommended board layout. These documents are available at www.ti.com <http://www.ti.com>.
E. Laser cutting apertures with trapezoidal walls and also rounding corners will offer better paste release. Customers should contact their board assembly site for stencil design recommendations. Refer to IPC 7525 for stencil design considerations.
F. Customers should contact their board fabrication site for recommended solder mask tolerances and via tenting options for vias placed in the thermal pad.

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